1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

J44 MLB-4GB SCHEMATIC
08/20/2013

ALIASES RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-0052	1	SCHEM,MLB-4GB,J44	SCH	CRITICAL	
820-3536	1	PCBF,MLB-4GB,J44	PCB	CRITICAL	

DRAWING TITLE <PART_DESCRIPTION>

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DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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BOM Groups

BOM GROUP	BOM OPTIONS
J44_COMMON	ALTERNATE,COMMON,J44_COMMON1,J44_COMMON2,J44_COMMON3,J44_COMMON4,J44_PROGPARTS
J44_COMMON1	TBTHV:P15V,SKIP_5V3V3:AUDIBLE,SPI:DUAL_IO
J44_COMMON2	EDP,EDP_LS_CAP,CAMERA_3V3:S0,CAM_WAKE:NO,CAM_XTAL:NO,MEM_ODT:PU,VCORE_FETS
J44_COMMON3	XDP,LPCPLUS,BKLT:PROD,CPUTHRM:ALRT,LOADRC:NO,OTHERRC:NO,DDRRC:NO,TBTRC:NO,BMONRC:NO
J44_PROGPARTS	SMC_PROG:PVT,BOOTROM:PVT,TBTROM:PVT,TPAD_PSOC:PROG
ENGISNS	LOADISNS,OTHERISNS,DDRISNS,TBTISNS,BMONISNS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4596	1	HSWULT,SR18A,PRQ,C0,2.4,28W,2+3,3M,BGA	U0500	CRITICAL	CPU_HSW:2.4G
337S4597	1	HSWULT,SR189,PRQ,C0,2.6,28W,2+3,3M,BGA	U0500	CRITICAL	CPU_HSW:2.6G
337S4598	1	HSWULT,SR188,PRQ,C0,2.8,28W,2+3,4M,BGA	U0500	CRITICAL	CPU_HSW:2.8G
338S1247	1	IC,TBT,FR-4C,A0,PRQ,CIO,SR1JC,FCBGA288	U2800	CRITICAL	
338S1186	1	IC,BCM15700A2,S2 PCIE CAMERA PROCESSOR	U3900	CRITICAL	
376S1194	2	MOSFET,N-CH,30V,15.3A,12M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:VSHY
376S1193	2	MOSFET,N-CH,30V,22A,6.0M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:VSHY
376S0964	2	MOSFET,N-CH,25V,30A,9.6M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:REN
376S1104	2	MOSFET,N-CH,25V,30A,6.1M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:REN

Programmables (All Builds)

TBT

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S3918	1	EPROM,FALCON RIDGE (V13.7) J44	U2890	CRITICAL	TBTROM:PVT

SMC

341S3922	1	IC,SMC-B1,EXT(V2.16F39),PVT,J44	U5000	CRITICAL	SMC_PROG:PVT

EFI ROM

341S3924	1	IC,EFI ROM (V0116),PVT,J44	U6100	CRITICAL	BOOTROM:PVT

PSOC

341S3862	1	IC,TRKPD/KYBD PSOC,CU ONLY(V224) J44	U4801	CRITICAL	TPAD_PSOC:PROG

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	Diodes alt to Fairchild
128S0311	128S0329		ALL	NEC alt to Sanyo
138S0739	138S0706		ALL	Samsung alt to Murata
197S0481	197S0480		ALL	Epson alt to NDK
152S0461	152S1645		ALL	Cyntec alt to Vishay
376S1080	376S0820		ALL	Diodes alt to On Semi
155S0667	155S0583		ALL	Panasonic alt to TDK
138S0725	138S0724		ALL	Samsung alt to Murata
376S1032	376S0855		ALL	Toshiba alt for Diodes Dual
376S1129	376S0855		ALL	NXP Alt for Diodes Dual
376S1089	376S1128		ALL	NXP Alt for Diodes Single
353S3452	353S1286		ALL	Maxim alt to Microchip
376S1180	376S0761		ALL	Renesas alt to Vishay
128S0364	128S0264		ALL	Sanyo 2nd Factory alt
107S0254	107S0241		ALL	Cyntec alt to TFT
138S0843	138S0674		ALL	Samsung alt to Murata (BKLT)
138S0803	138S0639		ALL	Samsung alt to Murata (BKLT)
138S0846	138S0811		ALL	Samsung alt to Murata (BKLT)
197S0542	197S0544		ALL	NDK alt to TXC
197S0545	197S0544		ALL	Epson alt to TXC
152S1876	152S1804		ALL	TDK alt to Toko
107S0255	107S0240		ALL	Cyntec alt to TFT
107S0250	107S0248		ALL	Cyntec alt to TFT
127S0164	127S0162		ALL	Rohm alt to Vishay
353S4070	353S4069		ALL	Pericom alt to TI DP Mux U9750
353S4068	353S4069		ALL	NXP alt to TI DP Mux U9750
353S3814	353S3812		ALL	TI alt to NXP
311S0649	311S0541		ALL	ONsemi alt to Toshiba
128S0436	128S0392		ALL	Kemet alt to Sanyo

SYNC_MASTER=J44 SYNC_DATE=08/20/2013

PAGE TITLE

BOM Configuration

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-0054	COMMON,MLB-4GB,J44	J44_COMMON
985-0053	DEV,MLB-4GB,J44	XDP_CONN
639-4878	PCBA,MLB-4GB,2.4G,4GB-HYNIX,J44	BASE_BOM,CPU_HSW:2.4G,RAM_4G_HYNIX_H,CAMDRAM:HYNIX_H
639-4879	PCBA,MLB-4GB,2.4G,4GB-ELPIDA,J44	BASE_BOM,CPU_HSW:2.4G,RAM_4G_ELPIDA,CAMDRAM:ELPIDA
639-4880	PCBA,MLB-4GB,2.4G,4GB-MICRON,J44	BASE_BOM,CPU_HSW:2.4G,RAM_4G_MICRON,CAMDRAM:MICRON
639-5272	PCBA,MLB-4GB,2.6G,4GB-HYNIX,J44	BASE_BOM,CPU_HSW:2.6G,RAM_4G_HYNIX_H,CAMDRAM:HYNIX_H
639-5273	PCBA,MLB-4GB,2.6G,4GB-ELPIDA,J44	BASE_BOM,CPU_HSW:2.6G,RAM_4G_ELPIDA,CAMDRAM:ELPIDA
639-5274	PCBA,MLB-4GB,2.6G,4GB-MICRON,J44	BASE_BOM,CPU_HSW:2.6G,RAM_4G_MICRON,CAMDRAM:MICRON
639-5275	PCBA,MLB-4GB,2.8G,4GB-HYNIX,J44	BASE_BOM,CPU_HSW:2.8G,RAM_4G_HYNIX_H,CAMDRAM:HYNIX_H
639-5276	PCBA,MLB-4GB,2.8G,4GB-ELPIDA,J44	BASE_BOM,CPU_HSW:2.8G,RAM_4G_ELPIDA,CAMDRAM:ELPIDA
639-5277	PCBA,MLB-4GB,2.8G,4GB-MICRON,J44	BASE_BOM,CPU_HSW:2.8G,RAM_4G_MICRON,CAMDRAM:MICRON
685-0074	VCORE,FET,VSHY,J44	VCORE_FET:VSHY
685-0075	VCORE,FET,REN,J44	VCORE_FET:REN

DEVELOPMENT/BASE BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-0054	1	J44 MLB COMMON BOM	BASE	CRITICAL	BASE_BOM
985-0053	1	J44 MLB DEVEL BOM	DEVEL	CRITICAL	DEVEL_BOM

SUB-BOMS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-0074	1	VCORE,FET,VSHY,J44	VCOREFETS	CRITICAL	VCORE_FETS

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-0075	685-0074		ALL	RENESAS ALT TO VISHAY

DRAM PARTS

333S0704	8	IC,SDRAM,4GBIT,256MX16,DDR3-1600,F DIE,96FBGA	U2300,U2320,U2340,U2360,U2500,U2520,U2540,U2560	CRITICAL	4G_ELPIDA
333S0700	8	IC,SDRAM,4GBIT,256MX16,DDR3-1600,HUMA,96FBGA	U2300,U2320,U2340,U2360,U2500,U2520,U2540,U2560	CRITICAL	4G_HYNIX_H
333S0698	8	IC,SDRAM,4GBIT,256MX16,DDR3-1600,REV E,96FBGA	U2300,U2320,U2340,U2360,U2500,U2520,U2540,U2560	CRITICAL	4G_MICRON
333S0715	8	IC,SDRAM,4GBIT,256MX16,DDR3-1866,F DIE,96FBGA	U2300,U2320,U2340,U2360,U2500,U2520,U2540,U2560	CRITICAL	4G_ELPIDA_1866
333S0717	8	IC,SDRAM,4GBIT,256MX16,DDR3-1866,HUMA,96FBGA	U2300,U2320,U2340,U2360,U2500,U2520,U2540,U2560	CRITICAL	4G_HYNIX_H_1866
333S0720	8	IC,SDRAM,4GBIT,256MX16,DDR3-1866,REV E,96FBGA	U2300,U2320,U2340,U2360,U2500,U2520,U2540,U2560	CRITICAL	4G_MICRON_1866

DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM_4G_ELPIDA	4G_ELPIDA,RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:L,PPDDR:1V35
RAM_4G_HYNIX_H	4G_HYNIX_H,RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:H,PPDDR:1V35
RAM_4G_MICRON	4G_MICRON,RAMCFG3:L,RAMCFG2:L,RAMCFG1:H,RAMCFG0:L,PPDDR:1V35
RAM_4G_ELPIDA_1866	4G_ELPIDA_1866,RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:L,PPDDR:1V5
RAM_4G_HYNIX_H_1866	4G_HYNIX_H_1866,RAMCFG3:L,RAMCFG2:L,RAMCFG1:L,RAMCFG0:H,PPDDR:1V5
RAM_4G_MICRON_1866	4G_MICRON_1866,RAMCFG3:L,RAMCFG2:L,RAMCFG1:H,RAMCFG0:L,PPDDR:1V5

NOTE: 1866 PARTS BEING STRAPPED TO RUN AT 1600

13" MBP VARIABLE BOM GROUPS

BOM GROUP	BOM OPTIONS
J44_COMMON4	SMCBOARDID:8

DRAM SPD Straps

BOM GROUP	BOM OPTIONS
CAMDRAM:HYNIX_H	CAMDRAM_TYPE:HYNIX_H
CAMDRAM:ELPIDA	CAMDRAM_TYPE:ELPIDA
CAMDRAM:MICRON	CAMDRAM_TYPE:MICRON

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0700	1	IC,SDRAM,4GBIT,DDR3L-1600,HUMA,96B FBGA	U4000	CRITICAL	CAMDRAM_TYPE:HYNIX_H
333S0704	1	IC,SDRAM,4GBIT,DDR3L-1600,DIE F,96B FBGA	U4000	CRITICAL	CAMDRAM_TYPE:ELPIDA
333S0698	1	IC,SDRAM,4GBIT,DDR3L-1600,REV E,96B FBGA	U4000	CRITICAL	CAMDRAM_TYPE:MICRON

Shield Cans

SH0451 SM
SHLD-J44-MLB
USB Cage

SH0450 SM
SHLD-J44-MLB-T29
TBT Cage

Mounting Holes & Slots

OMIT
ZT0411
4P5R2P3-3P5B
ABOVE GUMSTICK CARD IN MIDDLE OF BOARD
(998-1195)

OMIT
ZT0413
6.19X4.60-SNOWMAN
SNOWMAN-SHAPED SLOT AT LEFT OF MEMORY BANK
(998-5879)

OMIT
ZT0414
6.19X4.60-SNOWMAN
SNOWMAN-SHAPED SLOT AT RIGHT OF MEMORY BANK
(998-5879)

TH0400
TH-NSP
SL-1.1X0.5-1.4x0.8
Upper TBT can Ground slot
(862-0118)

TH0403
TH-NSP
SL-1.1X0.5-1.4x0.8
Lower TBT can Ground slot
(862-0118)

TH0404
TH-NSP
SL-1.1X0.45-1.4x0.75
USB can Ground slot
(998-3975)

TH0405
TH-NSP
SL-1.1X0.45-1.4x0.75
USB can Ground slot
(998-3975)

Rubber Mount Standoffs (860-1448)

SH0460
2.9OD1.2ID-1.35H-SM

SH0461
2.9OD1.2ID-1.35H-SM

SH0462
2.9OD1.2ID-1.35H-SM

SH0465
2.9OD1.2ID-1.35H-SM

SH0464
2.9OD1.2ID-1.35H-SM

SH0463
2.9OD1.2ID-1.35H-SM

SH0466
2.9OD1.2ID-1.35H-SM

SH0467
2.9OD1.2ID-1.35H-SM

THERMAL MODULE STANDOFF (860-1645)

SH0420
THERMAL-4.50-J44-SM

SH0421
THERMAL-4.50-J44-SM

SH0426
THERMAL-4.50-J44-SM

SH0427
THERMAL-4.50-J44-SM

RIO FLEX BRACKET BOSSES (860-2354)

SH0443
3.5OD2.0H-SM

SH0444
3.5OD2.0H-SM

SSD STANDOFF (806-5375)

SH0440
5.0OD2.0H-SM

FAN STANDOFF (806-5376)

SH0441
STDOFF-4.5OD1.73H-SM-1.33-3.2

POGO PINS (870-2451)

SH0435 & SH0436 removed.

SH0432
POGO-2.3OD-5.5H-SM-LOW-FORCE
SM

SH0433
POGO-2.3OD-5.5H-SM-LOW-FORCE
SM

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
PD Parts

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
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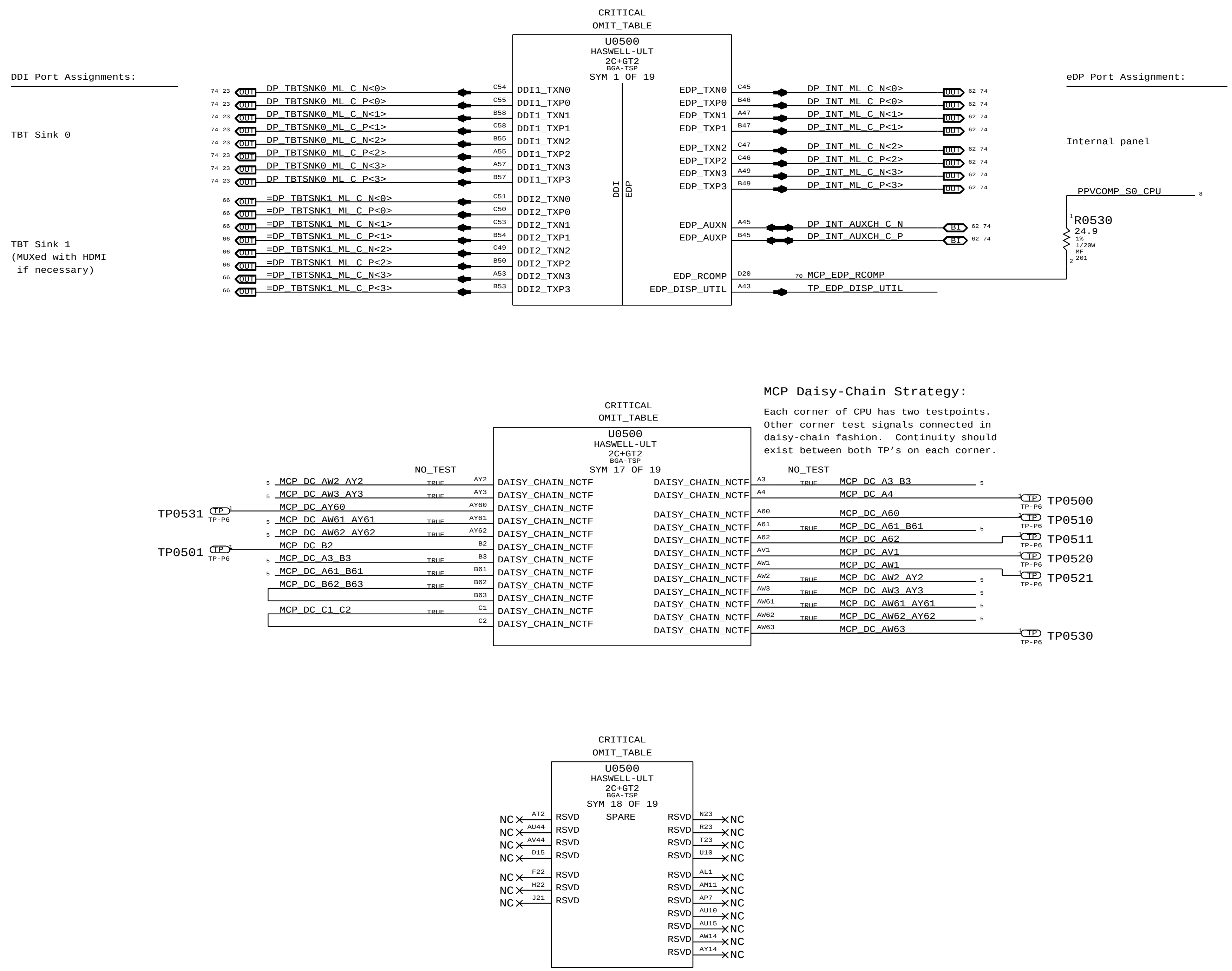

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PAGE TITLE

CPU GFX/NCTF/RSVD

Apple Inc.

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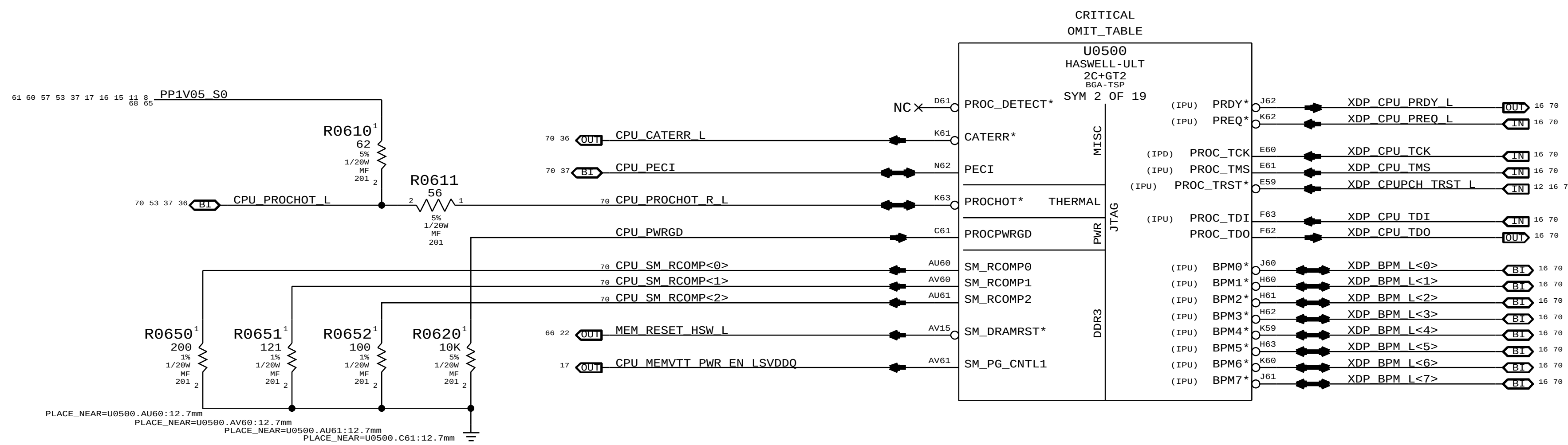
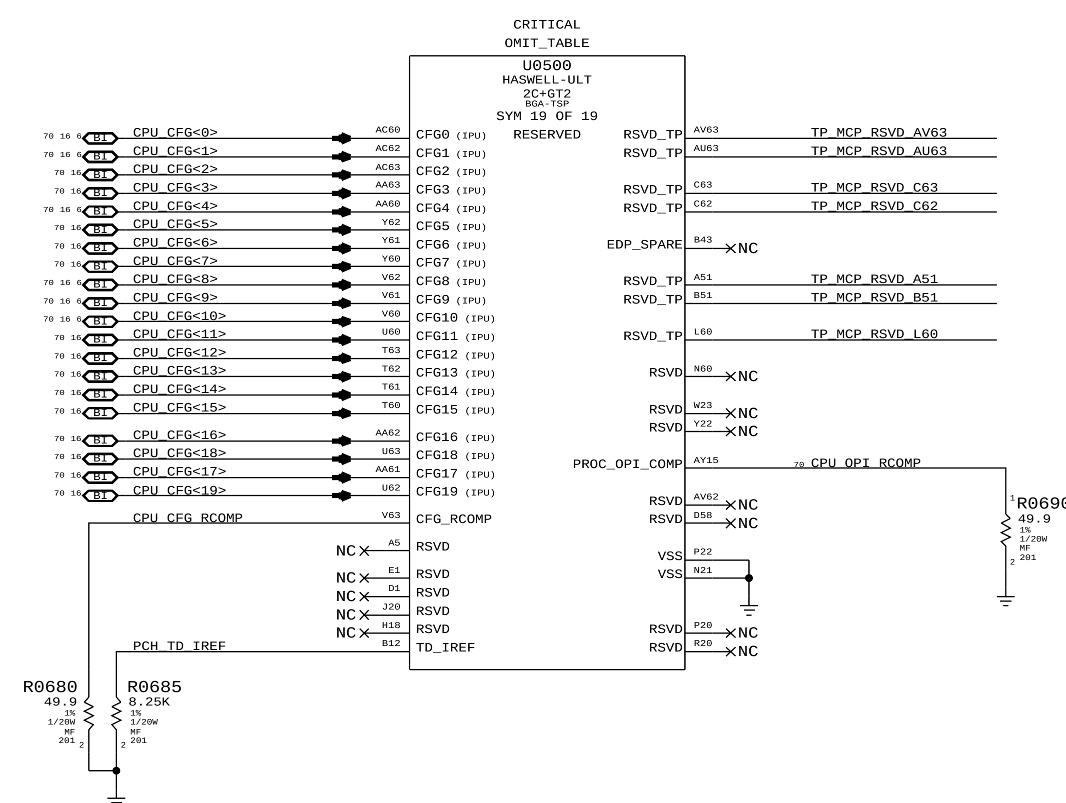
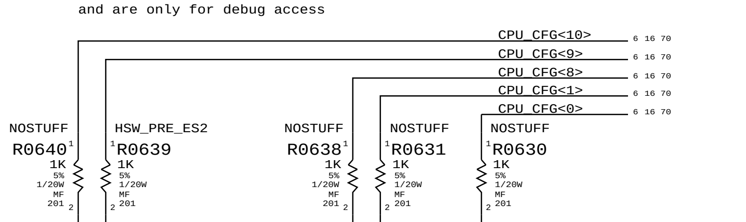


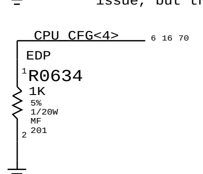


CFG<10>:SAFE MODE BOOT	1 = NORMAL OPERATION	0 = POWER FEATURES NOT ACTIVE
CFG<9> :NO SVID-CAPABLE VR	1 = VR SUPPORTS SVID	0 = VR DOES NOT SUPPORT SVID
CFG<8> :ALLOW NOA ON LOCKED UNITS	1 = NORMAL OPERATION	0 = NOA ALWAYS UNLOCKED
CFG<4> :eDP ENABLE/DISABLE	1 = DISABLED	0 = ENABLED
CFG<1> :PCH-LESS MODE	1 = NORMAL OPERATION	0 = PCH-LESS MODE
CFG<0> :RESET SEQUENCE STALL	1 = NORMAL OPERATION	0 = STALL AFTER PCU PLL LOCK

These can be placed close to J1800
and are only for debug access



NOTE: Pre-ES2 CPUs have issue with Sx cycling, must set CFG<9> low to avoid
issue, but this locks CPU VR at 1.7V Vboot (CPU Sighting #4391569).



SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

CPU Misc/JTAG/CFG/RSVD

Apple Inc.

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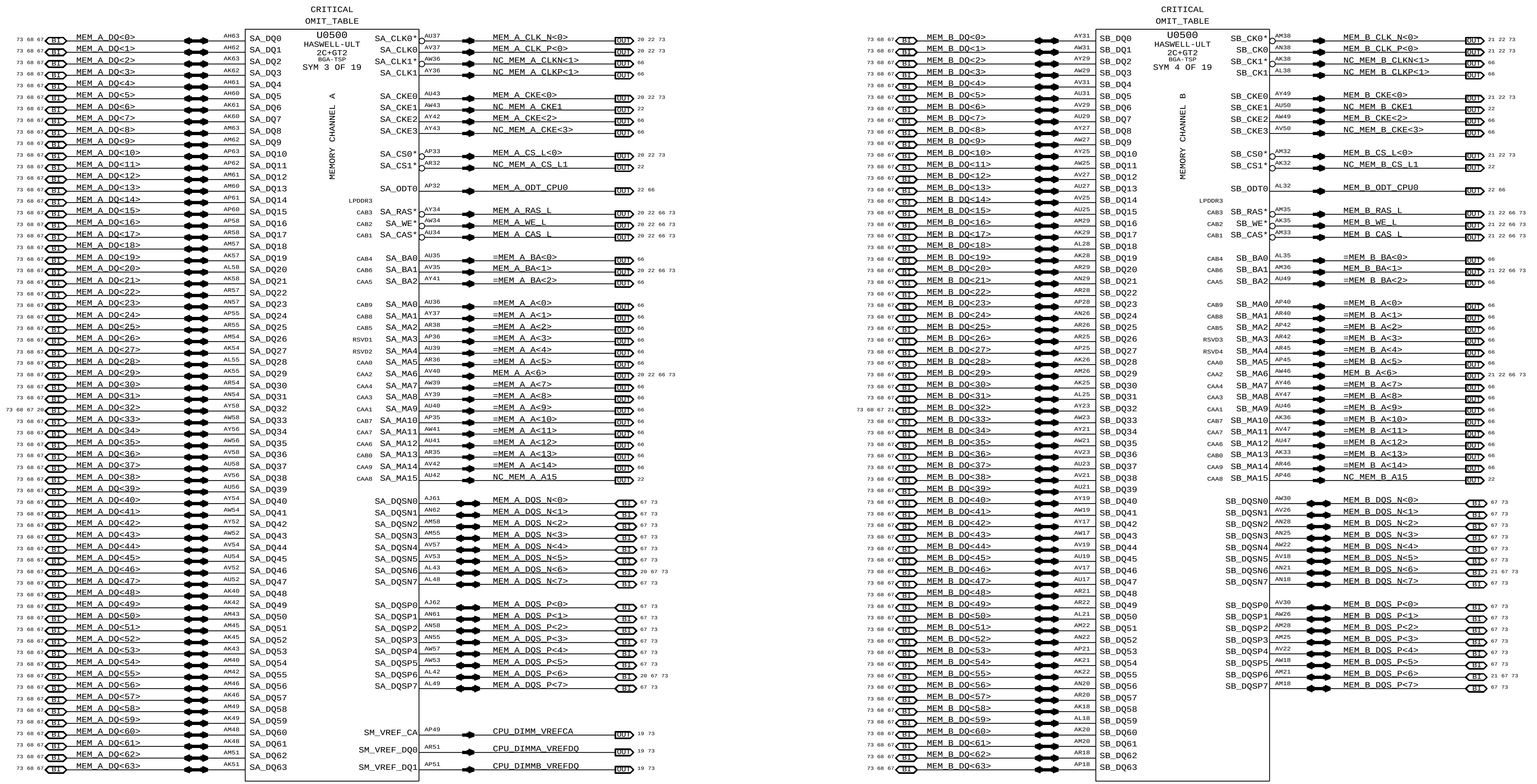

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

CPU DDR3/LPDDR3 Interfaces

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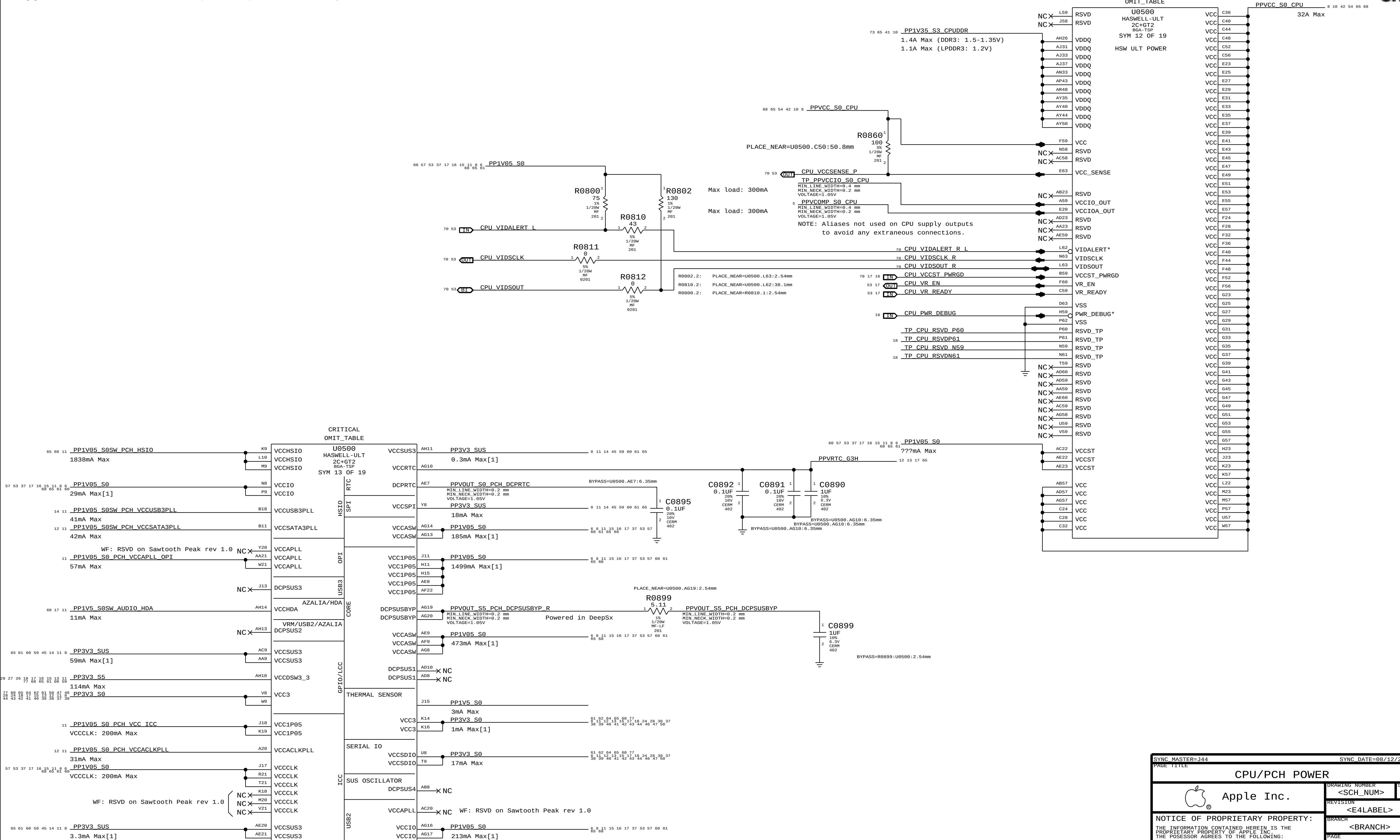
HSW-ULT current estimates from Haswell Mobile ULT Processor EDS vol 1, doc #502406, v0.9.
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0.
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.



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PAGE TITLE: CPU/PCH POWER

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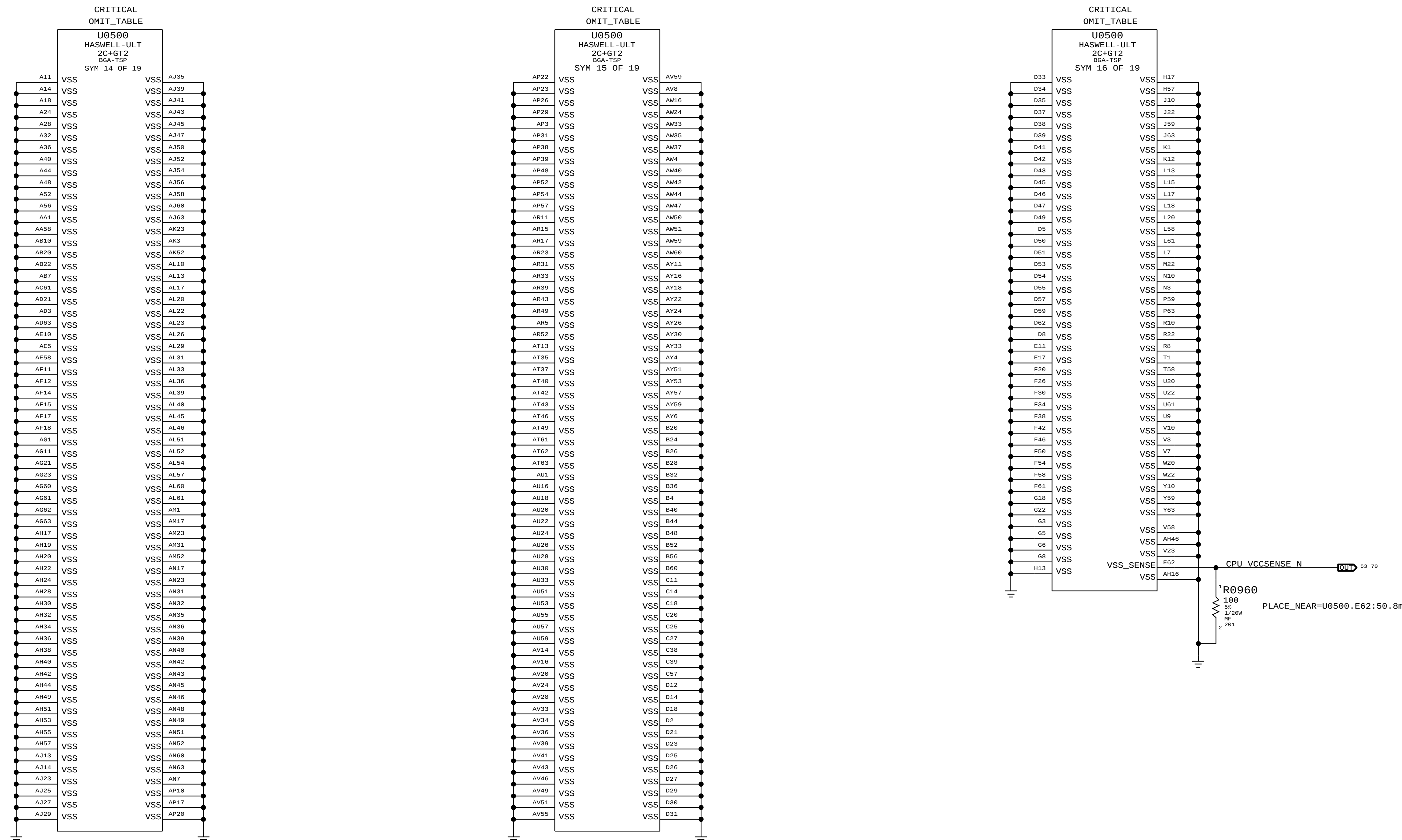

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PAGE TITLE

CPU/PCH GROUNDS

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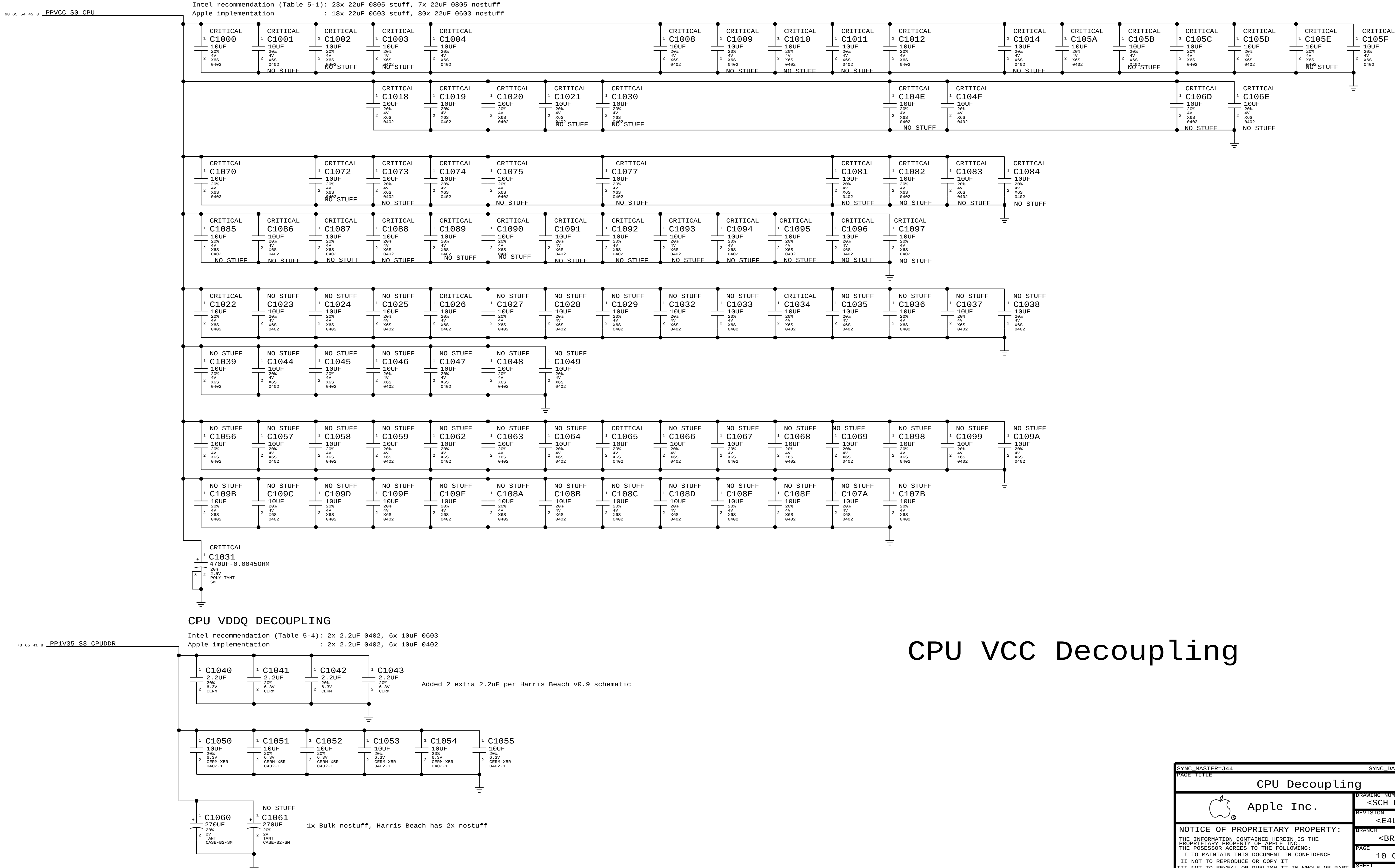
All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 0.9 unless stated otherwise

CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff
Apple implementation : 18x 22uF 0603 stuff, 80x 22uF 0603 nostuff



CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 2x 2.2uF 0402, 6x 10uF 0603
Apple implementation : 2x 2.2uF 0402, 6x 10uF 0402

Added 2 extra 2.2uF per Harris Beach v0.9 schematic

1x Bulk nostuff, Harris Beach has 2x nostuff

CPU VCC Decoupling

SYNC_MASTER=J44 SYNC_DATE=08/12/2013
PAGE TITLE
CPU Decoupling
Apple Inc.
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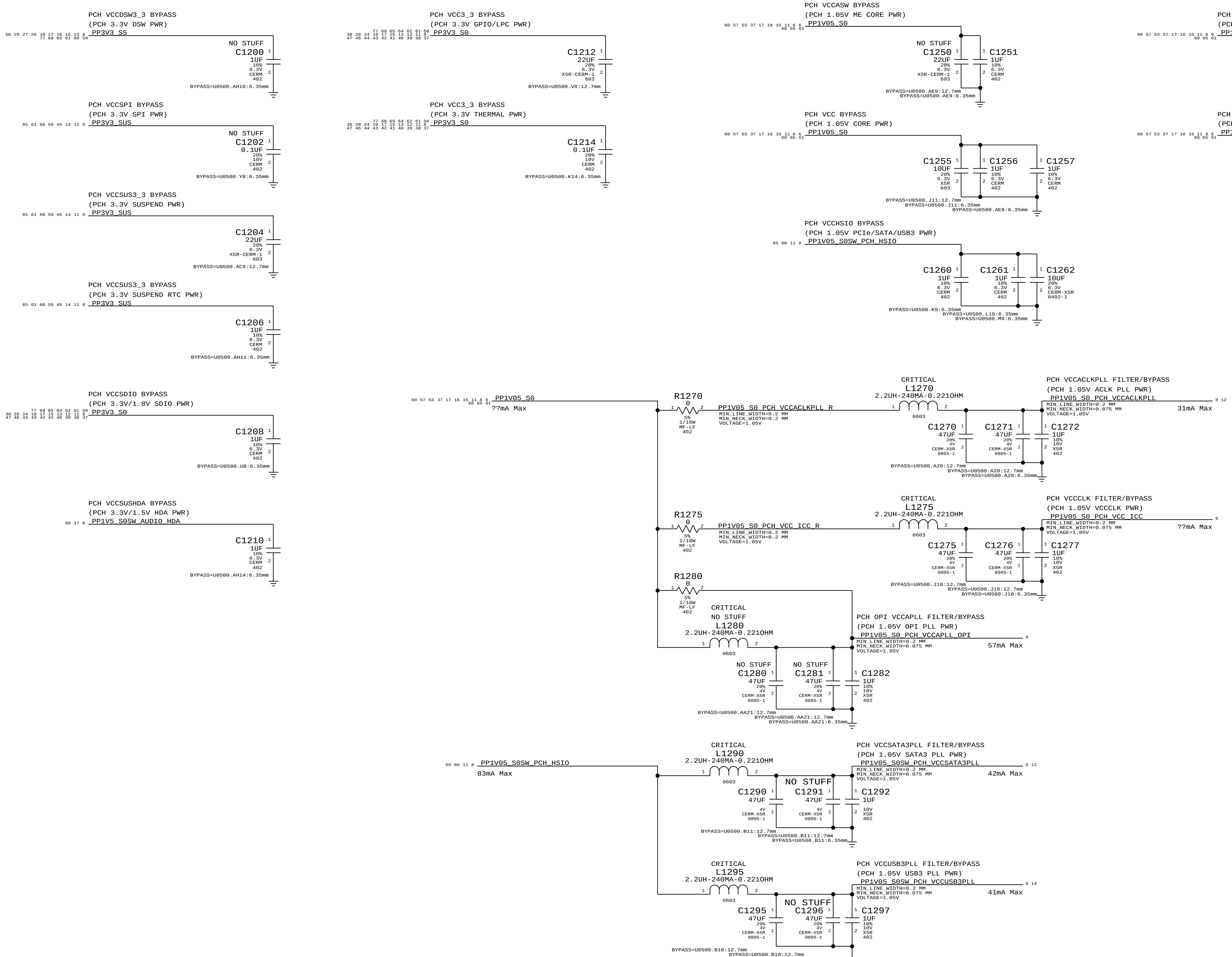

LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Srini, dated 9/10/2012 2:11pm.

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

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PCH Decoupling

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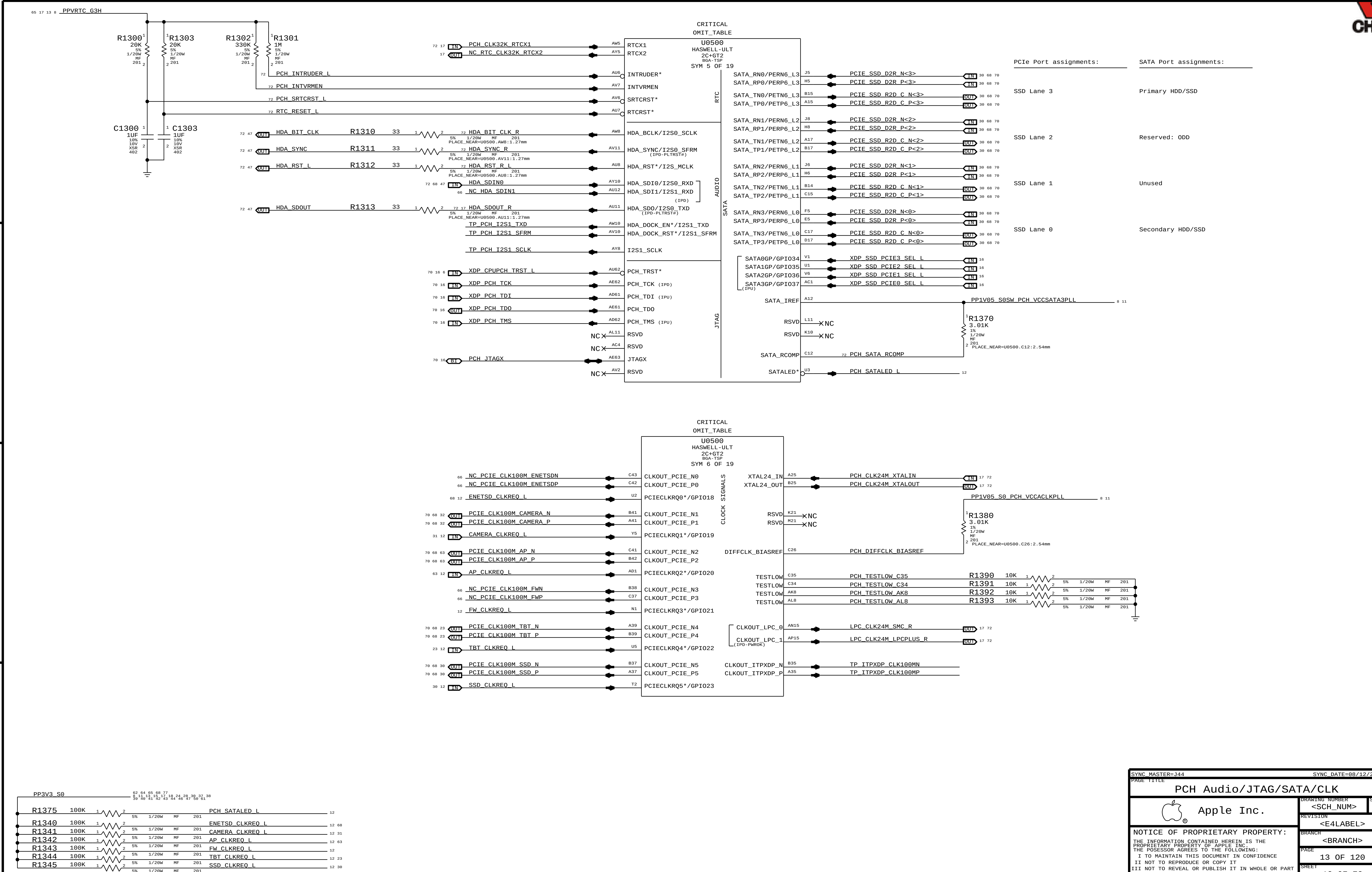

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PAGE TITLE

PCH Audio/JTAG/SATA/CLK

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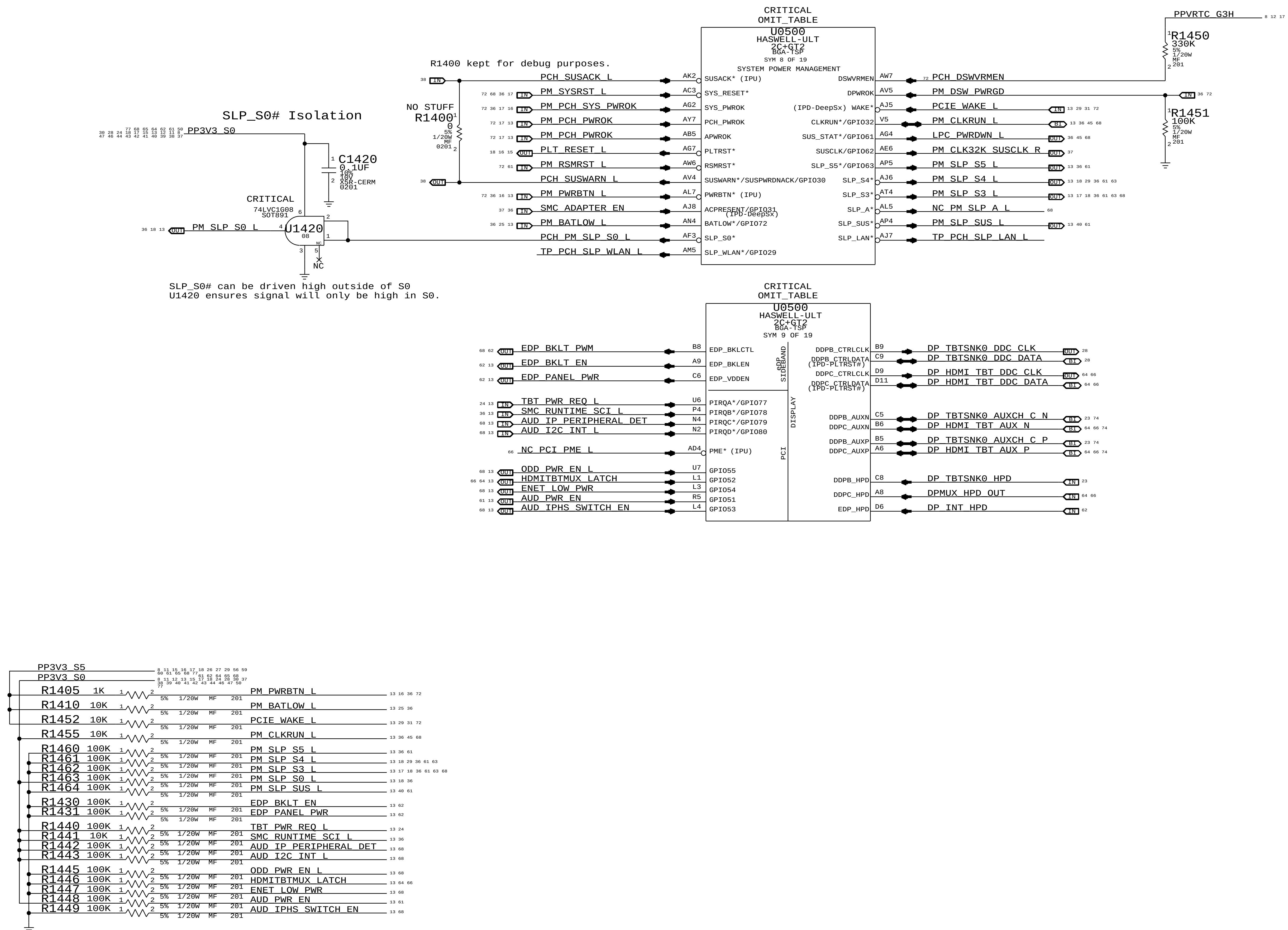

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PAGE TITLE

PCH PM/PCI/GFX

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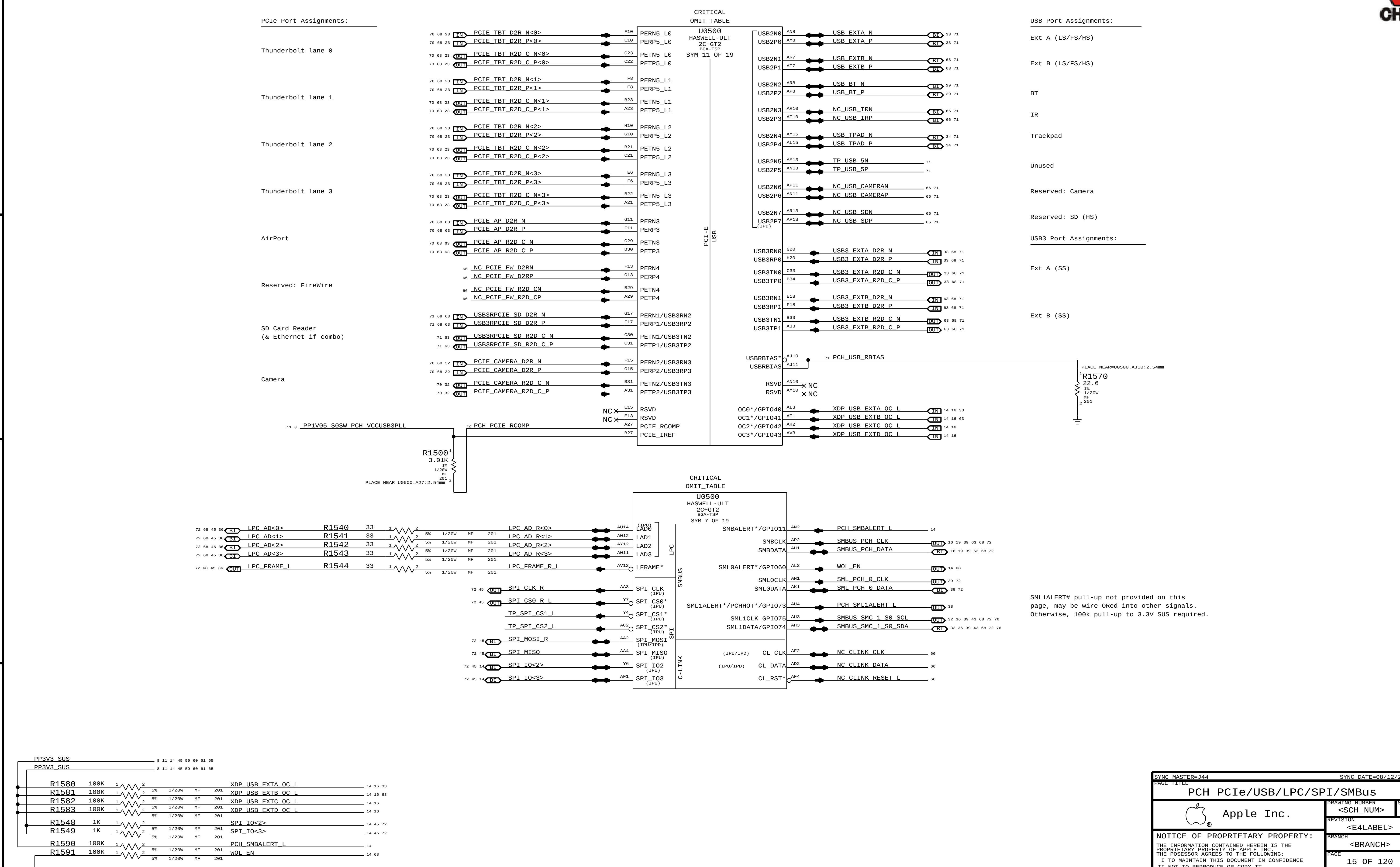

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

PCH PCIe/USB/LPC/SPI/SMBus

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DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H

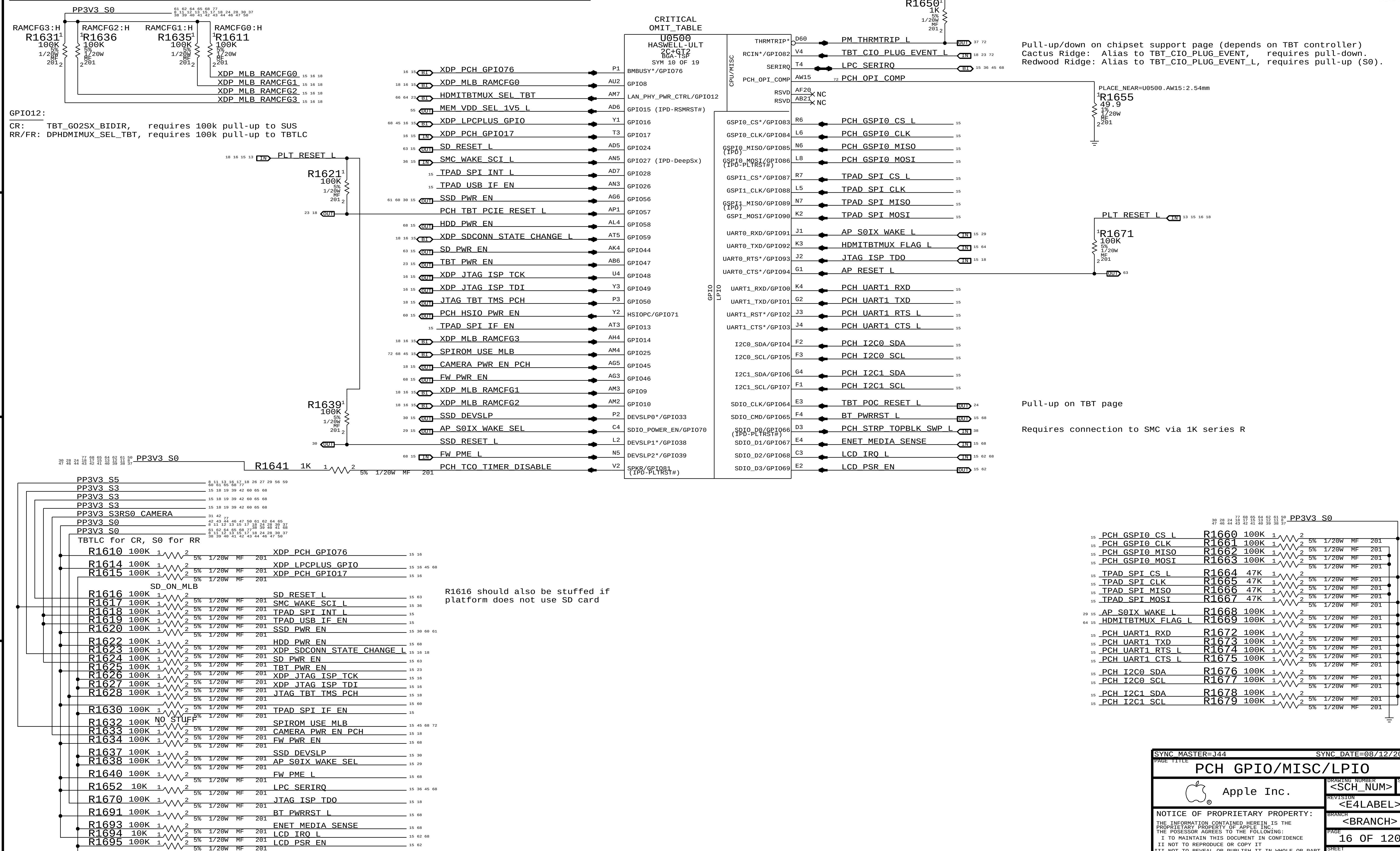
GPIO12:
CR: TBT_GO2SX_BIDIR, requires 100k pull-up to SUS
RR/FR: DPHDMIMUX_SEL_TBT, requires 100k pull-up to TBTLC

Pull-up/down on chipset support page (depends on TBT controller)
Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.
Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).

Pull-up on TBT page

Requires connection to SMC via 1K series R

R1616 should also be stuffed if platform does not use SD card



SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
PCH GPIO/MISC/LPIO

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
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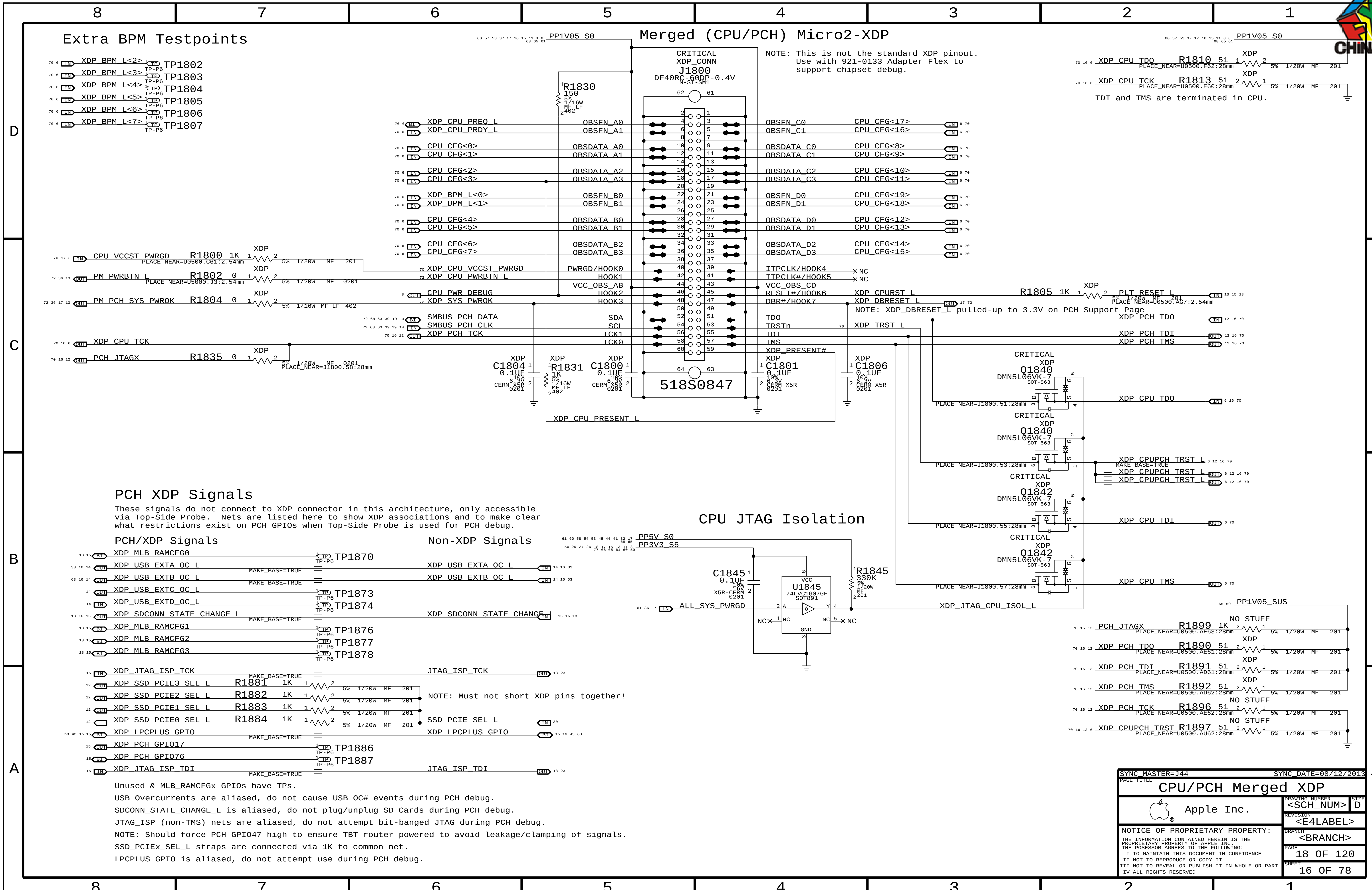

Extra BPM Testpoints
XDP_BPM_L<2> TP1802
XDP_BPM_L<3> TP1803
XDP_BPM_L<4> TP1804
XDP_BPM_L<5> TP1805
XDP_BPM_L<6> TP1806
XDP_BPM_L<7> TP1807
PP1V05_S0
Merged (CPU/PCH) Micro2-XDP
CRITICAL
XDP_CONN
J1800
DF40RC-60DP-0.4V
M-ST-SM1
NOTE: This is not the standard XDP pinout.
Use with 921-0133 Adapter Flex to
support chipset debug.
XDP_CPU_TDO R1810 51
XDP_CPU_TCK R1813 51
TDI and TMS are terminated in CPU.
R1830 150
XDP_CPU_PREQ_L OBSFN_A0 OBSFN_C0 CPU_CFG<17>
XDP_CPU_PRDY_L OBSFN_A1 OBSFN_C1 CPU_CFG<16>
CPU_CFG<0> OBSDATA_A0 OBSDATA_C0 CPU_CFG<8>
CPU_CFG<1> OBSDATA_A1 OBSDATA_C1 CPU_CFG<9>
CPU_CFG<2> OBSDATA_A2 OBSDATA_C2 CPU_CFG<10>
CPU_CFG<3> OBSDATA_A3 OBSDATA_C3 CPU_CFG<11>
XDP_BPM_L<0> OBSFN_B0 OBSFN_D0 CPU_CFG<19>
XDP_BPM_L<1> OBSFN_B1 OBSFN_D1 CPU_CFG<18>
CPU_CFG<4> OBSDATA_B0 OBSDATA_D0 CPU_CFG<12>
CPU_CFG<5> OBSDATA_B1 OBSDATA_D1 CPU_CFG<13>
CPU_CFG<6> OBSDATA_B2 OBSDATA_D2 CPU_CFG<14>
CPU_CFG<7> OBSDATA_B3 OBSDATA_D3 CPU_CFG<15>
CPU_VCCST_PWRGD R1800 1K
PM_PWRBTN_L R1802 0
PM_PCH_SYS_PWROK R1804 0
XDP_CPU_VCCST_PWRGD PWRGD/HOOK0 ITPCLK/HOOK4
XDP_CPU_PWRBTN_L HOOK1 ITPCLK#/HOOK5
VCC_OBS_AB VCC_OBS_CD
CPU_PWR_DEBUG HOOK2 RESET#/HOOK6 XDP_CPURST_L R1805 1K PLT_RESET_L
XDP_SYS_PWROK HOOK3 DBR#/HOOK7 XDP_DBRESET_L
NOTE: XDP_DBRESET_L pulled-up to 3.3V on PCH Support Page
SMBUS_PCH_DATA SDA TDO XDP_PCH_TDO
SMBUS_PCH_CLK SCL TRSTn XDP_TRST_L
XDP_PCH_TCK TCK1 TDI XDP_PCH_TDI
XDP_CPU_TCK TCK0 TMS XDP_PCH_TMS
PCH_JTAGX R1835 0
XDP_PRESENT#
C1804 0.1UF R1831 1K C1800 0.1UF C1801 0.1UF C1806 0.1UF
518S0847
XDP_CPU_PRESENT_L
Q1840 DMN5L06VK-7 XDP_CPU_TDO
Q1840 DMN5L06VK-7 XDP_CPUPCH_TRST_L
Q1842 DMN5L06VK-7 XDP_CPU_TDI
Q1842 DMN5L06VK-7 XDP_CPU_TMS
PCH XDP Signals
These signals do not connect to XDP connector in this architecture, only accessible
via Top-Side Probe. Nets are listed here to show XDP associations and to make clear
what restrictions exist on PCH GPIOs when Top-Side Probe is used for PCH debug.
PCH/XDP Signals
Non-XDP Signals
XDP_MLB_RAMCFG0 TP1870
XDP_USB_EXTA_OC_L XDP_USB_EXTA_OC_L
XDP_USB_EXTB_OC_L XDP_USB_EXTB_OC_L
XDP_USB_EXTC_OC_L TP1873
XDP_USB_EXTD_OC_L TP1874
XDP_SDCONN_STATE_CHANGE_L XDP_SDCONN_STATE_CHANGE_L
XDP_MLB_RAMCFG1 TP1876
XDP_MLB_RAMCFG2 TP1877
XDP_MLB_RAMCFG3 TP1878
XDP_JTAG_ISP_TCK JTAG_ISP_TCK
XDP_SSD_PCIE3_SEL_L R1881 1K
XDP_SSD_PCIE2_SEL_L R1882 1K
XDP_SSD_PCIE1_SEL_L R1883 1K
XDP_SSD_PCIE0_SEL_L R1884 1K
NOTE: Must not short XDP pins together!
SSD_PCIE_SEL_L
XDP_LPCPLUS_GPIO XDP_LPCPLUS_GPIO
XDP_PCH_GPIO17 TP1886
XDP_PCH_GPIO76 TP1887
XDP_JTAG_ISP_TDI JTAG_ISP_TDI
CPU JTAG Isolation
PP5V_S0
PP3V3_S5
C1845 0.1UF
U1845 74LVC1G07GF SOT891
R1845 330K
ALL_SYS_PWRGD
XDP_JTAG_CPU_ISOL_L
PP1V05_SUS
PCH_JTAGX R1899 1K NO STUFF
XDP_PCH_TDO R1890 51
XDP_PCH_TDI R1891 51
XDP_PCH_TMS R1892 51
XDP_PCH_TCK R1896 51 NO STUFF
XDP_CPUPCH_TRST_L R1897 51 NO STUFF
Unused & MLB_RAMCFGx GPIOs have TPs.
USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.
SDCONN_STATE_CHANGE_L is aliased, do not plug/unplug SD Cards during PCH debug.
JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.
NOTE: Should force PCH GPIO47 high to ensure TBT router powered to avoid leakage/clamping of signals.
SSD_PCIEx_SEL_L straps are connected via 1K to common net.
LPCPLUS_GPIO is aliased, do not attempt use during PCH debug.
SYNC_MASTER=J44 SYNC_DATE=08/12/2013
CPU/PCH Merged XDP
Apple Inc.
<SCH_NUM> D
<E4LABEL>
<BRANCH>
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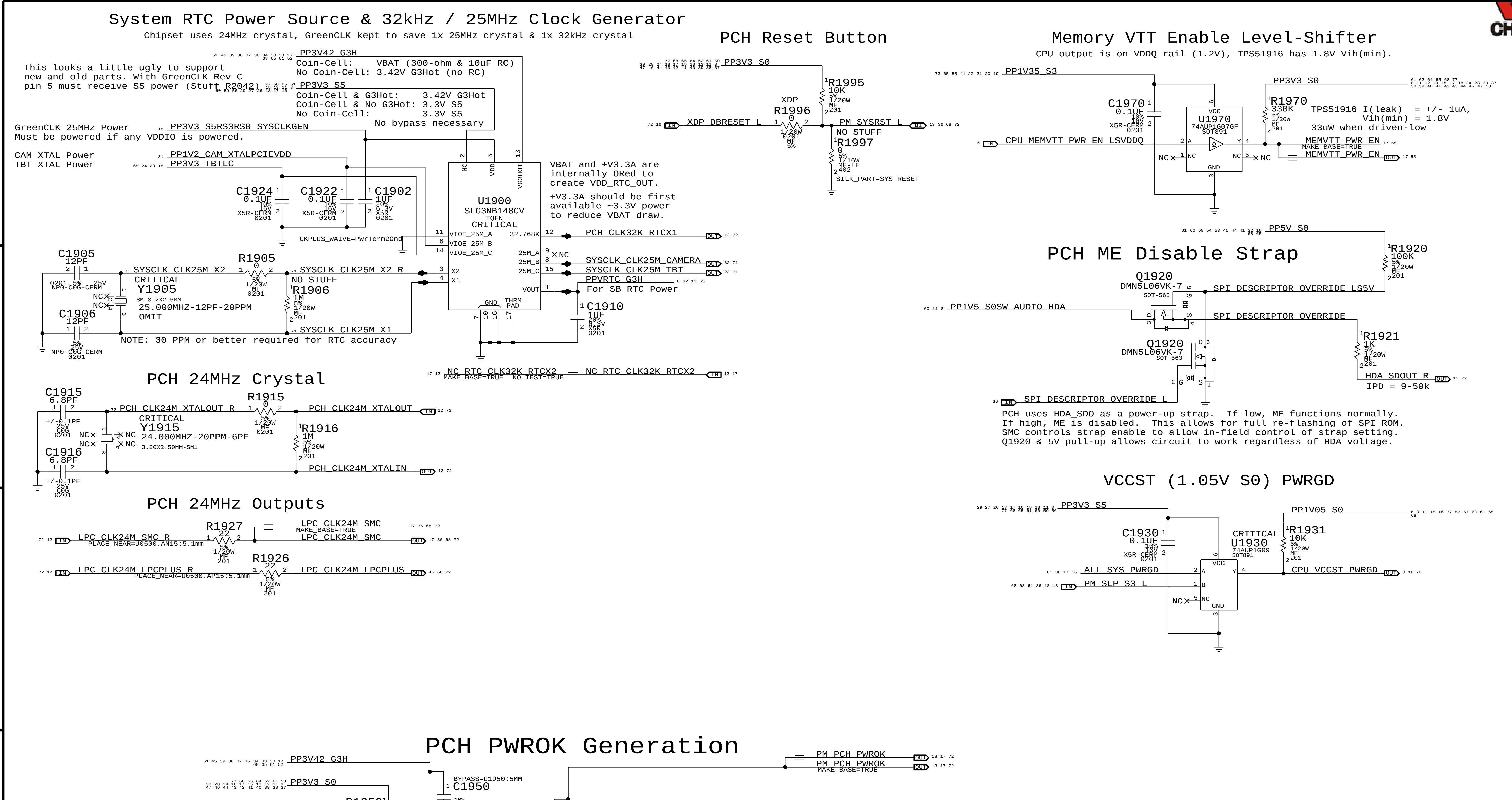

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
197S0480	1	XTAL,25MHZ,20PPM,12PF,3.2X2.5X.6MM,85C	Y1905		

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

Chipset Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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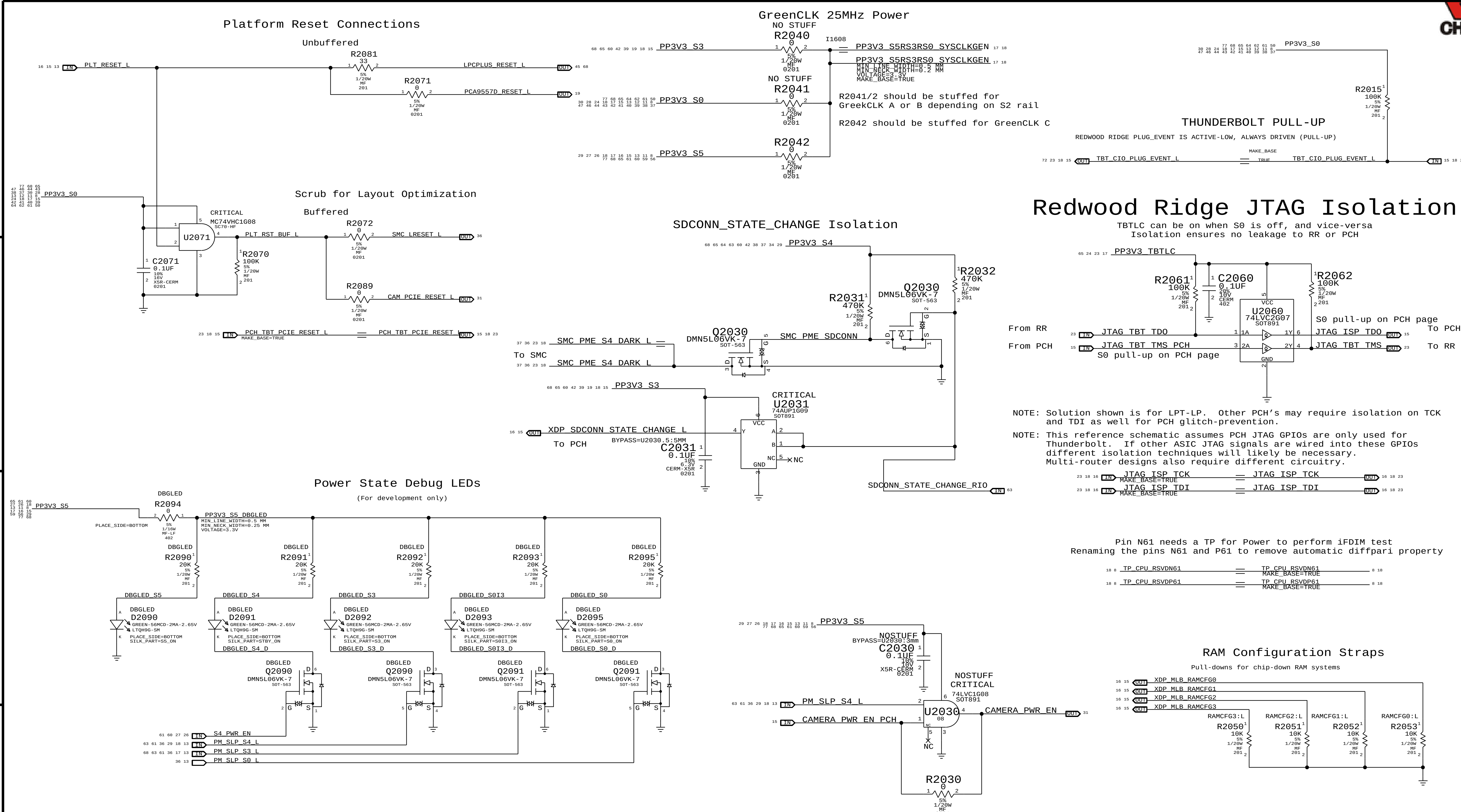

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Project Chipset Support

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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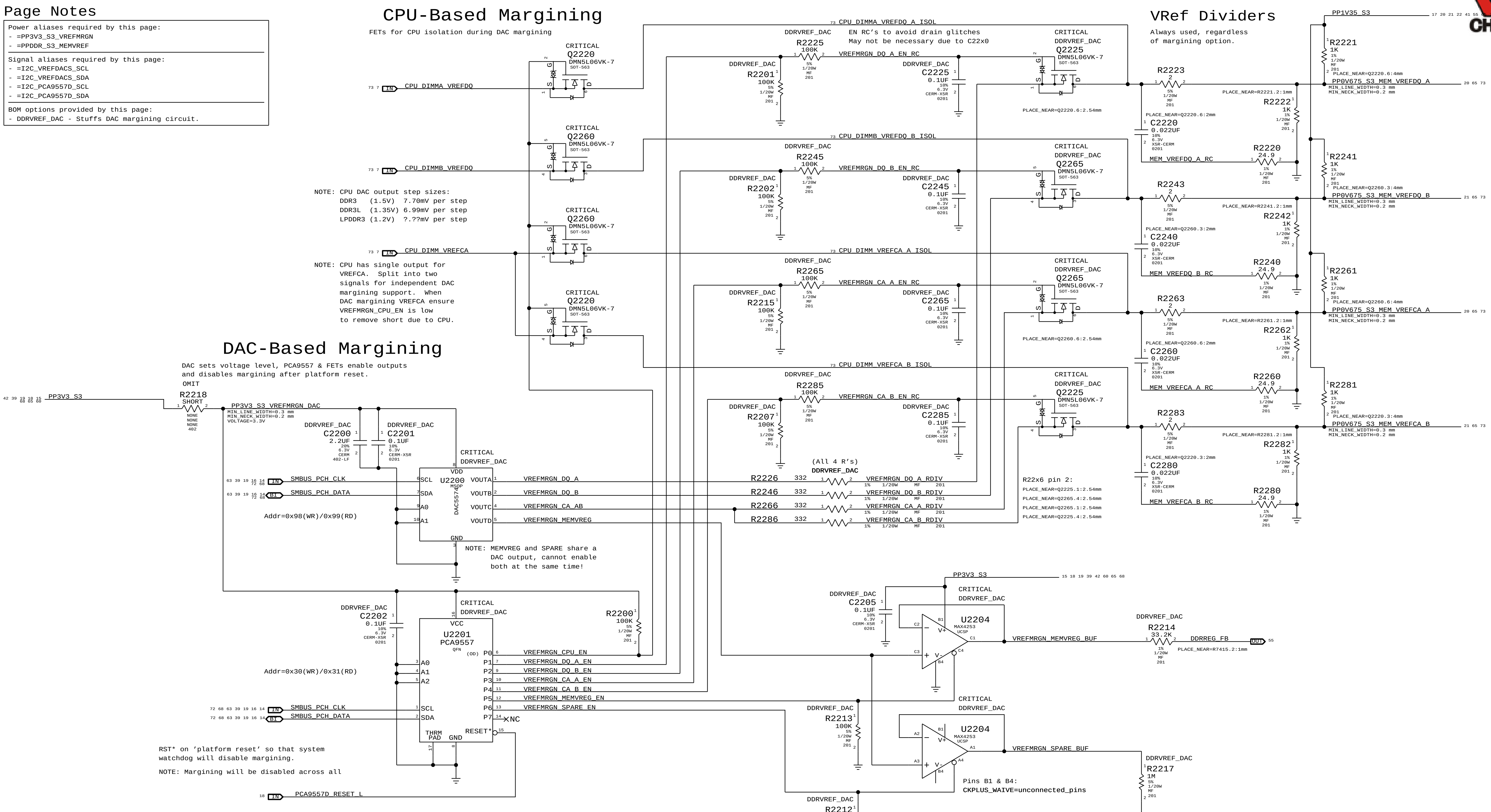
Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	D	D
PCA9557D Pin:	1	2	3	4	5

	LPDDR3 (1.2V)	DDR3L (1.35V)	LPDDR3 (1.2V)	DDR3L (1.35V)
Nominal value	0.600V (DAC: 0x2E.5)	0.675V (DAC: 0x34)	1.200V (DAC: 0x34)	1.343V (DAC: 0x68)
Margined target:	0.300V - 0.900V (+/- 300mV)	0.337V - 1.013V (+/- 337.5mV)	0.800V - 1.600V (+/- 400mV)	0.972V - 1.714V (+/- 371mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)	0.000V - 1.354V (0x00 - 0x69)	0.000V - 2.397V (0x00 - 0xBA)	0.000V - 2.694V (0x00 - 0xD1)
VRef current:	+73uA - -73uA (- = sourced)	+82uA - -82uA (- = sourced)	+21uA - -21uA (- = sourced)	+25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output	6.36mV / step @ output	4.28mV / step @ output	3.53mV / step @ output

NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
DDR3 VREF MARGINING

Apple Inc.

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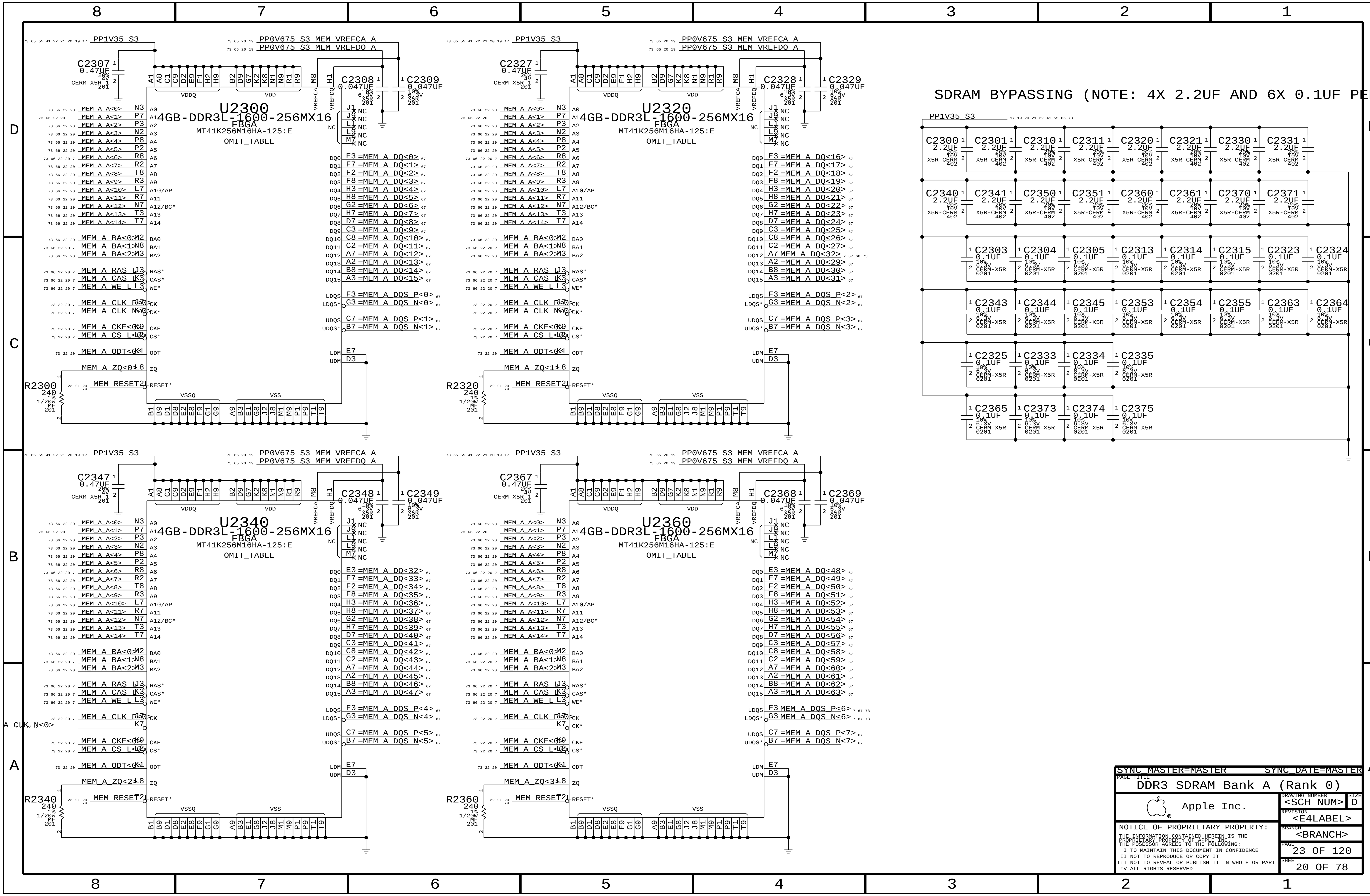

PP1V35_S3
PP0V675_S3_MEM_VREFCA_A
PP0V675_S3_MEM_VREFDQ_A
U2300
4GB-DDR3L-1600-256MX16
FBGA
MT41K256M16HA-125:E
OMIT_TABLE
U2320
4GB-DDR3L-1600-256MX16
FBGA
MT41K256M16HA-125:E
OMIT_TABLE
U2340
4GB-DDR3L-1600-256MX16
FBGA
MT41K256M16HA-125:E
OMIT_TABLE
U2360
4GB-DDR3L-1600-256MX16
FBGA
MT41K256M16HA-125:E
OMIT_TABLE
VDDQ
VDD
VSSQ
VSS
VREFCA
VREFDQ
C2307 0.47UF
C2327 0.47UF
C2347 0.47UF
C2367 0.47UF
C2308 0.047UF
C2309 0.047UF
C2328 0.047UF
C2329 0.047UF
C2348 0.047UF
C2349 0.047UF
C2368 0.047UF
C2369 0.047UF
R2300 240
R2320 240
R2340 240
R2360 240
MEM_A_RESET_L
MEM_A_ZQ
MEM_A_ODT
MEM_A_CKE
MEM_A_CS_L
MEM_A_CLK_P
MEM_A_CLK_N
MEM_A_RAS_L
MEM_A_CAS_L
MEM_A_WE_L
SDRAM BYPASSING (NOTE: 4X 2.2UF AND 6X 0.1UF PER CHIP)
C2300 C2301 C2310 C2311 C2320 C2321 C2330 C2331 2.2UF
C2340 C2341 C2350 C2351 C2360 C2361 C2370 C2371 2.2UF
C2303 C2304 C2305 C2313 C2314 C2315 C2323 C2324 0.1UF
C2343 C2344 C2345 C2353 C2354 C2355 C2363 C2364 0.1UF
C2325 C2333 C2334 C2335 0.1UF
C2365 C2373 C2374 C2375 0.1UF
SYNC_MASTER=MASTER
SYNC_DATE=MASTER
PAGE TITLE
DDR3 SDRAM Bank A (Rank 0)
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
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<E4LABEL>
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<BRANCH>
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20 OF 78
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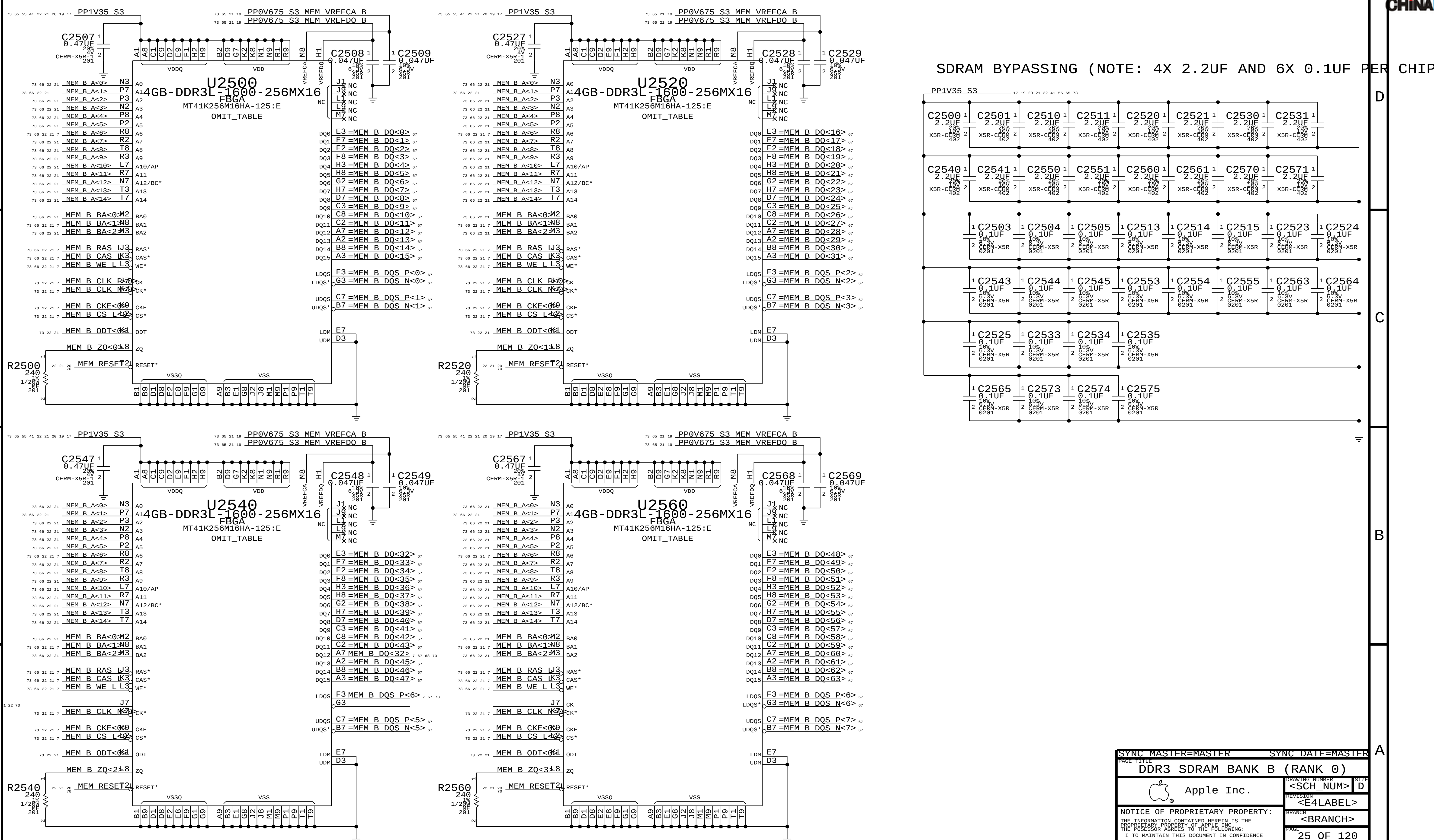

SYNC_MASTER=MASTER SYNC_DATE=MASTER

PAGE TITLE

DDR3 SDRAM BANK B (RANK 0)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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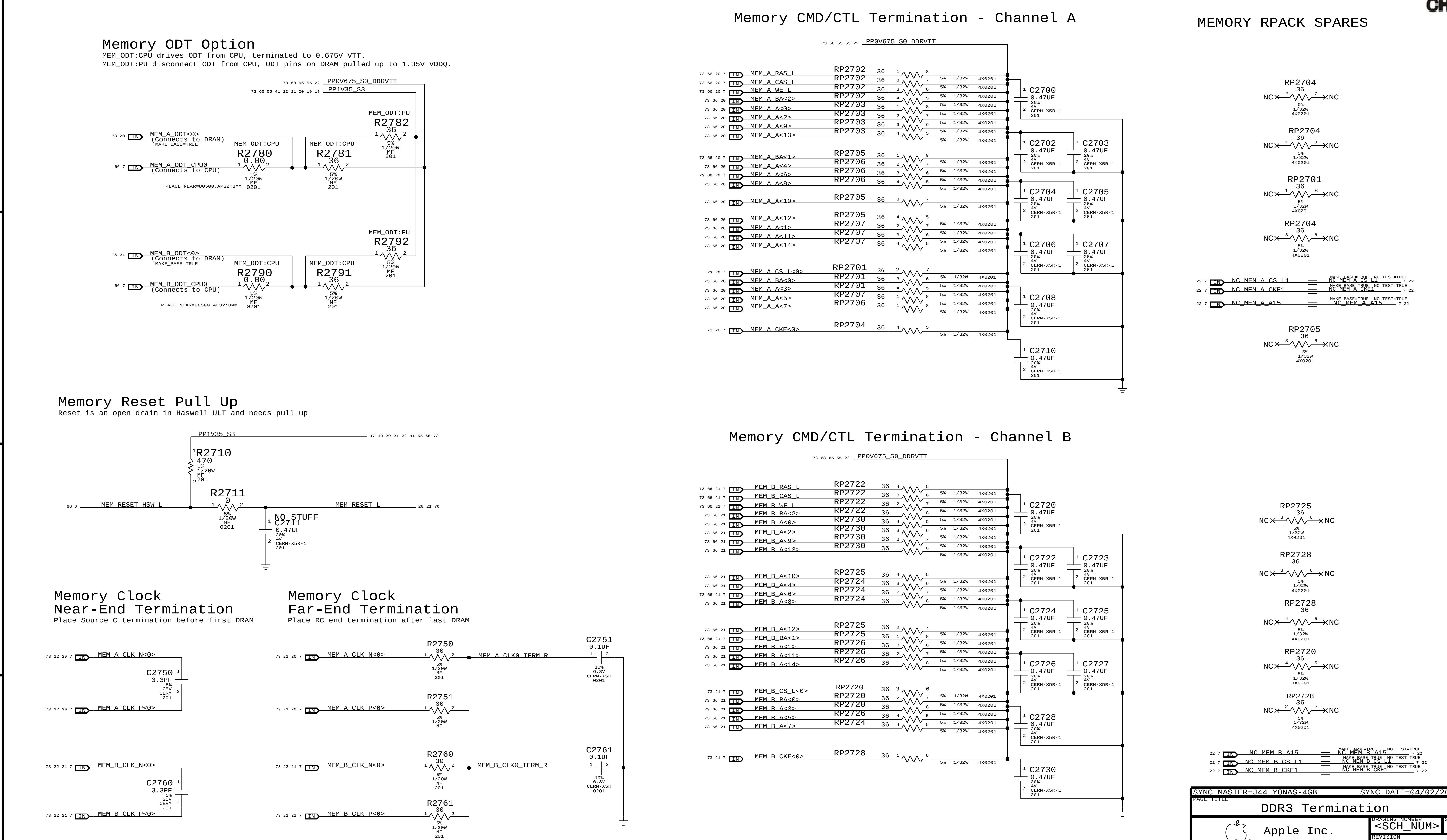

SYNC_MASTER=J44_YONAS-4GB SYNC_DATE=04/02/2013

PAGE TITLE

DDR3 Termination

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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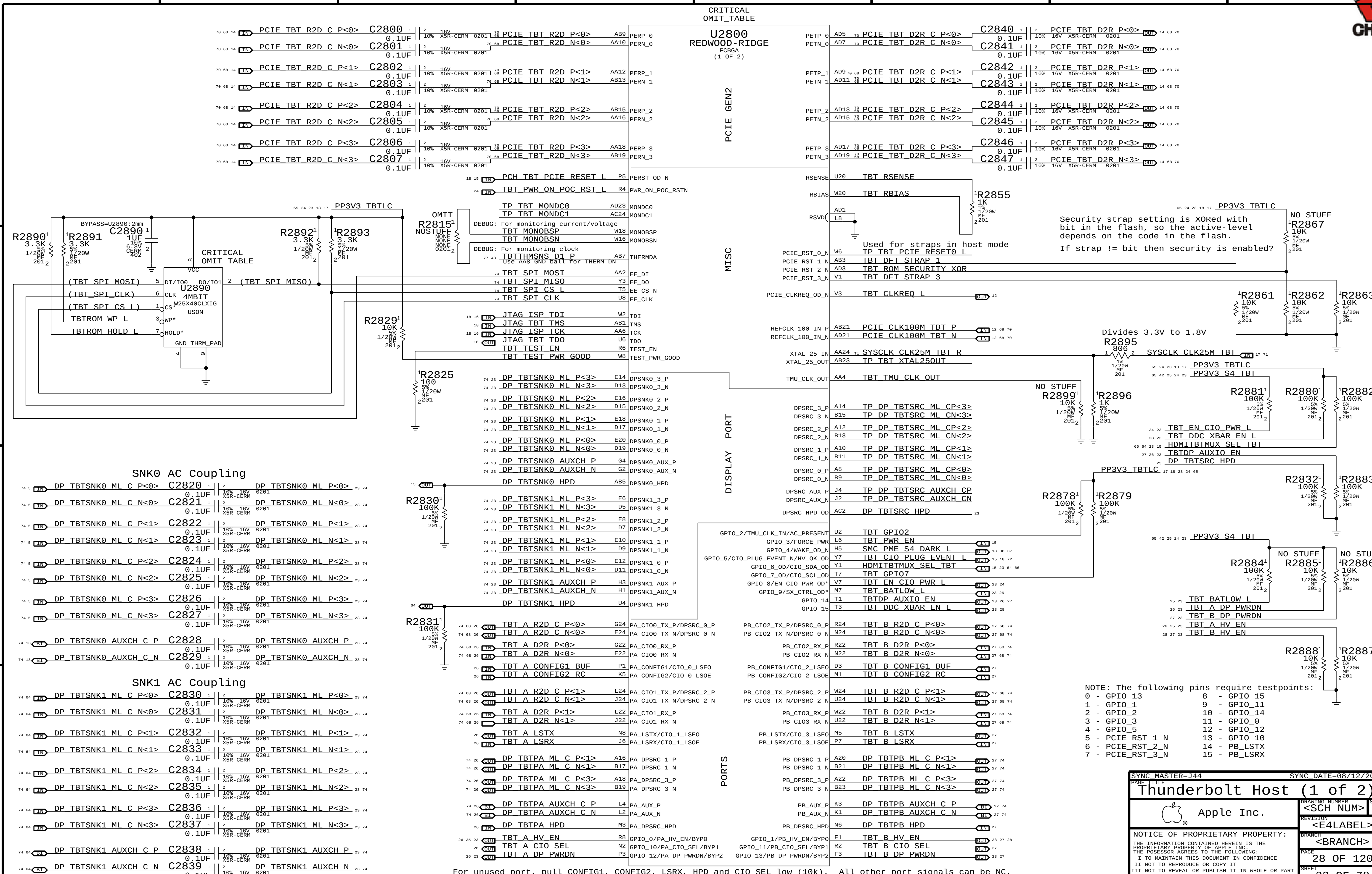

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

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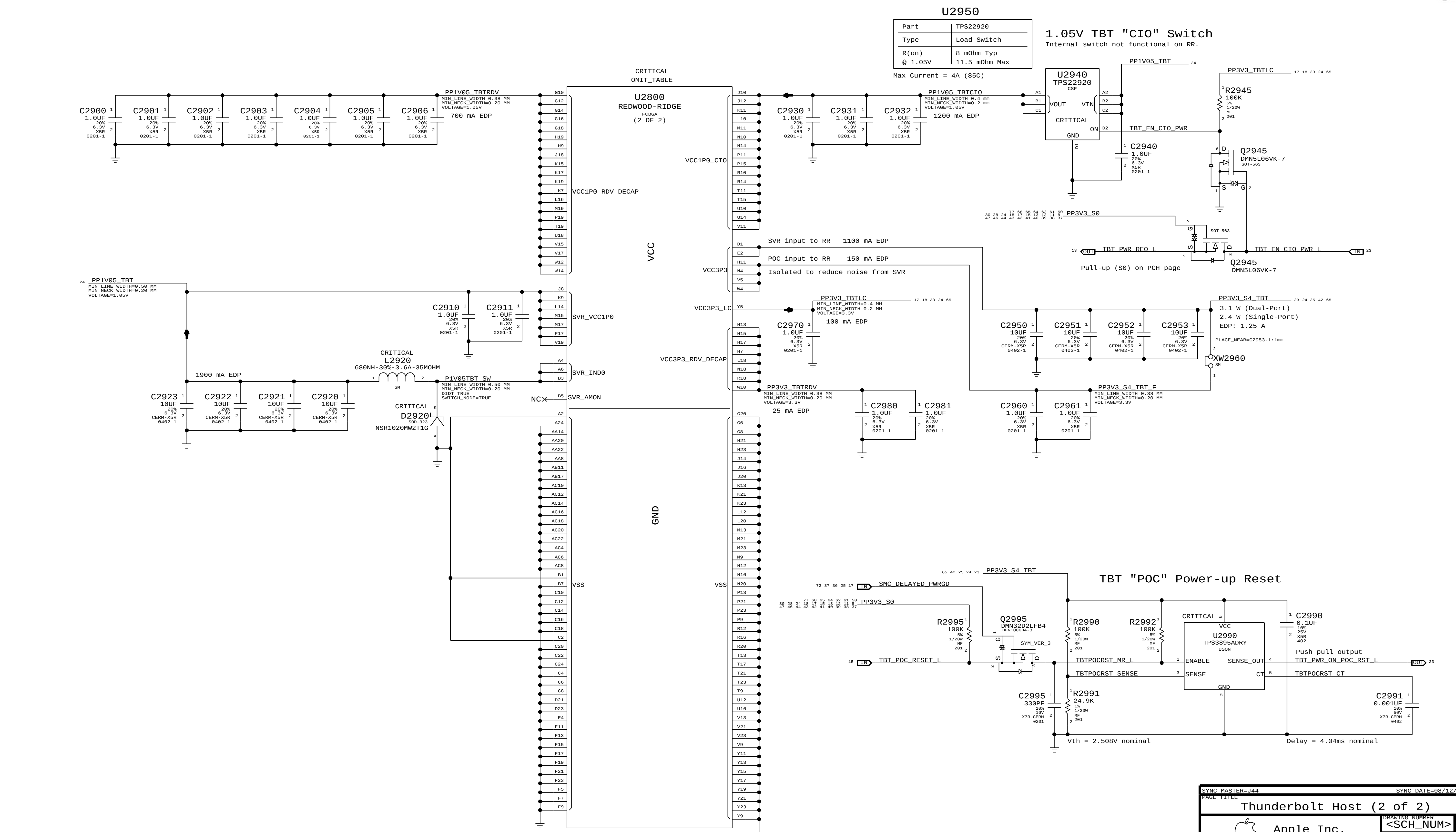

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

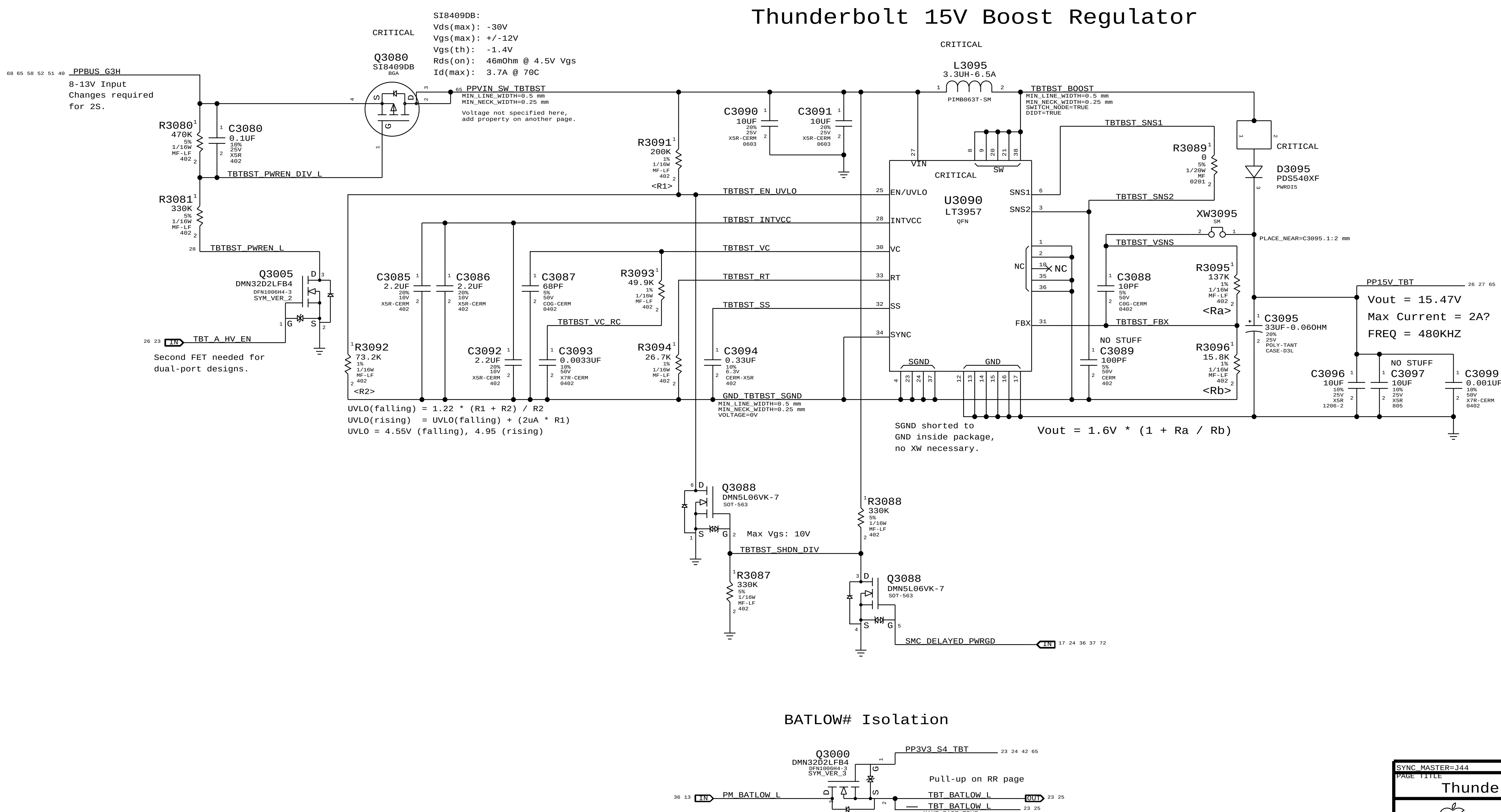
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Thunderbolt 15V Boost Regulator



BATLOW# Isolation

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Thunderbolt Mobile Support

Apple Inc.

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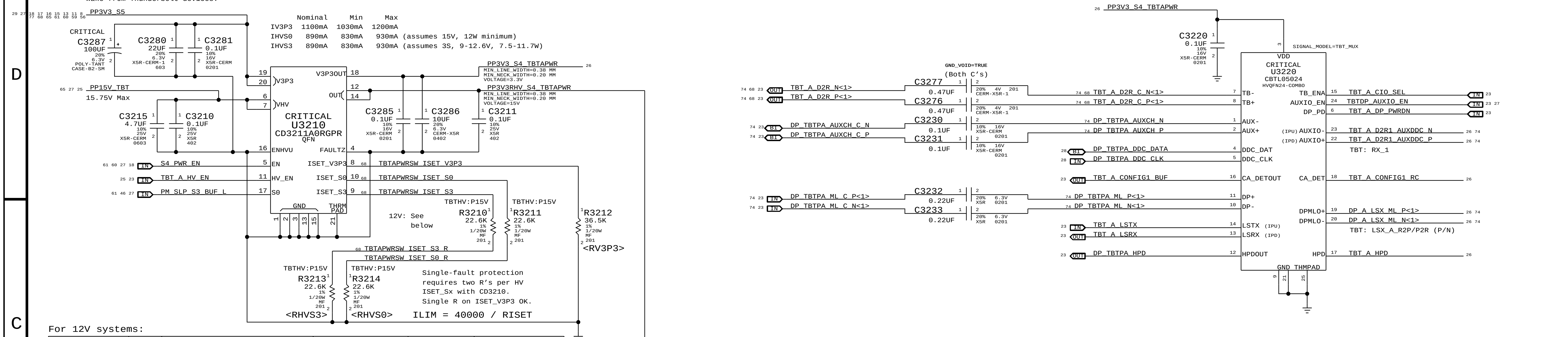
3.3V/HV Power MUX

V3P3 must be S4 to support
wake from Thunderbolt devices.

	Nominal	Min	Max	
IV3P3	1100mA	1030mA	1200mA	
IHVS0	890mA	830mA	930mA	(assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA	(assumes 3S, 9-12.6V, 7.5-11.7W)

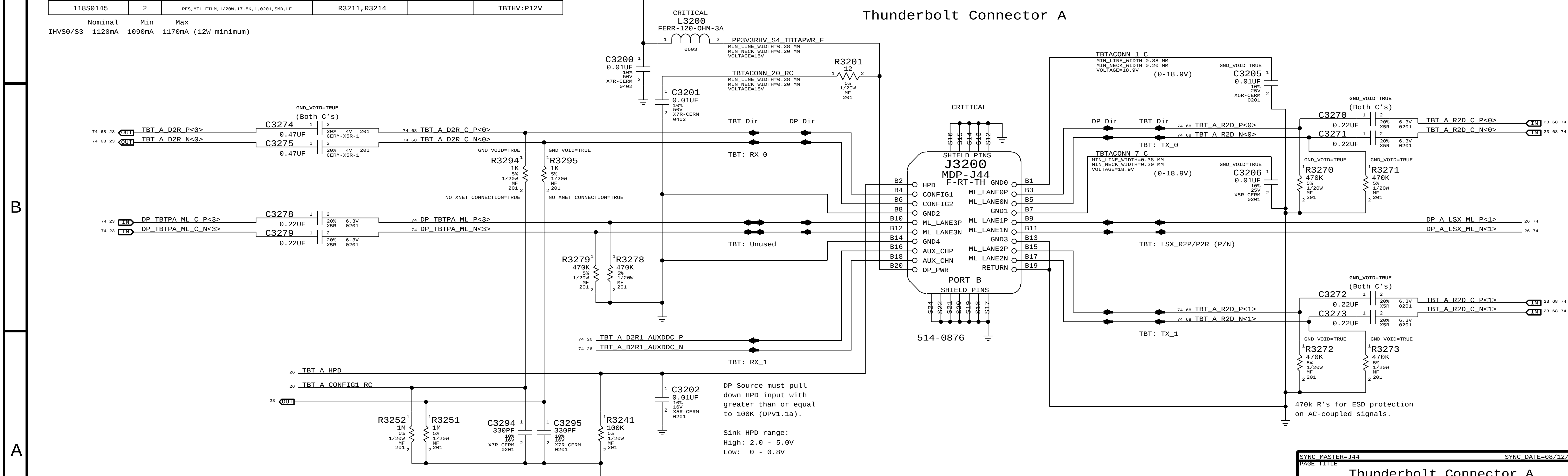


For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3210,R3213		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3211,R3214		TBTHV:P12V

	Nominal	Min	Max	
IHVS0/S3	1120mA	1090mA	1170mA	(12W minimum)

Thunderbolt Connector A



DP Source must pull
down HPD input with
greater than or equal
to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

470k R's for ESD protection
on AC-coupled signals.

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

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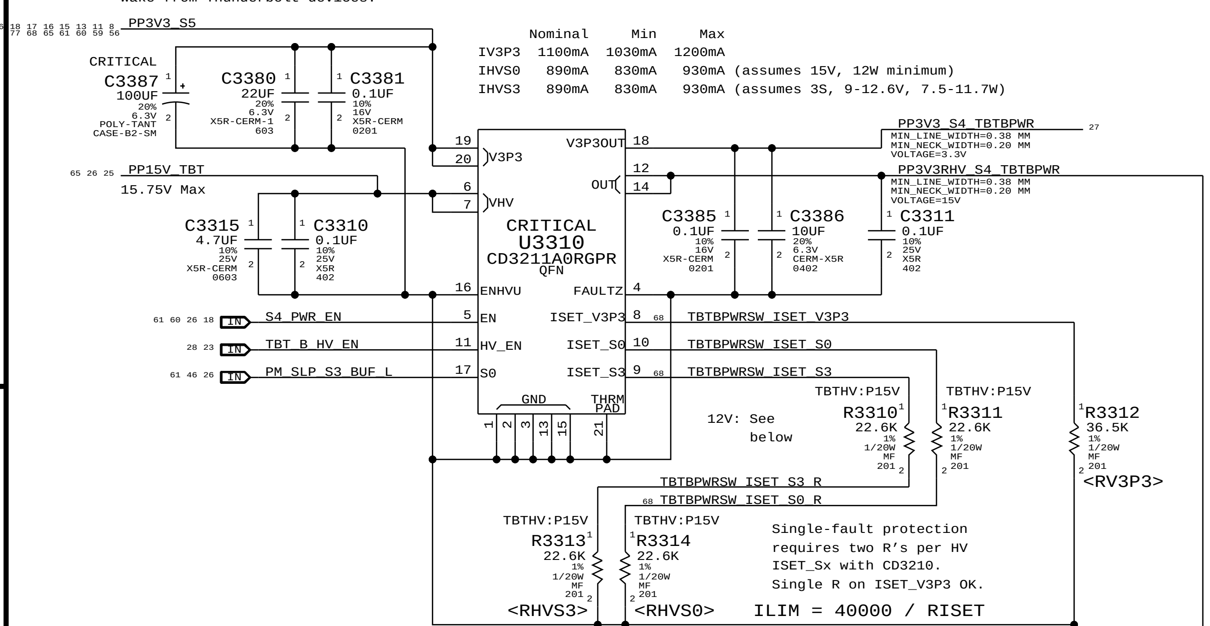
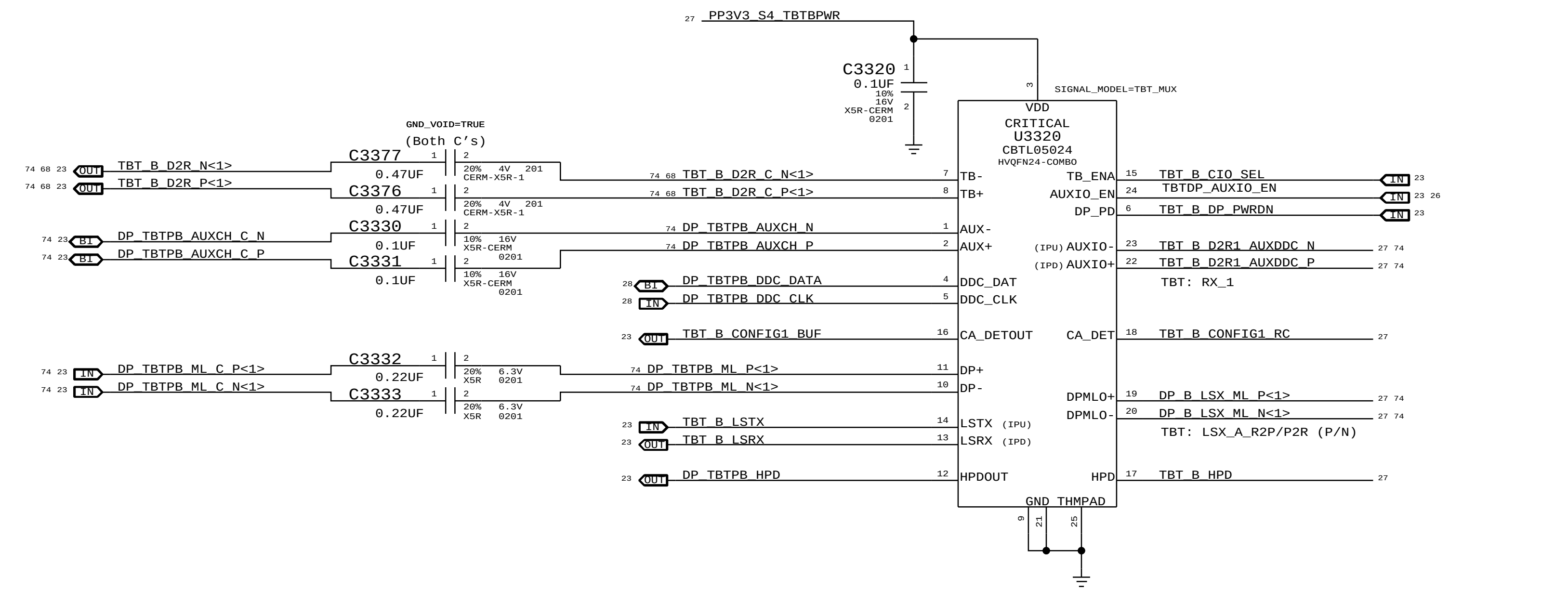
3.3V/HV Power MUX



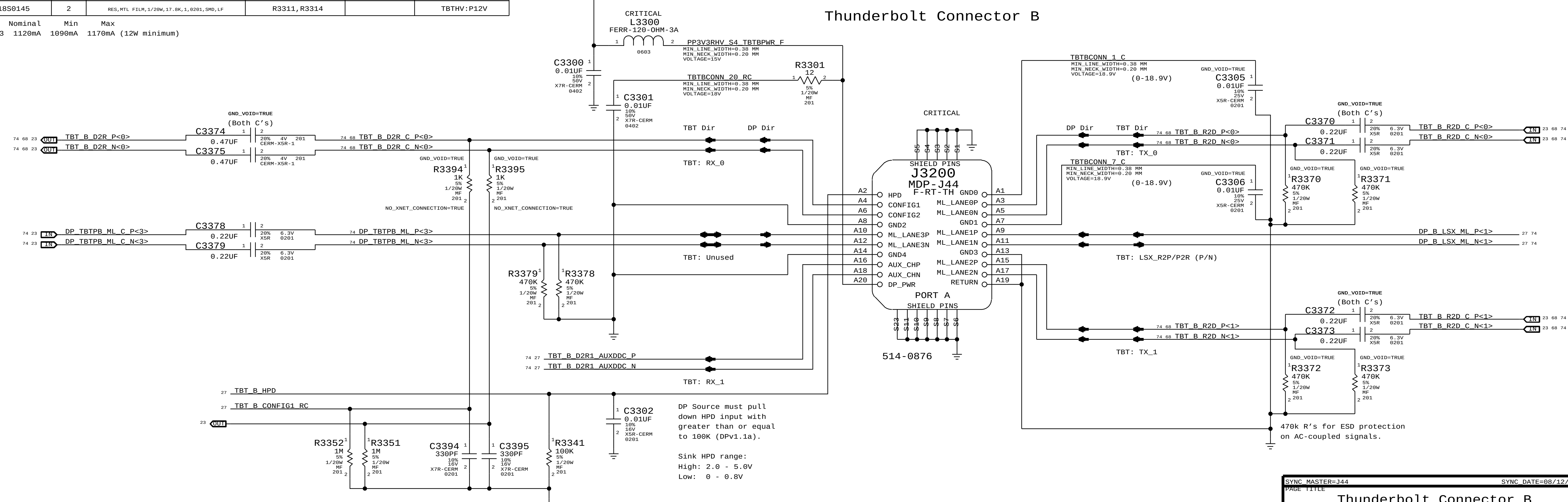


For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3310,R3313		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3311,R3314		TBTHV:P12V

Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)

Thunderbolt Connector B



SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

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BRANCH <BRANCH>

DDC Pull-Ups

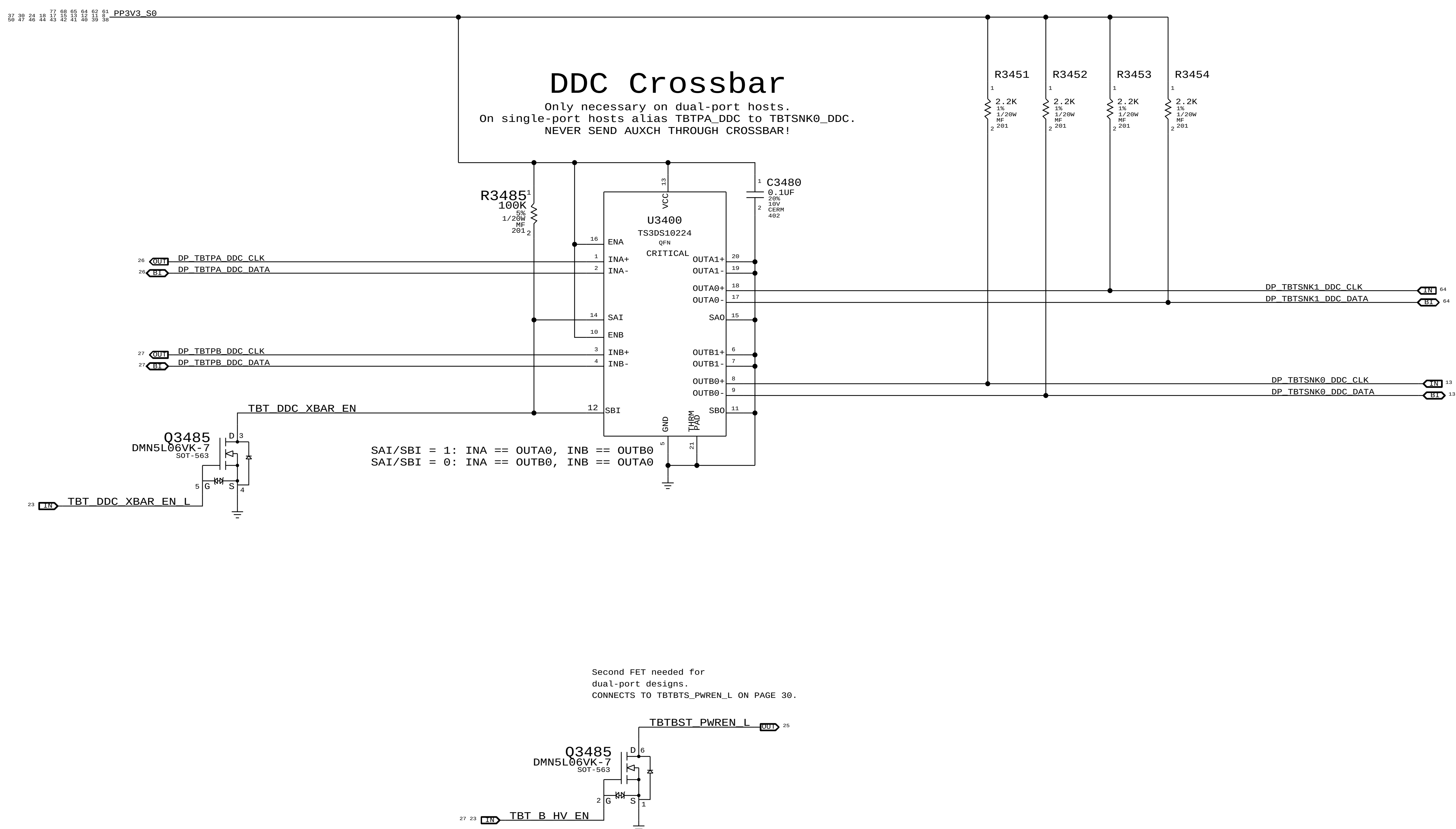
2.2k pull-ups are required by PCH
to indicate active display interface.

DP++ spec violation, should remove!

NOTE: Only DDC_DATA is sensed, so DDC_CLK
pull-ups are unstuffed.

DDC Crossbar

Only necessary on dual-port hosts.
On single-port hosts alias TBTPA_DDC to TBTSNK0_DDC.
NEVER SEND AUXCH THROUGH CROSSBAR!



SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: DDC Crossbar

Apple Inc.

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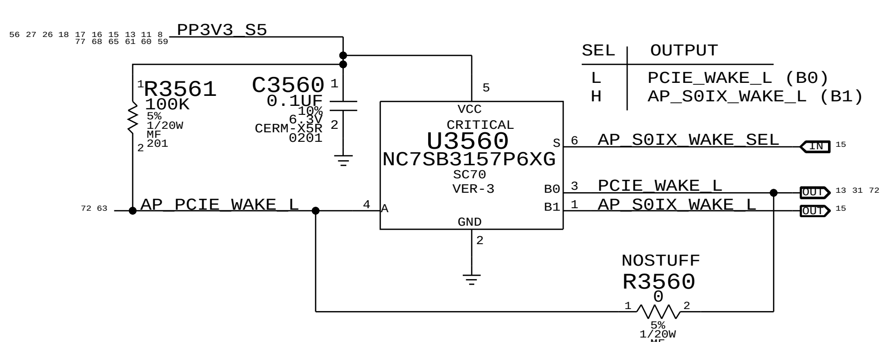
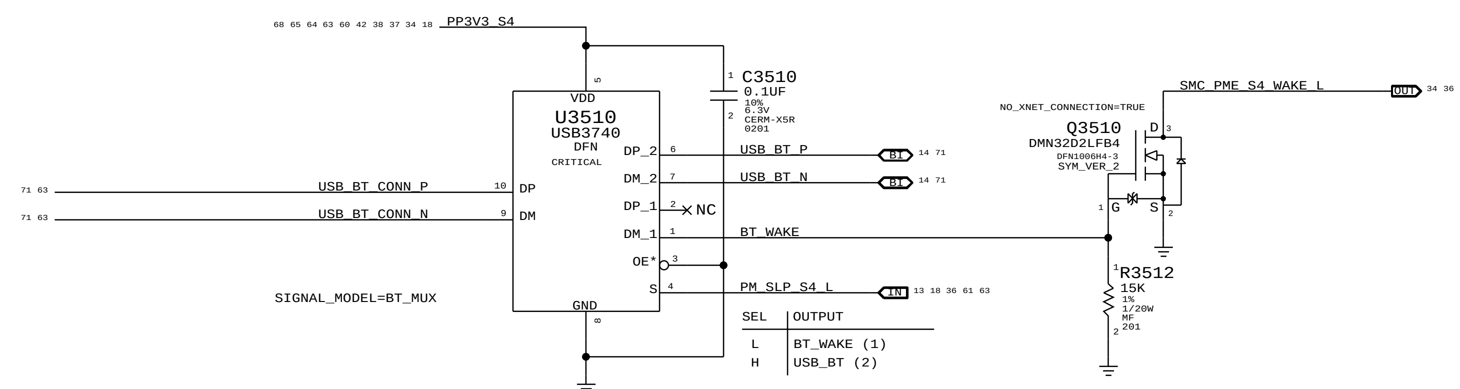
PCIe Wake Muxing



SEL	OUTPUT
L	PCIE_WAKE_L (B0)
H	AP_S0IX_WAKE_L (B1)

BLUETOOTH



SEL	OUTPUT
L	BT_WAKE (1)
H	USB_BT (2)

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: WIRELESS SUPPORT

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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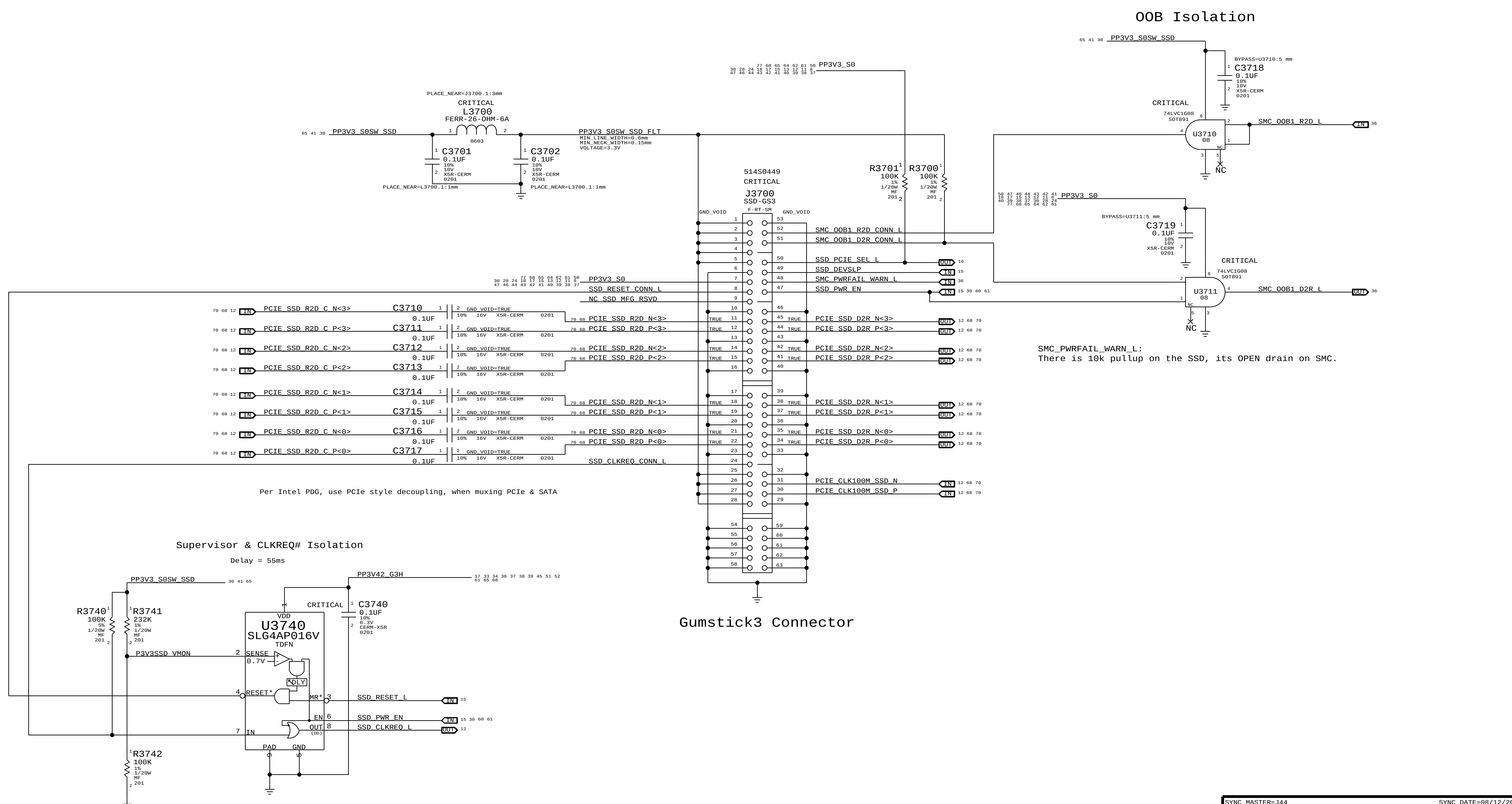

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: SSD Connector

Apple Inc.

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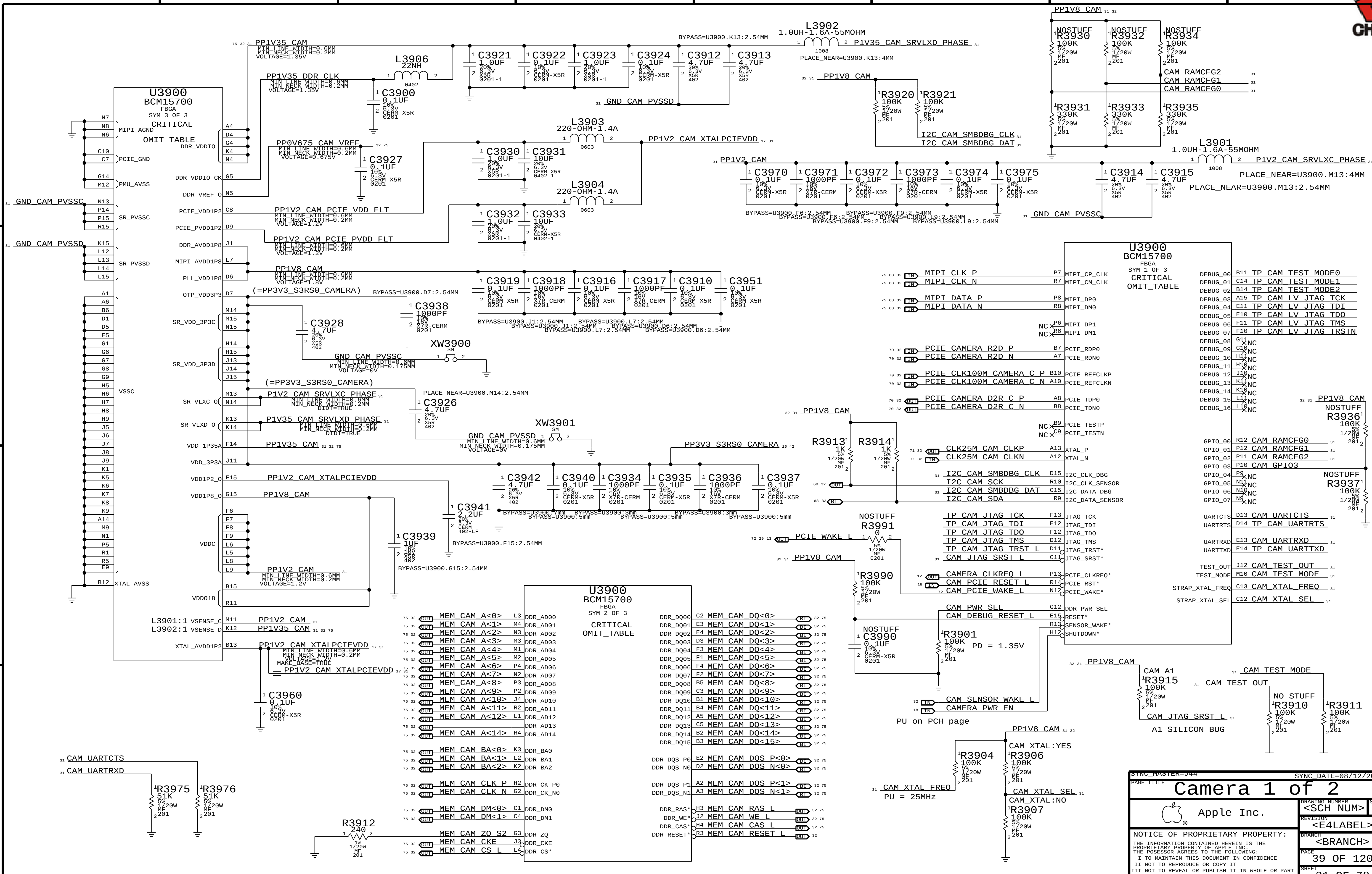

SYNC_MASTER=J44
SYNC_DATE=08/12/2013

PAGE TITLE
Camera 1 of 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
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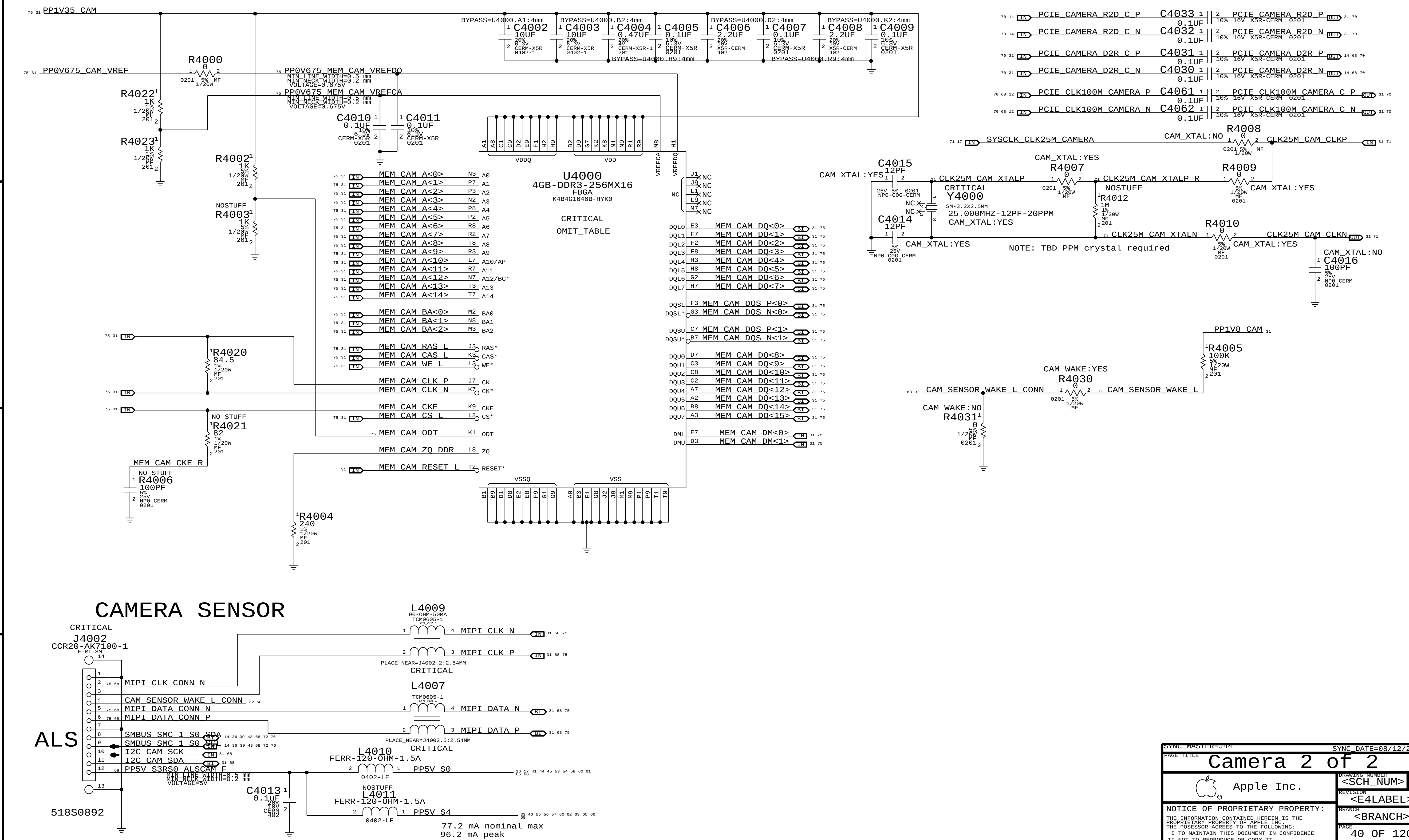

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PAGE TITLE

Camera 2 of 2

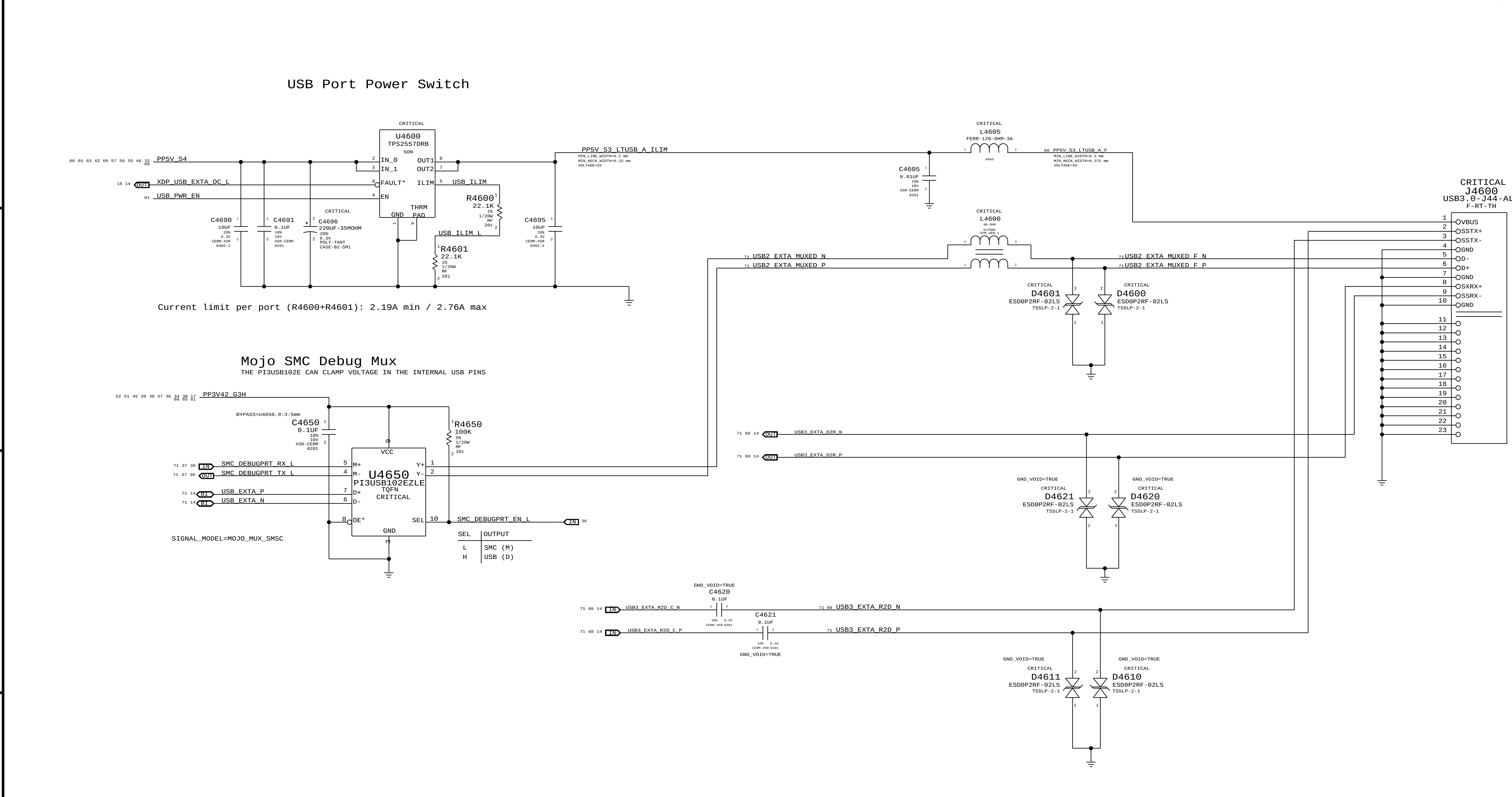
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

RIGHT USB PORT A



SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: External A USB3 Connector

Apple Inc.

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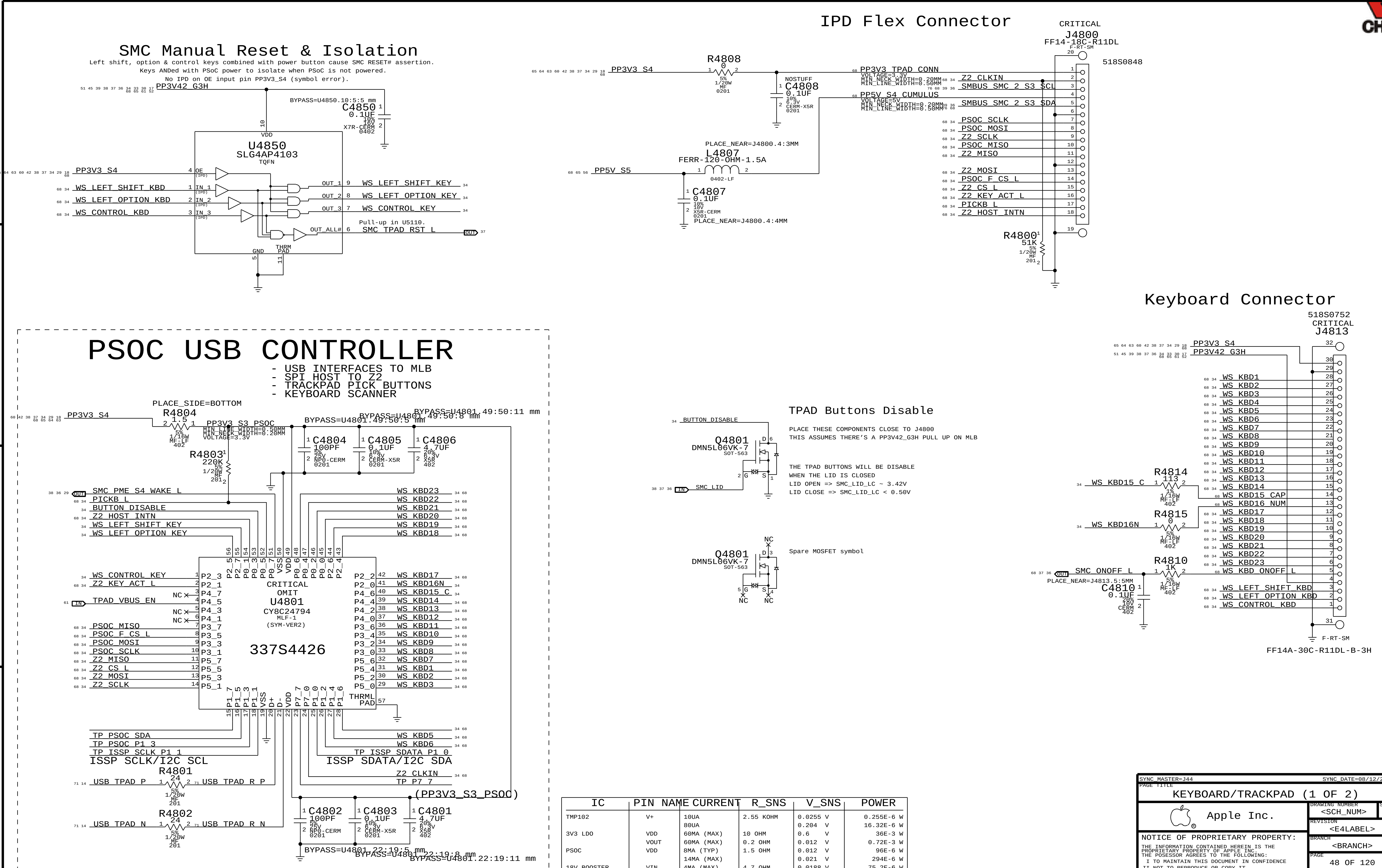
PAGE: 46 OF 120

SHEET: 33 OF 78



IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA	2.55 KOHM	0.0255 V	0.255E-6 W
		80UA		0.204 V	16.32E-6 W
3V3 LDO	VDD	60MA (MAX)	10 OHM	0.6 V	36E-3 W
	VOUT	60MA (MAX)	0.2 OHM	0.012 V	0.72E-3 W
PSOC	VDD	8MA (TYP)	1.5 OHM	0.012 V	96E-6 W
		14MA (MAX)		0.021 V	294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
KEYBOARD/TRACKPAD (1 OF 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
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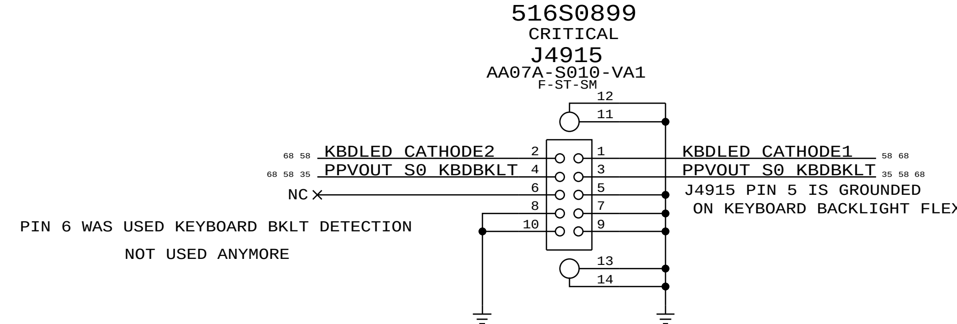
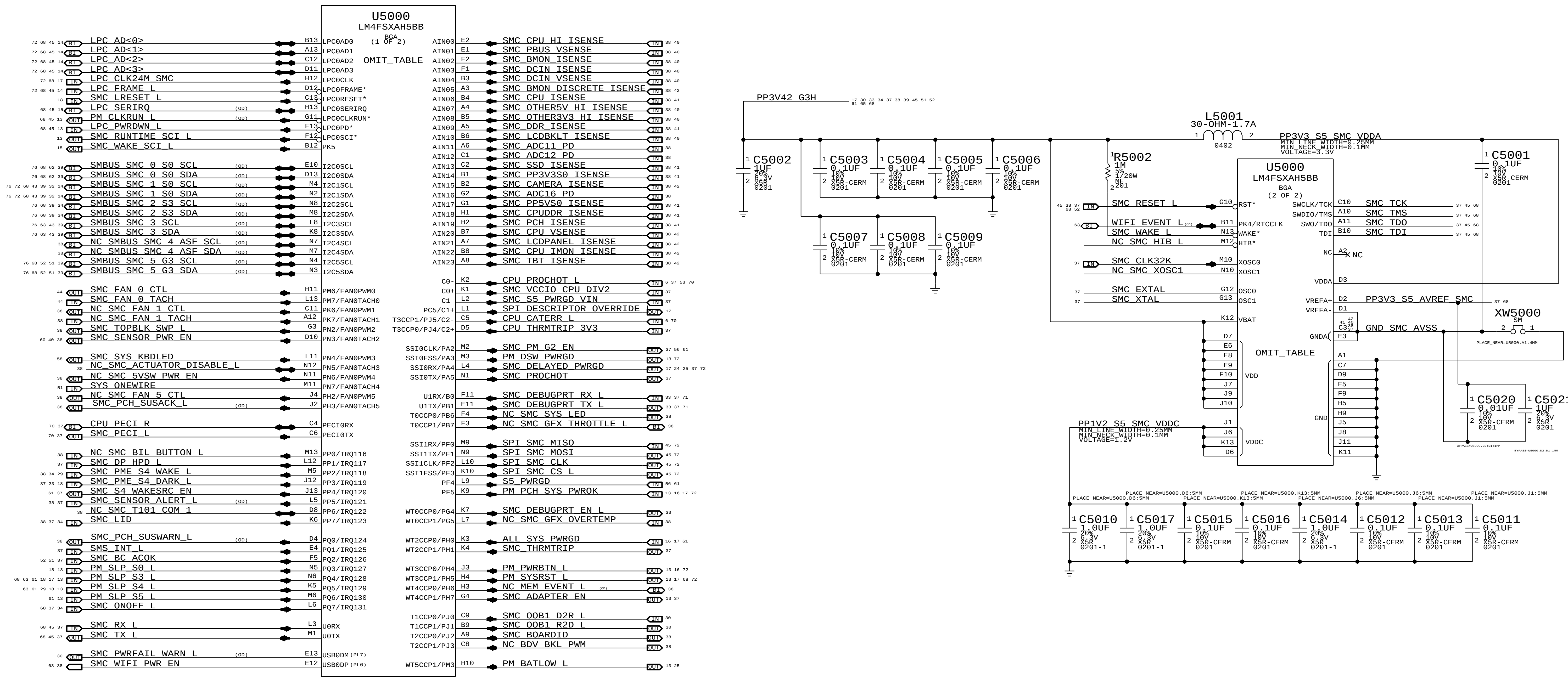
Keyboard Backlight Connector





NOTE:
SMS INTERRUPT IS NOT USED, PULL UP TO SMC RAIL.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
SMC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
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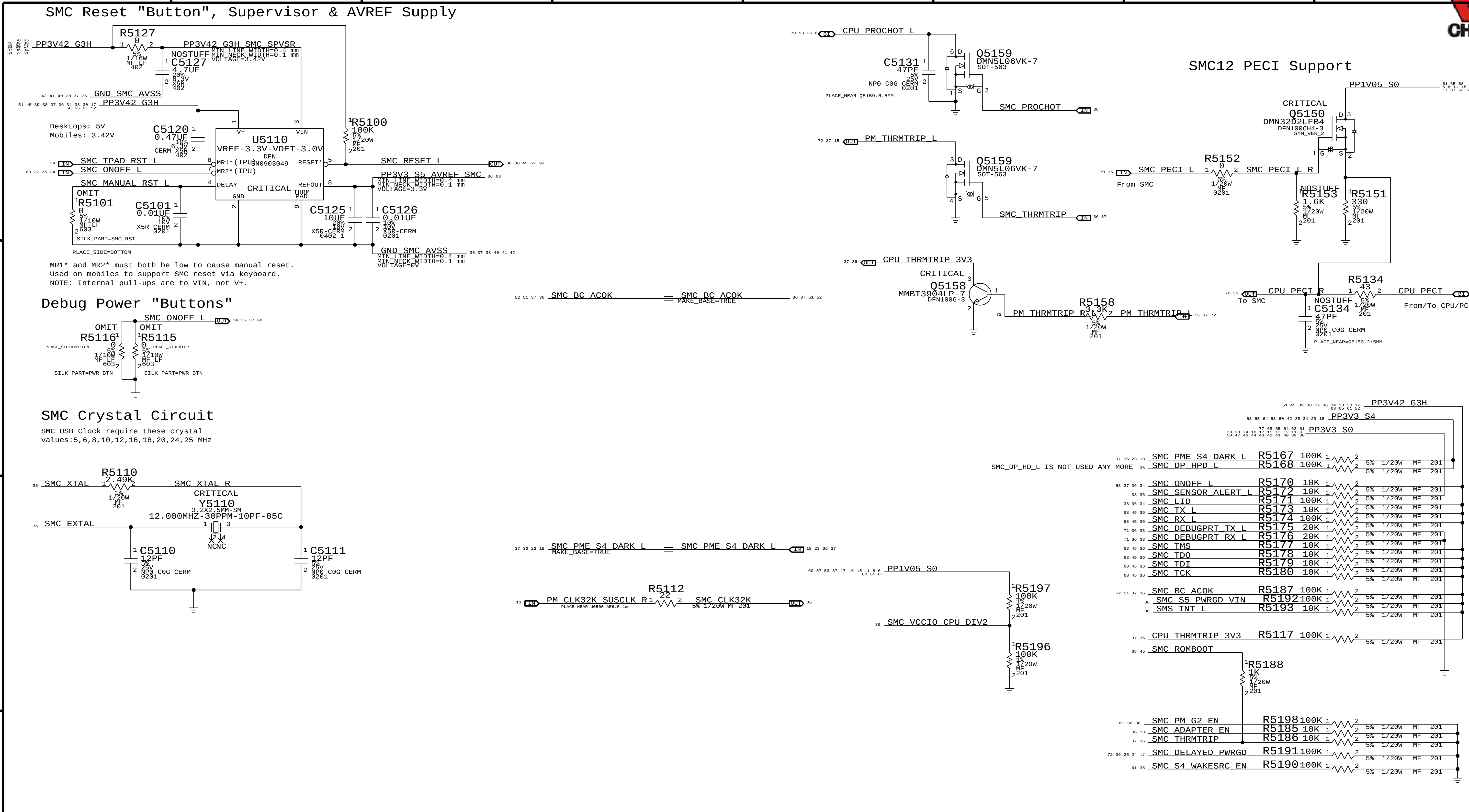

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

SMC Shared Support

Apple Inc.

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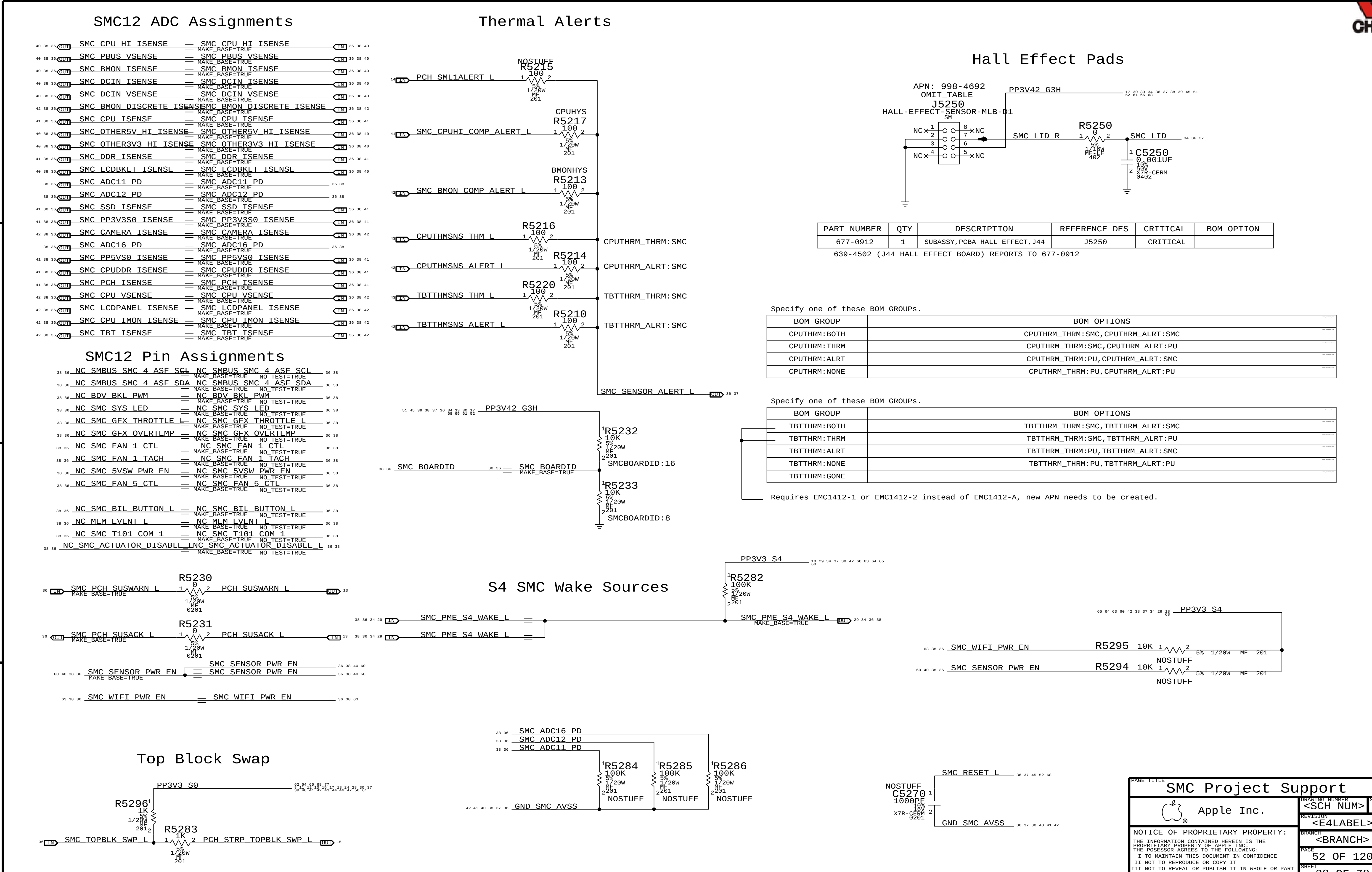
PAGE 51 OF 120

SHEET 37 OF 78



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
677-0912	1	SUBASSY,PCBA HALL EFFECT,J44	J5250	CRITICAL	

639-4502 (J44 HALL EFFECT BOARD) REPORTS TO 677-0912

Specify one of these BOM GROUPs.

BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:PU
CPUTHRM:ALRT	CPUTHRM_THRM:PU,CPUTHRM_ALRT:SMC
CPUTHRM:NONE	CPUTHRM_THRM:PU,CPUTHRM_ALRT:PU

Specify one of these BOM GROUPs.

BOM GROUP	BOM OPTIONS
TBTTHRM:BOTH	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:SMC
TBTTHRM:THRM	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:PU
TBTTHRM:ALRT	TBTTHRM_THRM:PU,TBTTHRM_ALRT:SMC
TBTTHRM:NONE	TBTTHRM_THRM:PU,TBTTHRM_ALRT:PU
TBTTHRM:GONE	

Requires EMC1412-1 or EMC1412-2 instead of EMC1412-A, new APN needs to be created.

SMC Project Support

Apple Inc.

<SCH_NUM> D

<E4LABEL>

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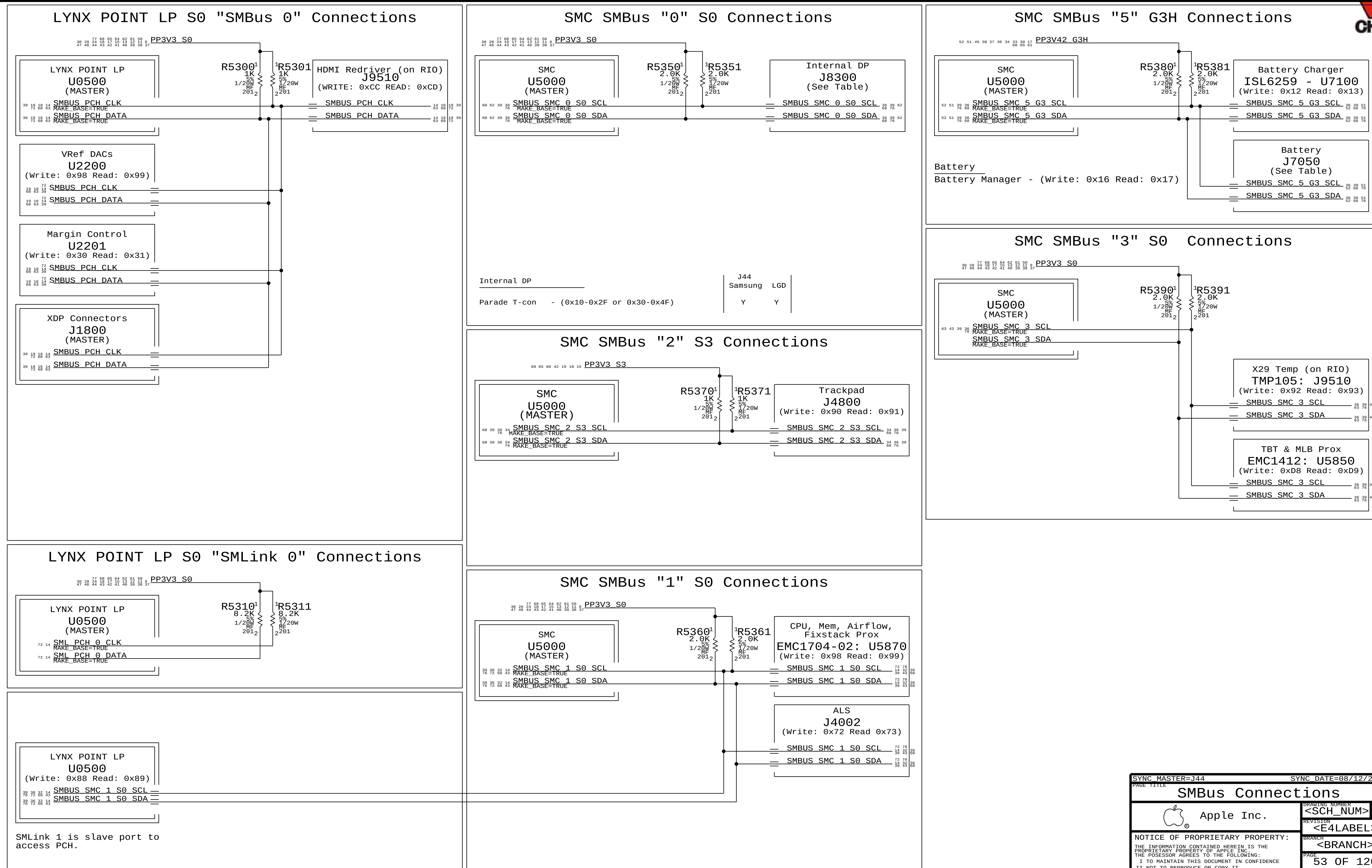

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
SMBus Connections

Apple Inc.

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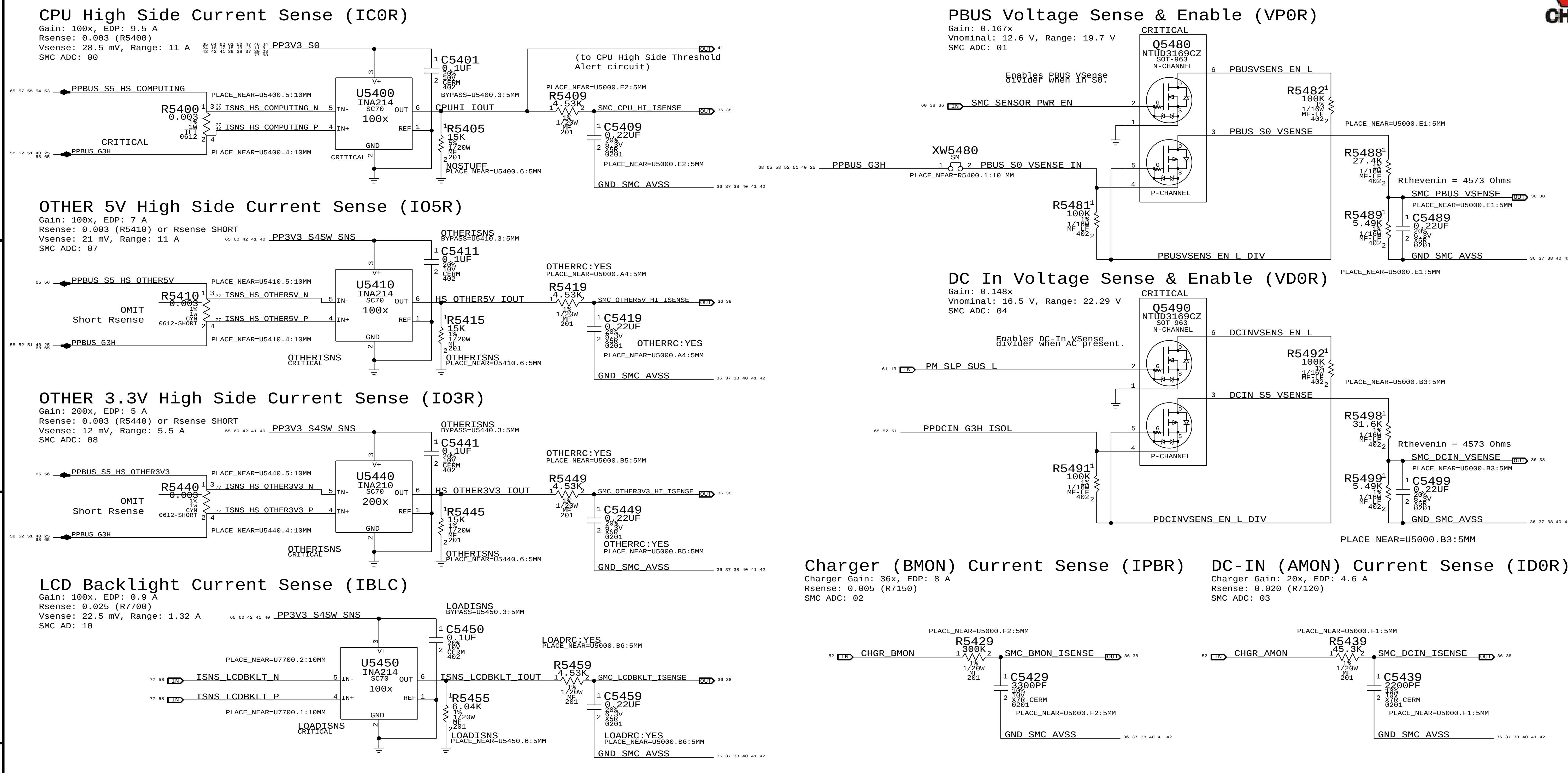

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5419,C5449		OTHERRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5459		LOADRC:NO

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
Power Sensors: High Side

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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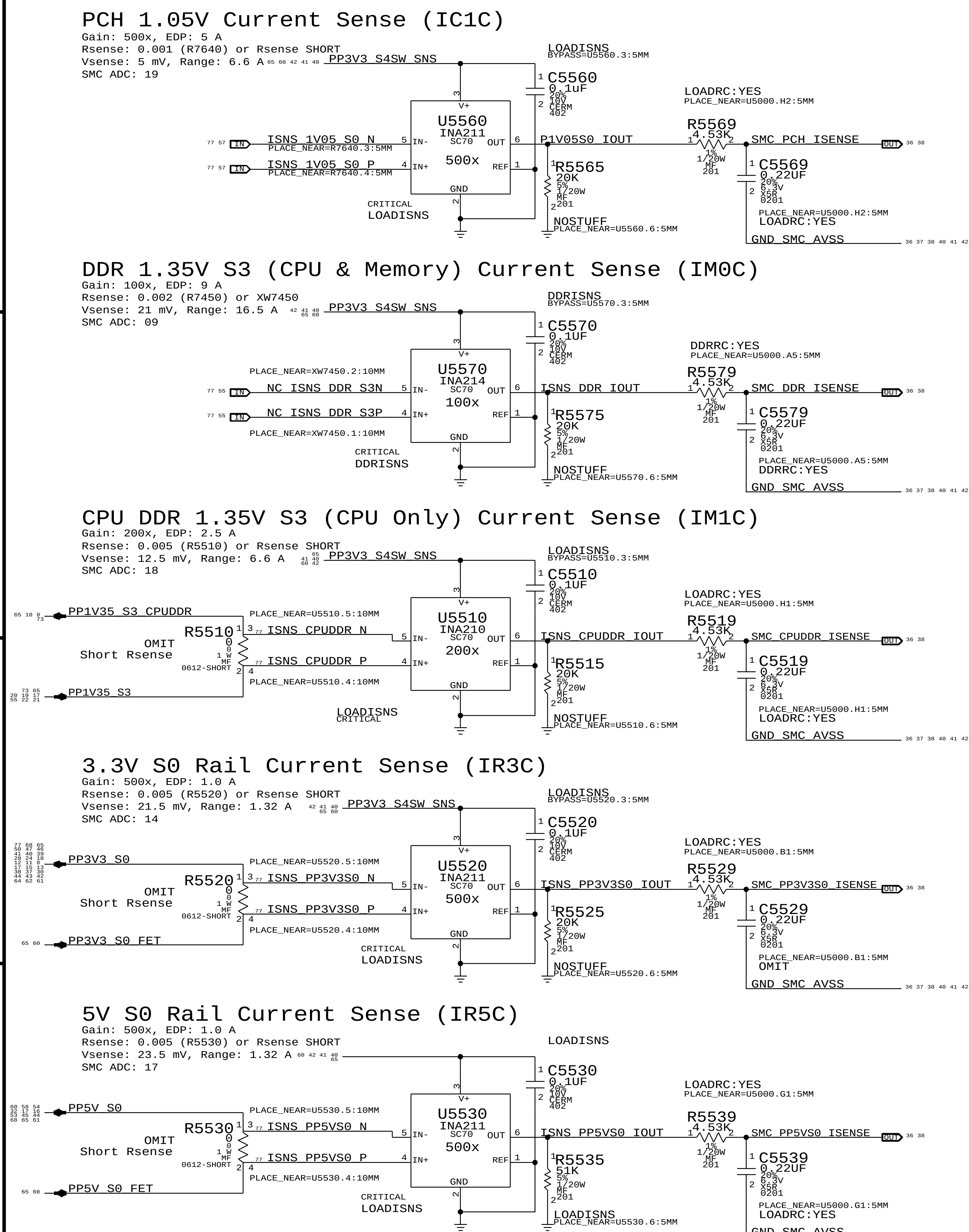

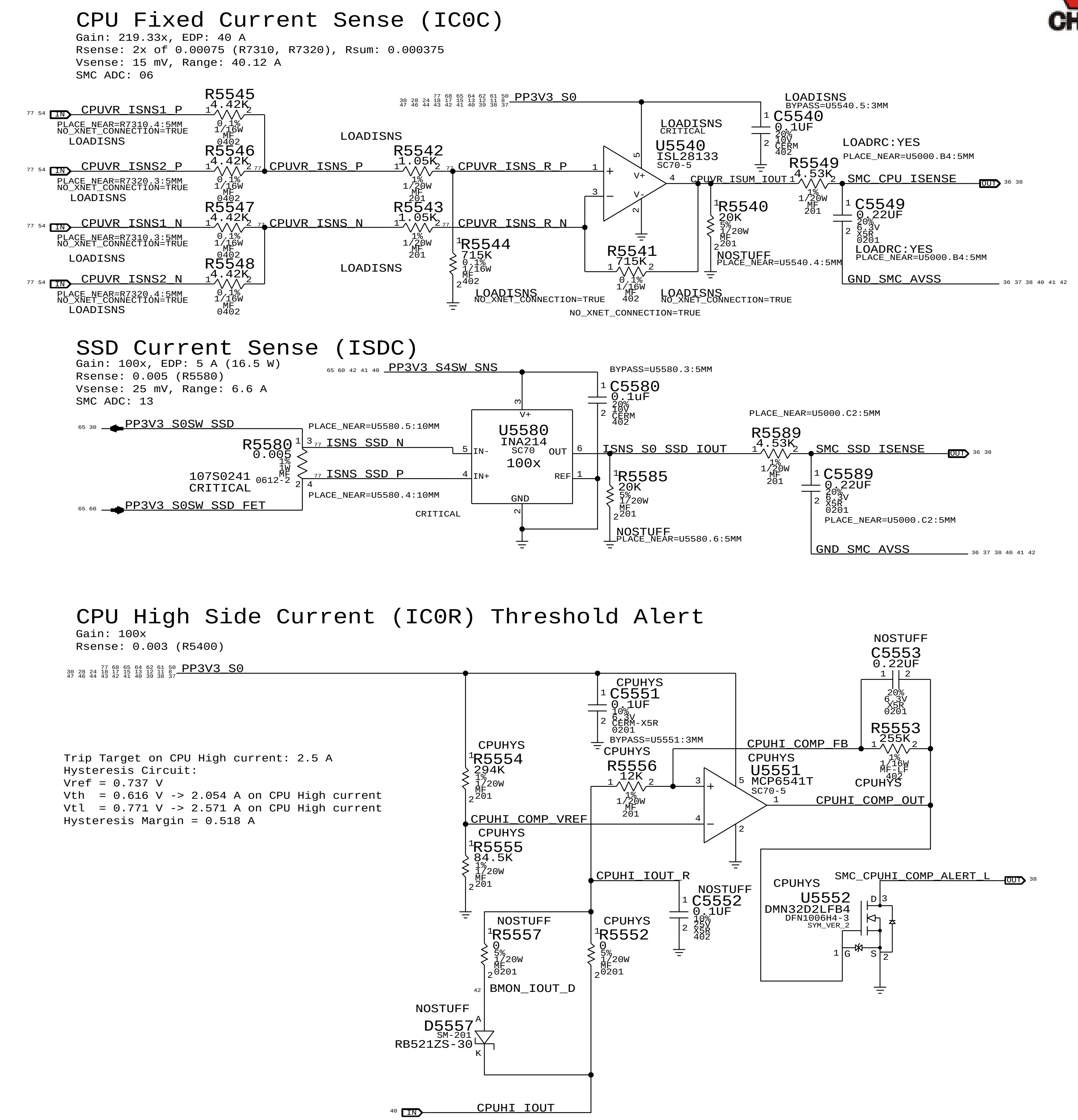

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5569,C5519		LOADRC:NO
117S0008	3	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5529,C5539,C5549		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5579		DDRRC:NO

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

Power Sensors: Load Side

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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CPU High Side (IC0R) Peak Detection Support

In battery discharge scenario negative voltage will be present on IN+/- pins with INA output voltage decreasing from 3.3V with increasing discharge current.

SENSE+ pins of EMC1704 sink 10-20uA current.
This deviation has been designed in our Peak Detection circuit.
With 10uA sink: 0.125A - 2.1A -> 13mV - 83 mV
With 20uA sink: 0.125A - 2.1A -> 23mV - 92 mV

Thunderbolt TBT Current/Voltage Sense (IHSC/VHSC)

Gain: 200x. EDP: 2.8 A
Rsense: 0.005 (R5640) or Rsense SHORT
Vsense: 14 mV, Range: 3.3 A
SMC AD: 23

Battery BMON Discrete Current Sense (IP0R) & Threshold Alert

Gain: 50x. EDP: 8 A
Rsense: 0.005 (R7150)
Vsense: 50 mV, Range: 13.2 A
SMC AD: 05

LCD Panel Current Sense (ILDC)

Gain: 500x. EDP: 1 A
RSENSE: 0.005 (R8320) or Rsense SHORT
Vsense: 5 mV, Range: 1.32 A
SMC AD: 21

CHGR_CSO_R_P/N are swapped on purpose
to measure Battery discharge power
into system.

Trip Target on Battery current: 3.5 A
Hysteresis Circuit:
Vref = 0.854 V
Vth = 0.758 V -> 3.031 A on Battery current
Vtl = 0.887 V -> 3.549 A on Battery current
Hysteresis Margin = 0.518 A

Camera (S2 Controller) Current Sense (ICMC)

Gain: 500x. EDP: 0.82 A
Rsense: 0.005 (R5610) or XW5610
Vsense: 4.1 mV, Range: 1.32 A
SMC AD: 15

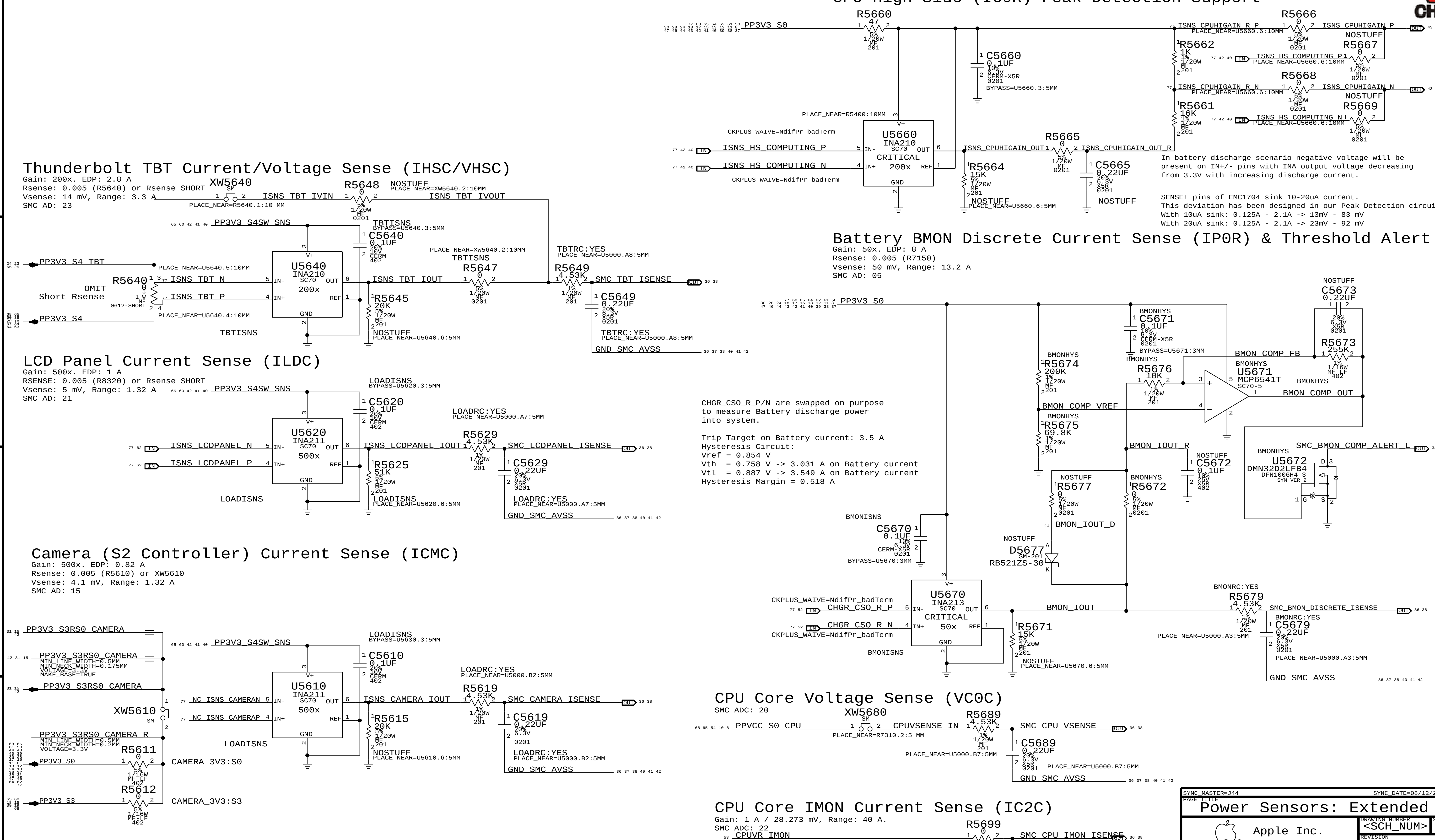
CPU Core Voltage Sense (VC0C)

SMC ADC: 20

CPU Core IMON Current Sense (IC2C)

Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 22

With R7210 (Ri) set to 316 Ohm,
R7310 (Rsen) set to 0.75 mOhm,
R7230 set to 95.3 kOhm,
Num Phases (N) is 2, and Io (ICCmax) is 40A,
then 1A of Io gives 28.273mV at the Vimon.



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5619,C5629,C5649		
117S0008	1	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5679		

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
Power Sensors: Extended

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>
PAGE 56 OF 120
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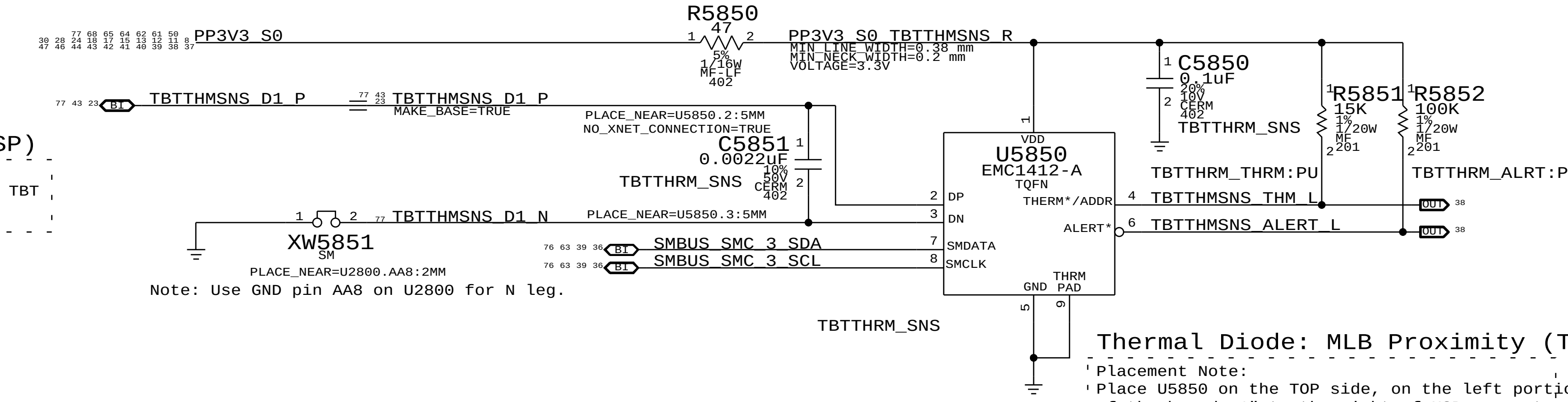
Thermal Sensor A:
Thunderbolt Die, MLB Proximity

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (THSP)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AA8.



U5850 I2C Address:
By setting R5851 to 15k, I2C address for U5850 is 0xD8/0xD9.

Thermal Diode: MLB Proximity (TMLB)

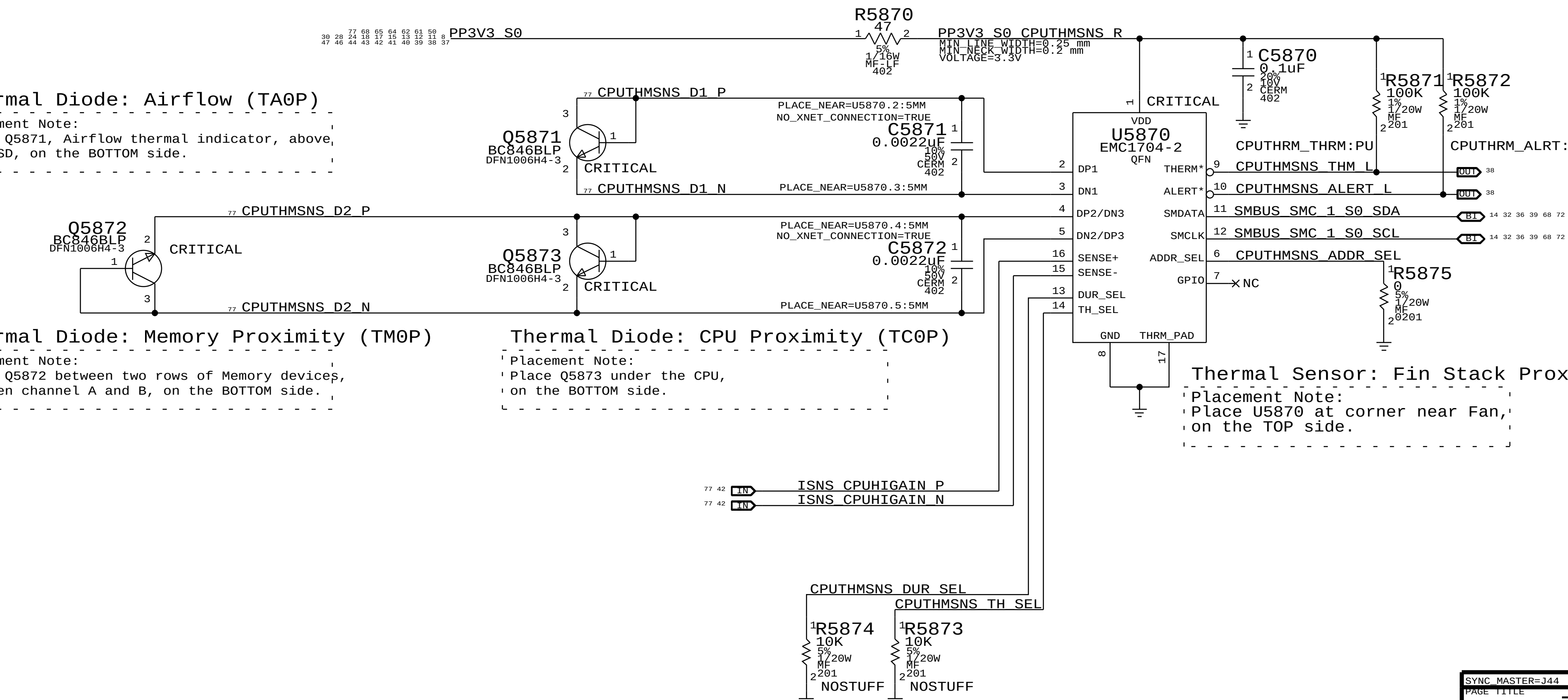
Placement Note:
Place U5850 on the TOP side, on the left portion of the board, 1" to the right of USB connector.

Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity

I2C Write: 0x98, I2C Read: 0x99

Thermal Diode: Airflow (TA0P)

Placement Note:
Place Q5871, Airflow thermal indicator, above the SSD, on the BOTTOM side.



Thermal Diode: Memory Proximity (TM0P)

Placement Note:
Place Q5872 between two rows of Memory devices, between channel A and B, on the BOTTOM side.

Thermal Diode: CPU Proximity (TC0P)

Placement Note:
Place Q5873 under the CPU, on the BOTTOM side.

Thermal Sensor: Fin Stack Proximity (Th1H)

Placement Note:
Place U5870 at corner near Fan, on the TOP side.

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Thermal Sensors

Apple Inc.

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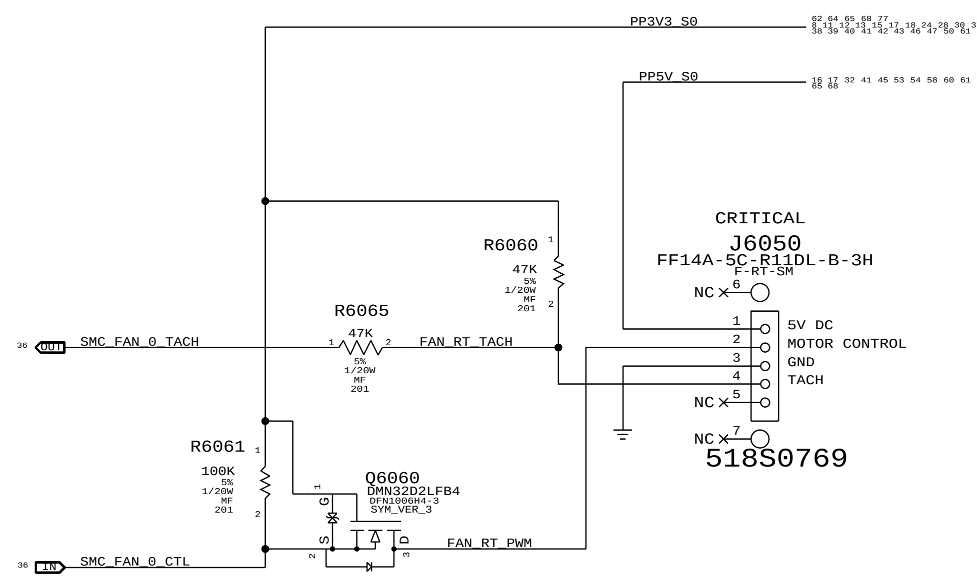
FAN CONNECTOR

KEEP THE 5 PIN CONNECTOR FROM D1



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Apple Inc.	DRAWING NUMBER: <SCH_NUM>	SIZE: D
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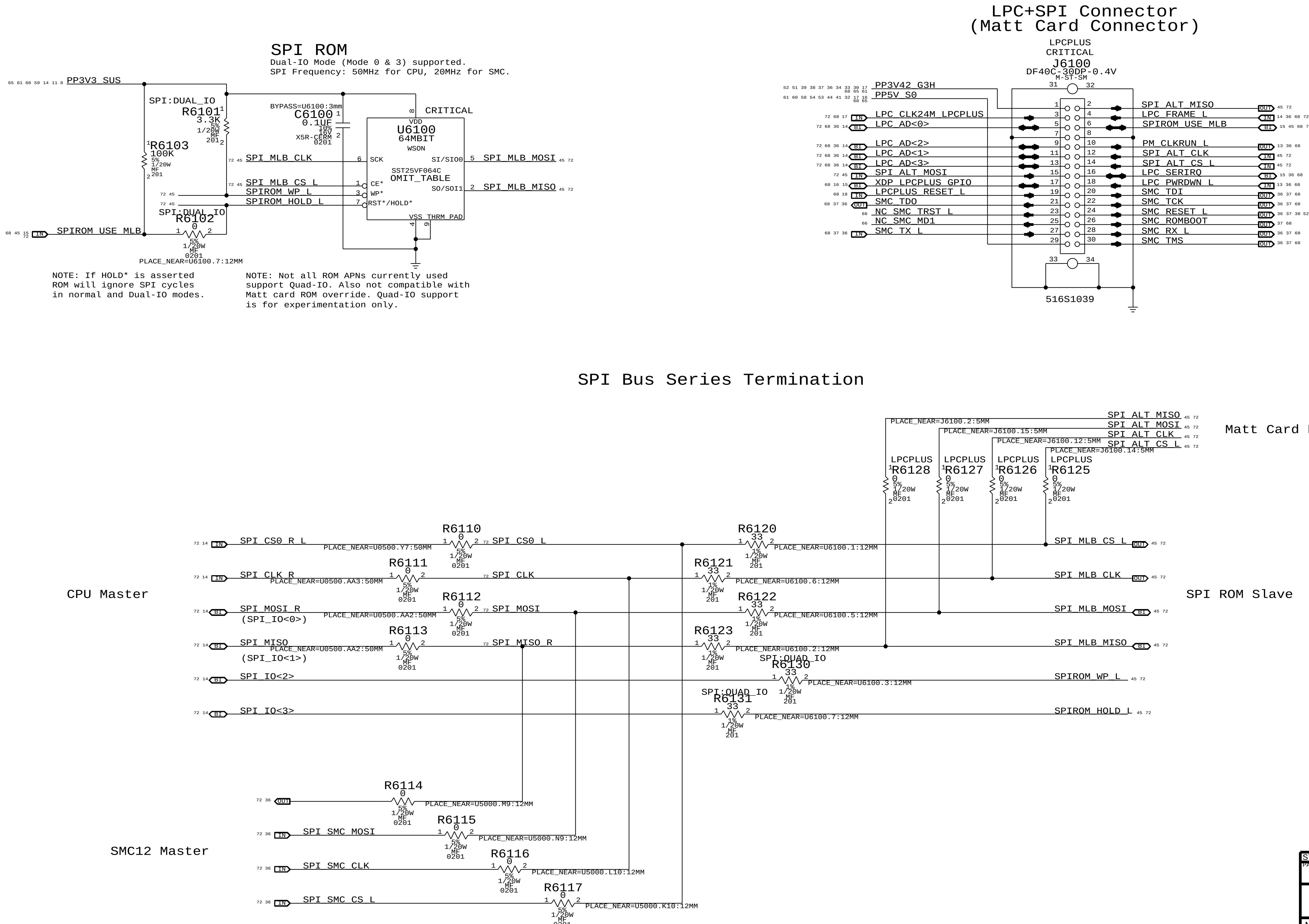
	PAGE: 60 OF 120	
	SHEET: 44 OF 78	



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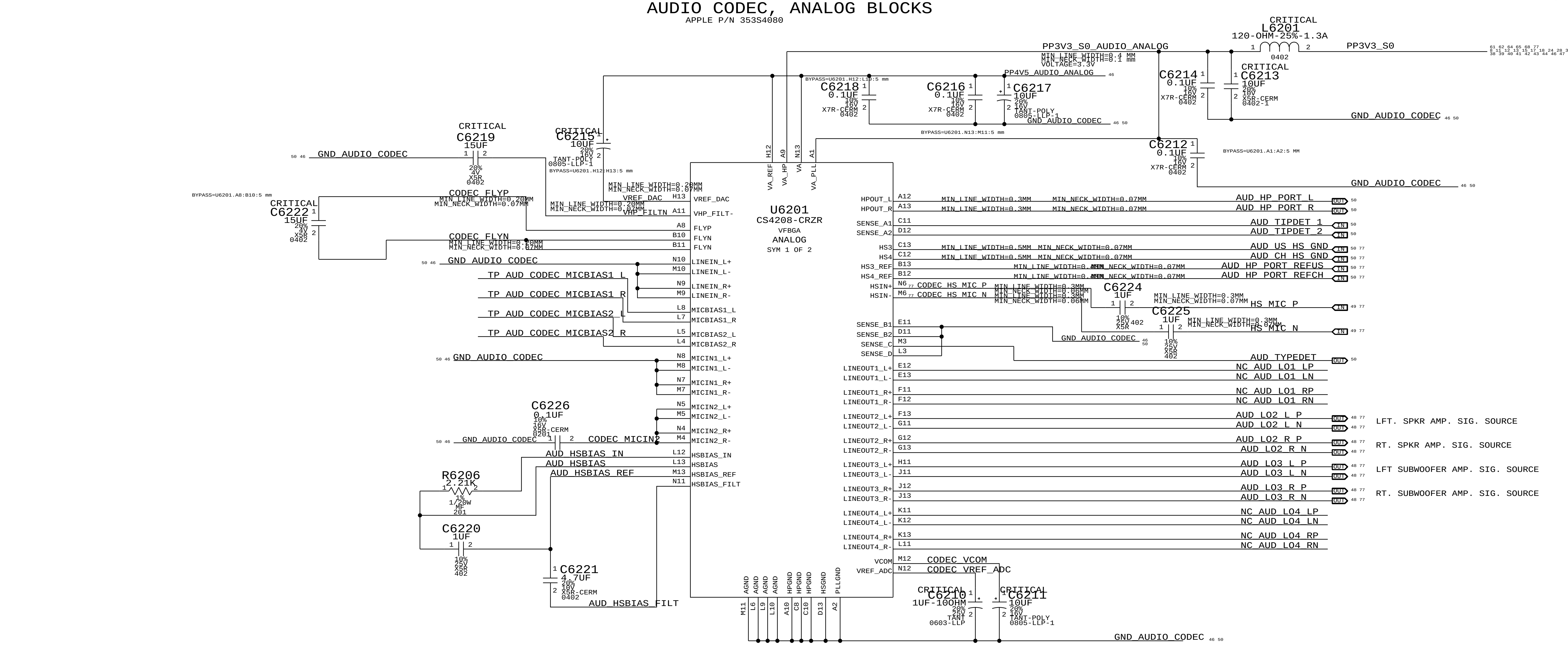
NOTICE OF PROPRIETARY PROPERTY:	BRANCH: <BRANCH>	
	PAGE: 61 OF 120	
	SHEET: 45 OF 78	

AUDIO CODEC, ANALOG BLOCKS

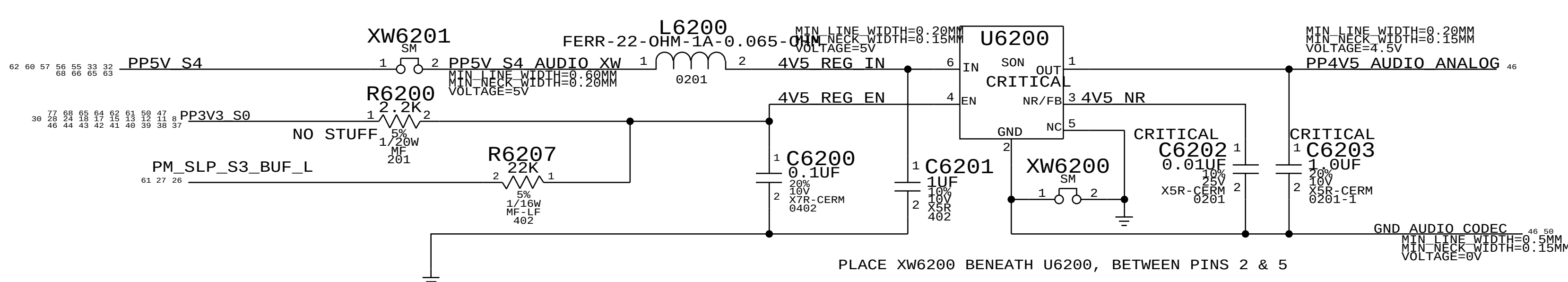
APPLE P/N 353S4080



4.5V POWER SUPPLY FOR CODEC

APPLE P/N 353S2456

PLACE XW6201 NEAR 5V SOURCE



PLACE XW6200 BENEATH U6200, BETWEEN PINS 2 & 5

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

AUDIO:CODEC, ANALOG

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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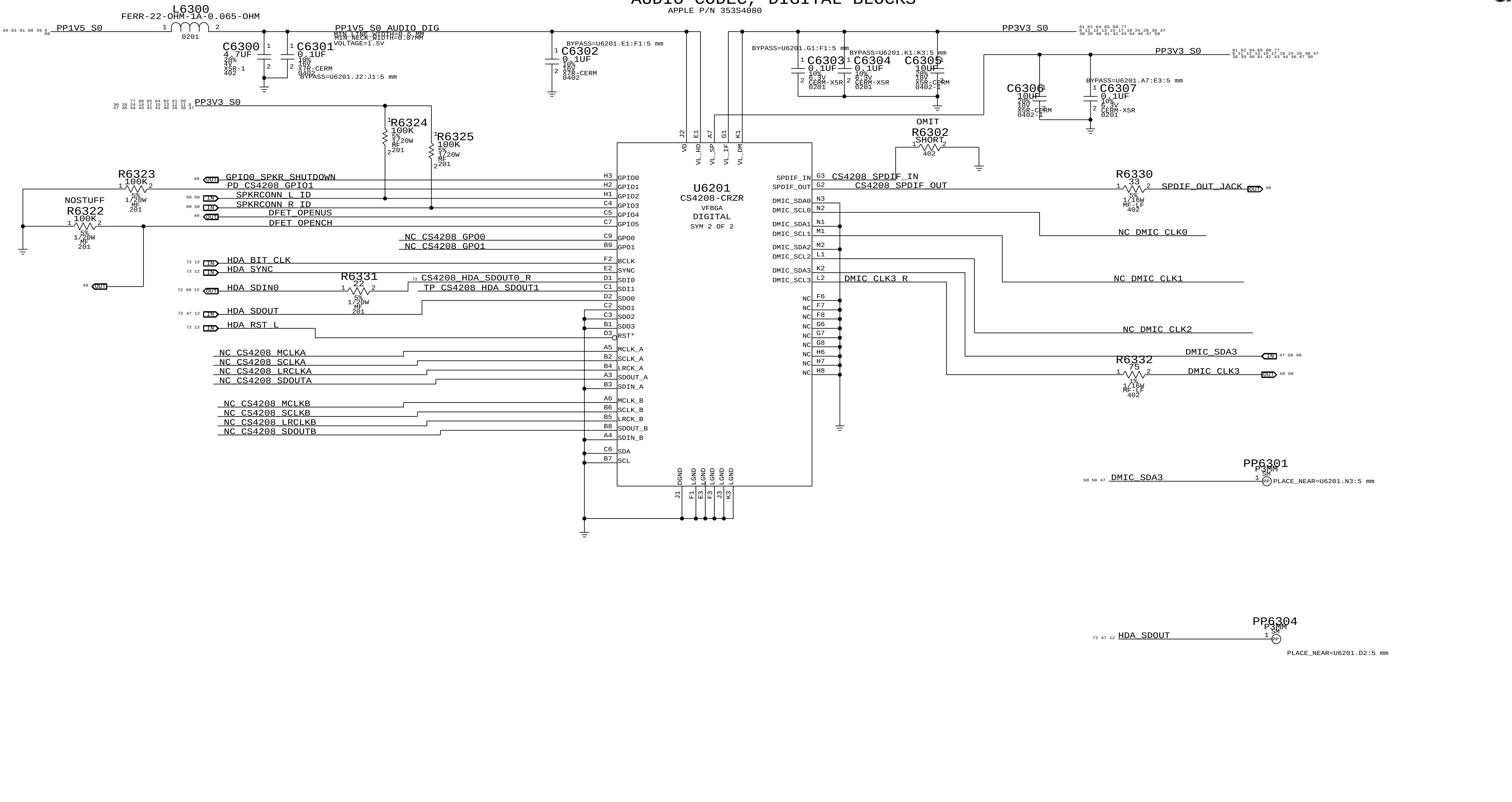
AUDIO CODEC, DIGITAL BLOCKS

APPLE P/N 353S4080



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PAGE TITLE

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Apple Inc.

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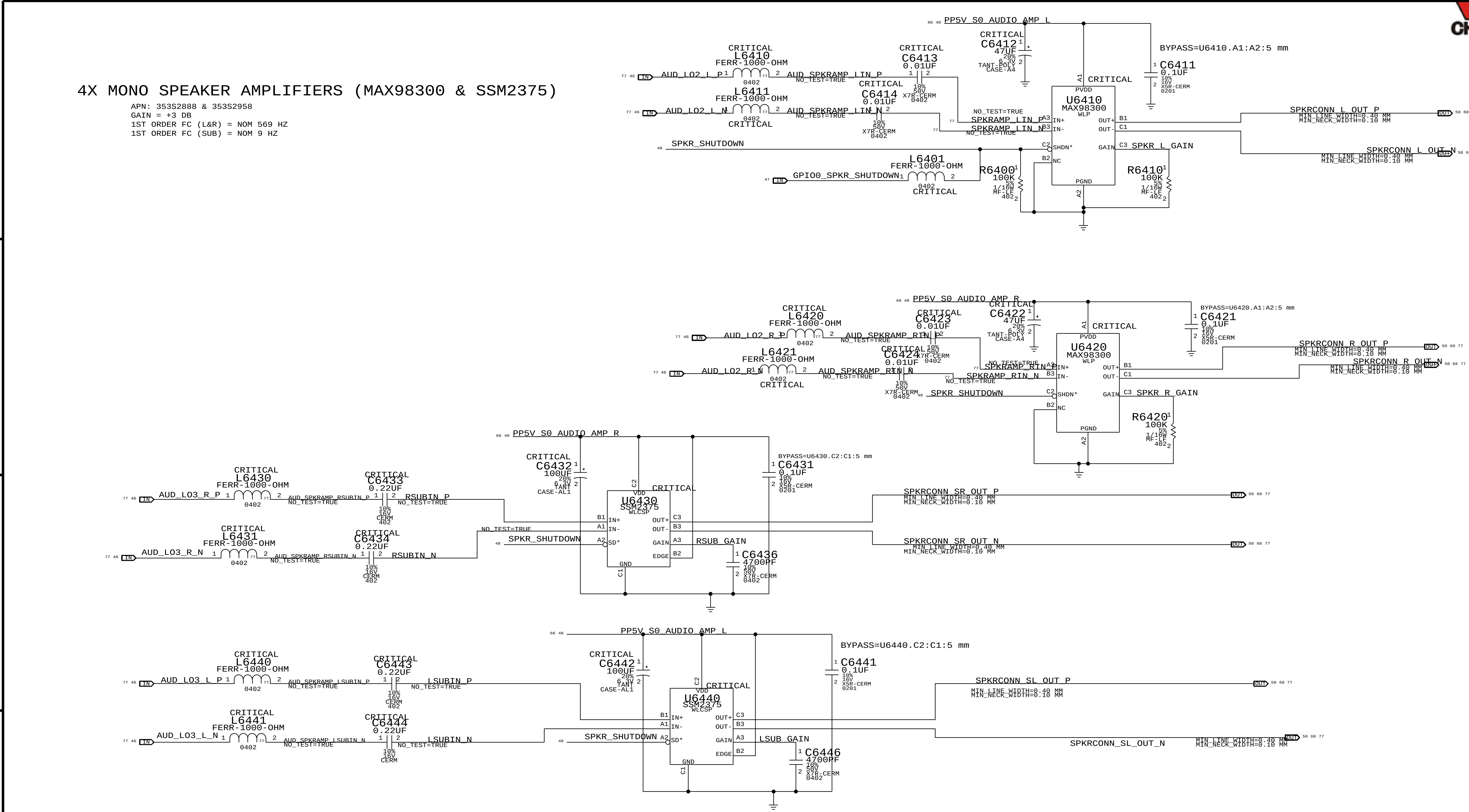

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PAGE TITLE

AUDIO: SPEAKER AMP

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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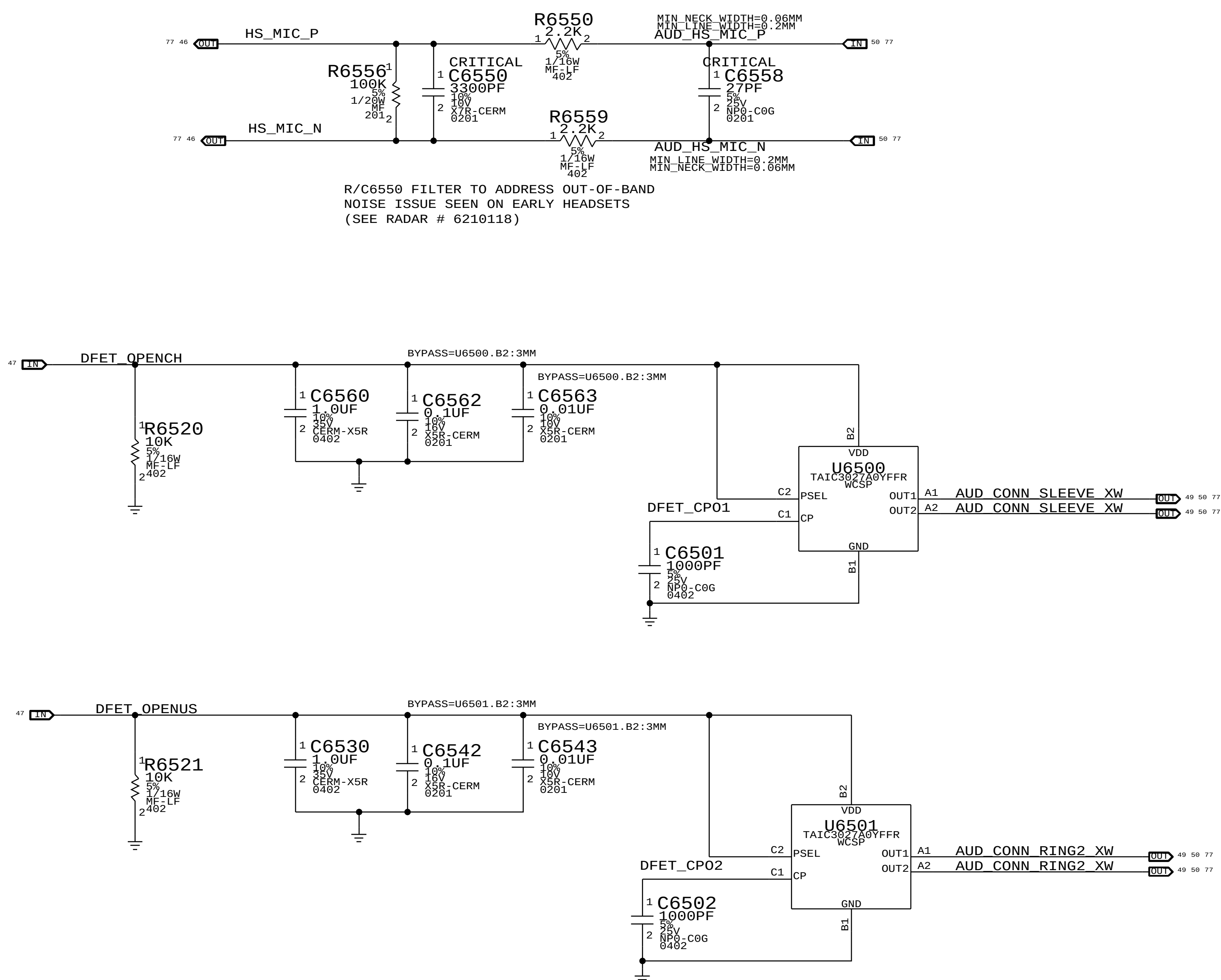

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AUDIO: JACK

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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CODEC OUTPUT SIGNAL PATHS

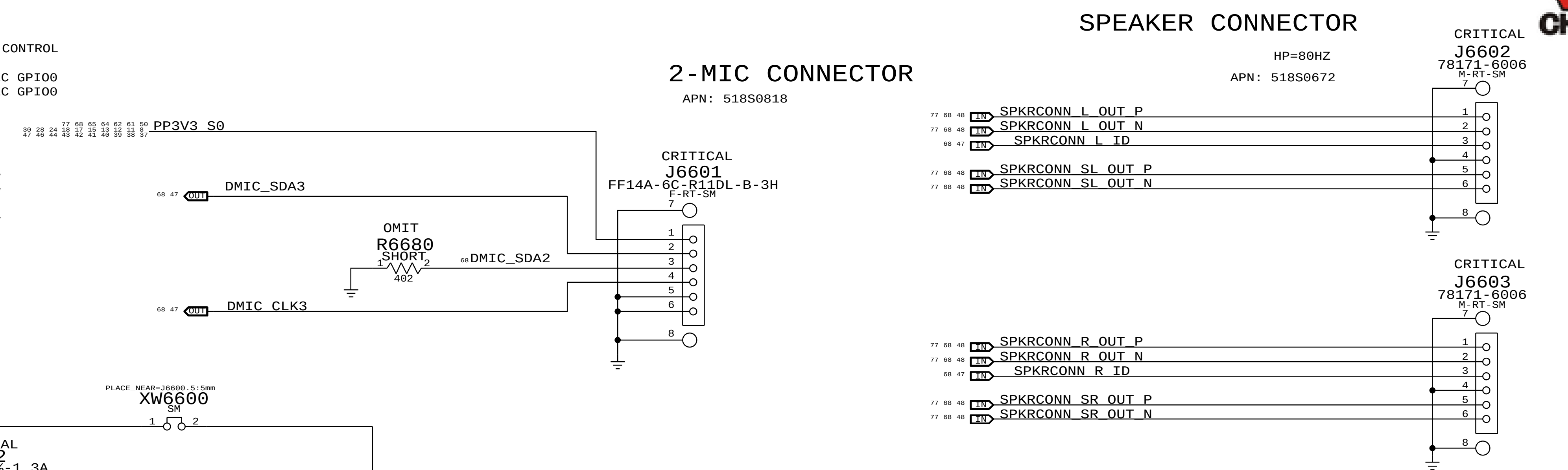
FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

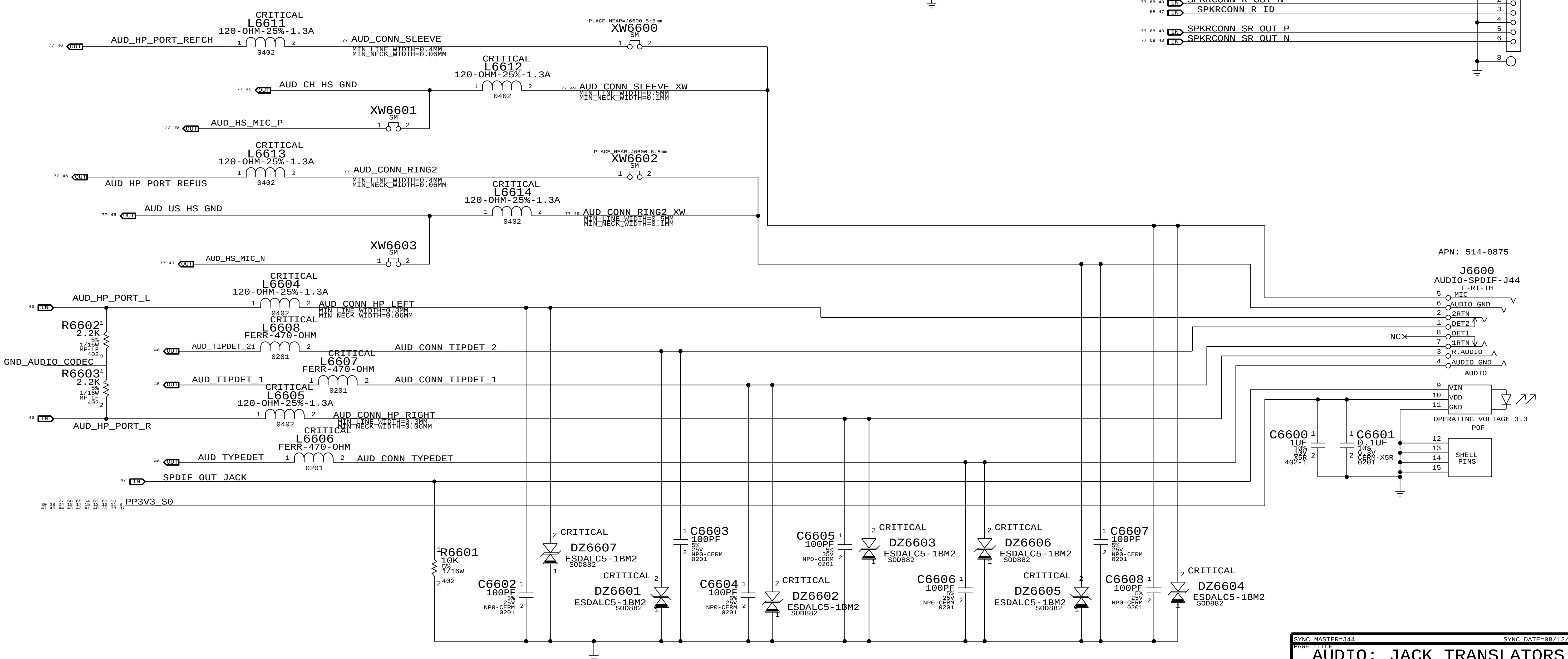
CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

OTHER CODEC GPIO LINES

LEFT SPEAKER ID	GPIO2 INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3 INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4 OUTPUT	HIGH = DFETs OPEN





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Apple Inc.

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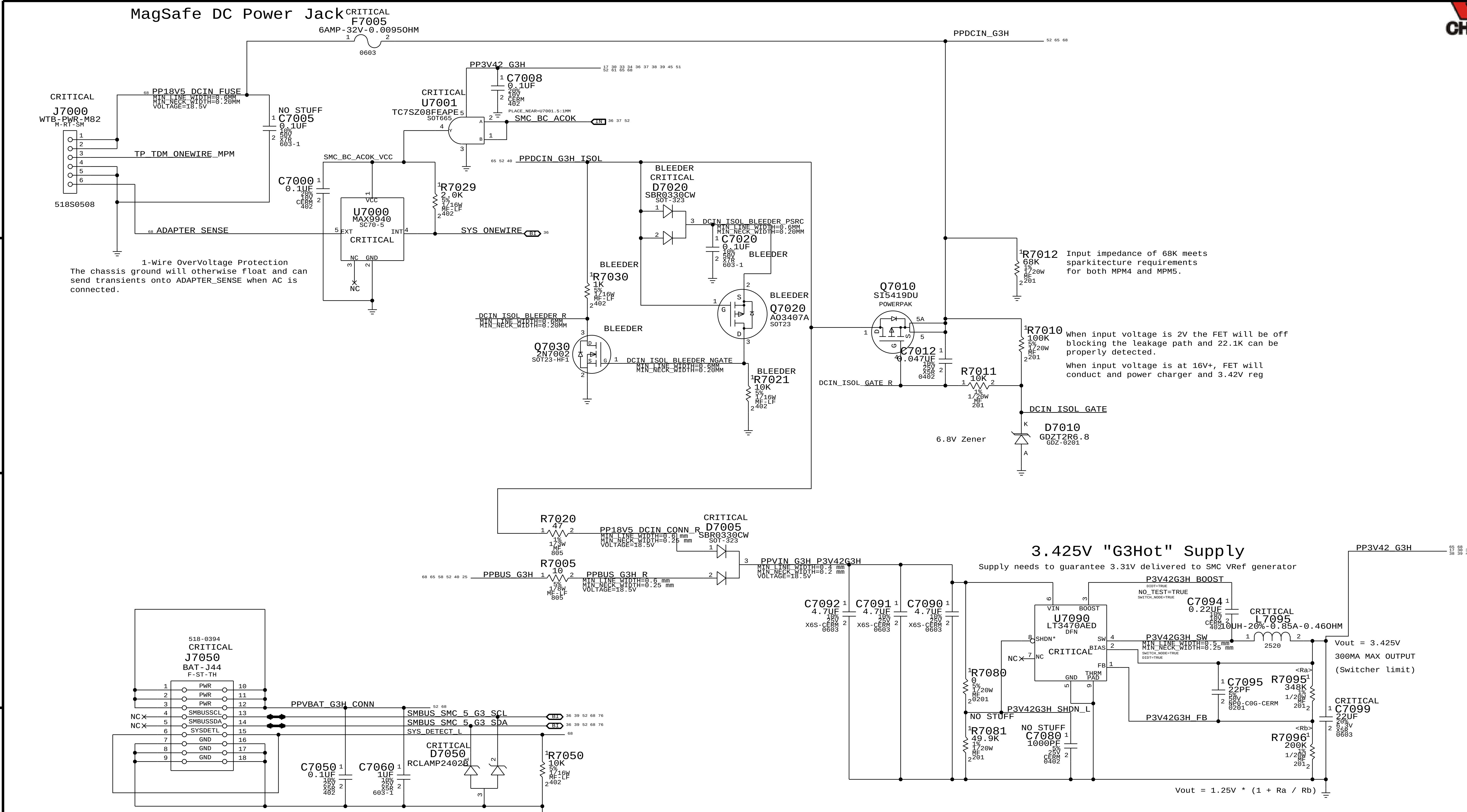

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PAGE TITLE
DC-In & Battery Connectors

Apple Inc.

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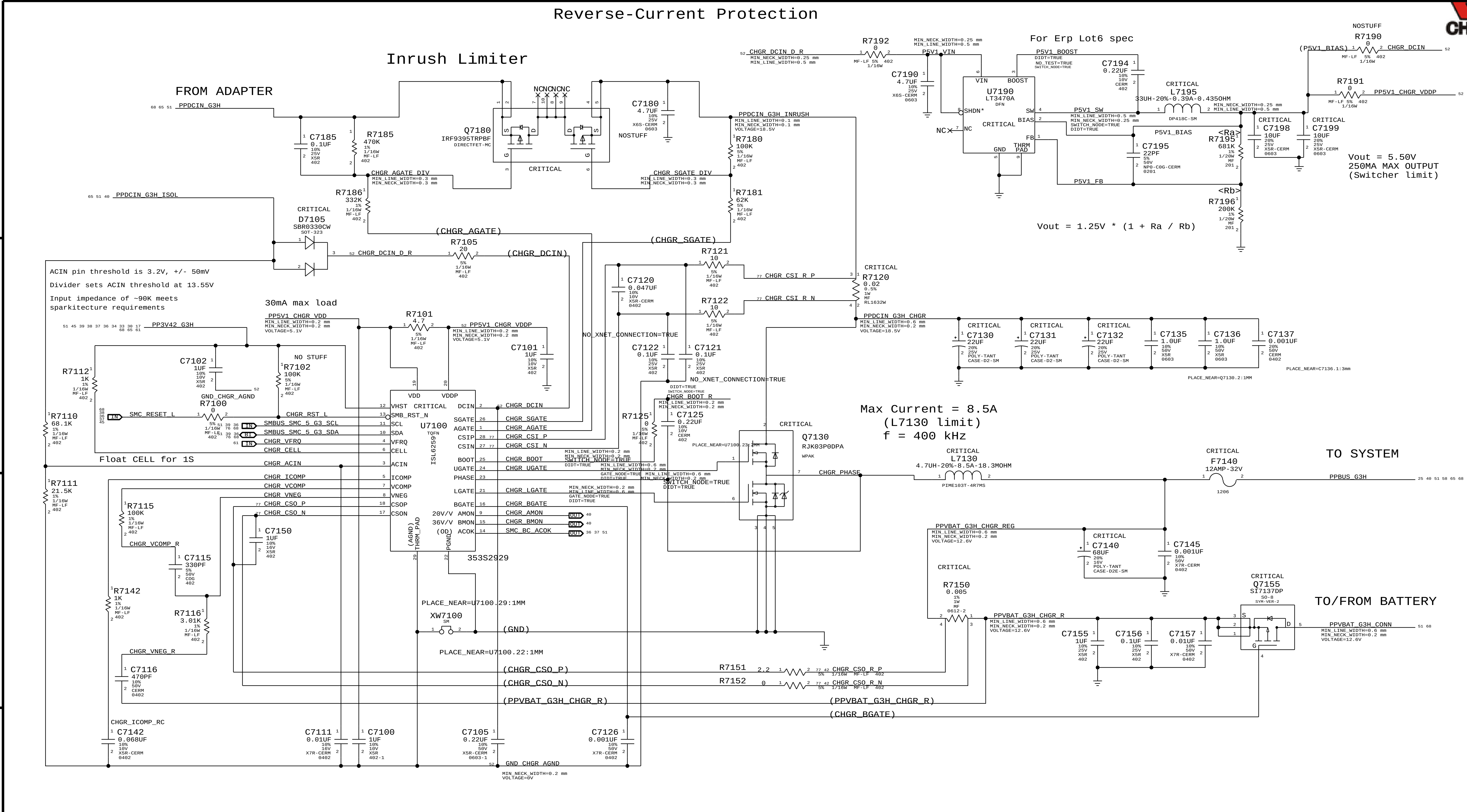

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PAGE TITLE

PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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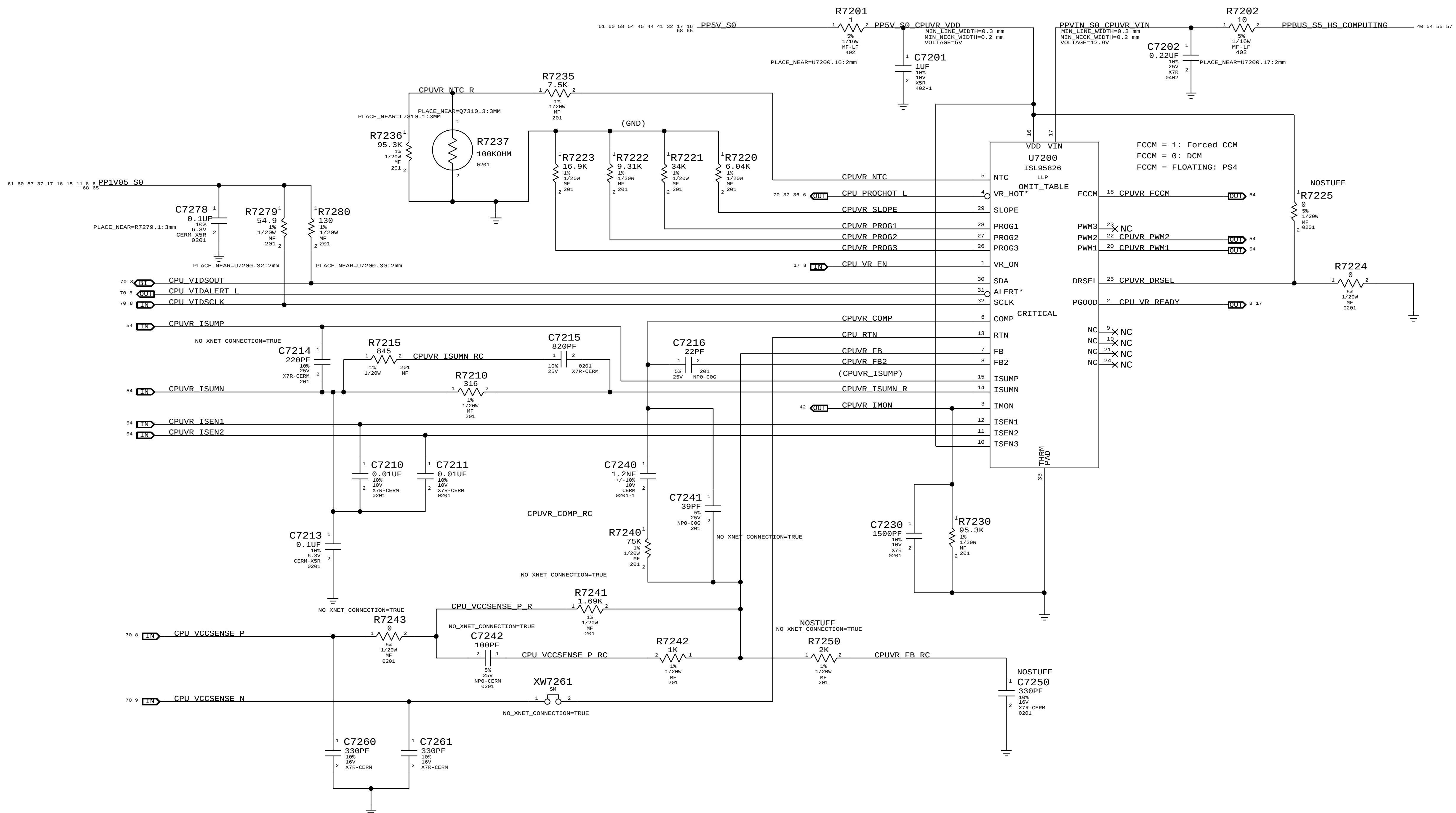
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S4170	1	IC,ISL95826R6200,PWM,PGOOD,SCREN,32P,QFN	U7200	CRITICAL	



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PAGE TITLE

CPU VR12.6 VCC Regulator IC

Apple Inc.

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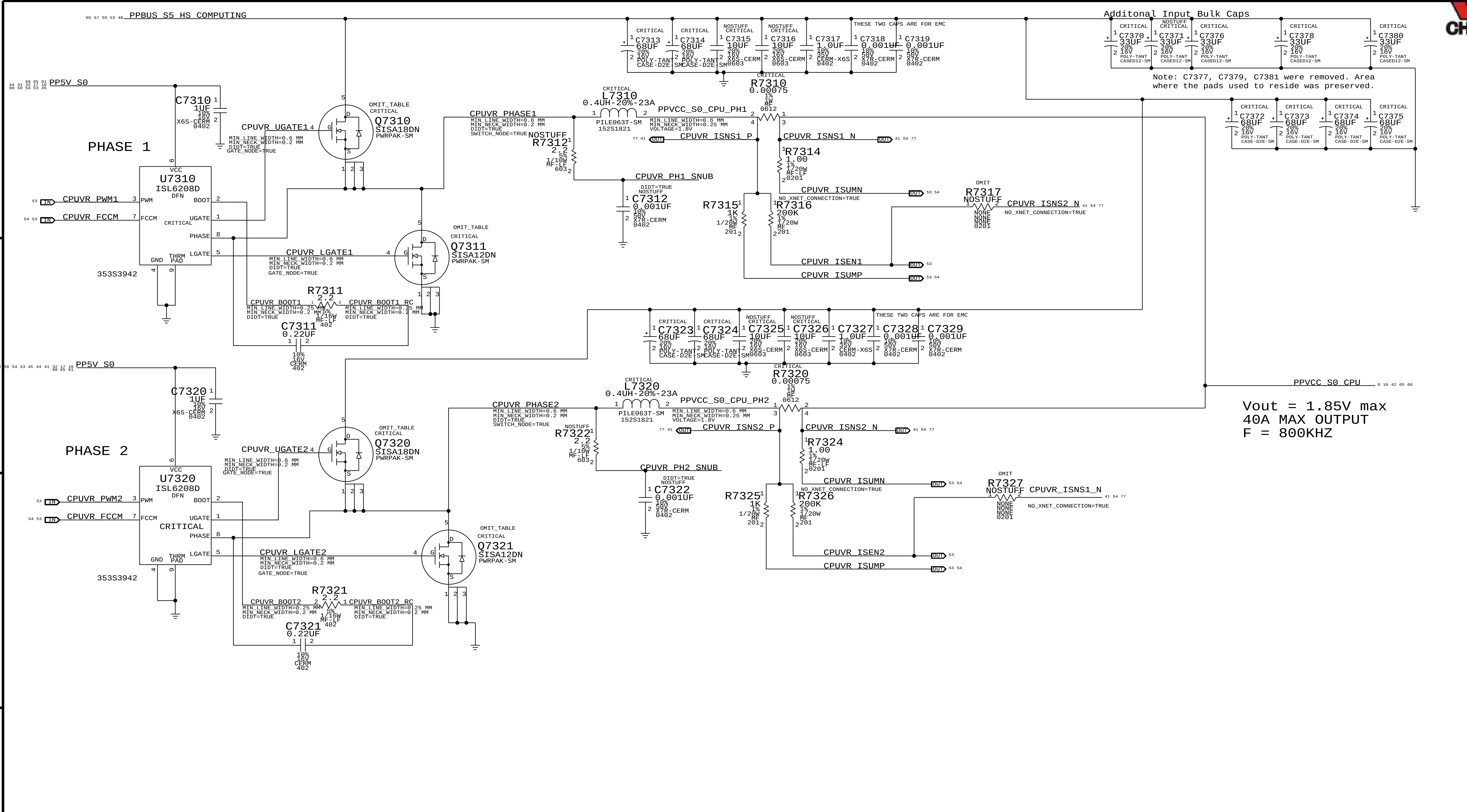

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
CPU VR12.5 VCC Power Stage

Apple Inc.

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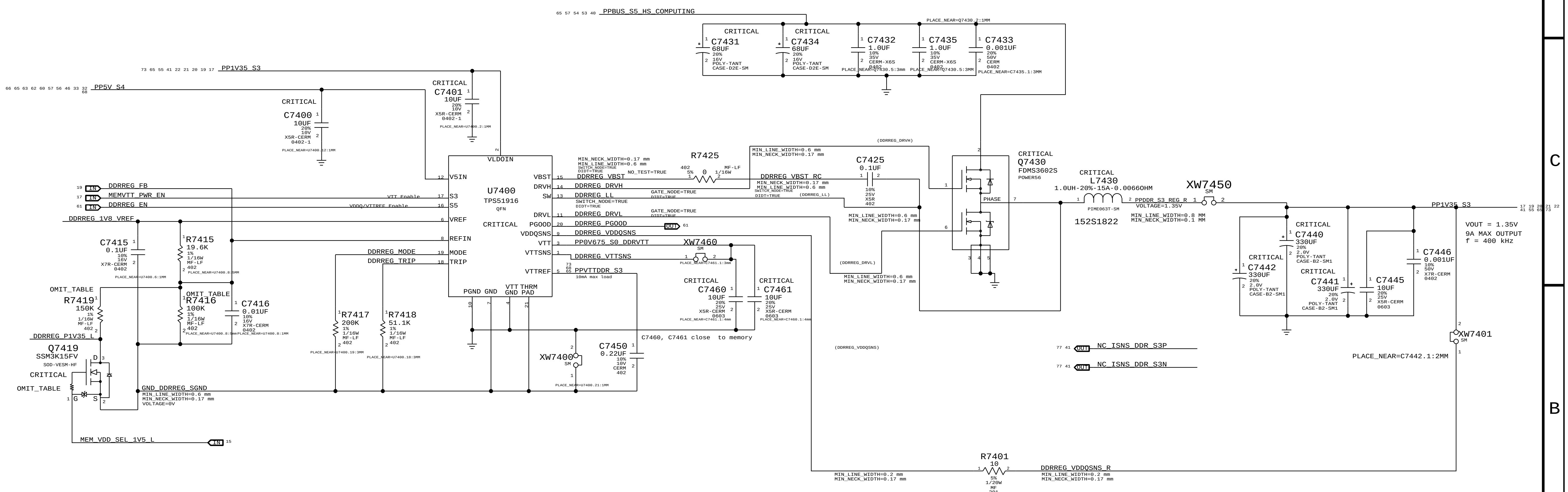
DDR3L (1V35 S3) REGULATOR



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0411	1	RES,MTL FILM,1/16W,100K,1,0402,SMD,LF	R7416	CRITICAL	PPDDR:1V5
114S0391	1	RES,MTL FILM,1/16W,60.4K,1,0402,SMD,LF	R7416	CRITICAL	PPDDR:1V35
376S0612	1	MOSFET,N-CH,30V,100MA,7.0OHM,SOT-723,HF	Q7419	CRITICAL	PPDDR:1V5
					PPDDR:1V5

SYNC_MASTER=J44 SYNC_DATE=08/12/2011

PAGE TITLE
1.35V DDR3 SUPPLY

Apple Inc.

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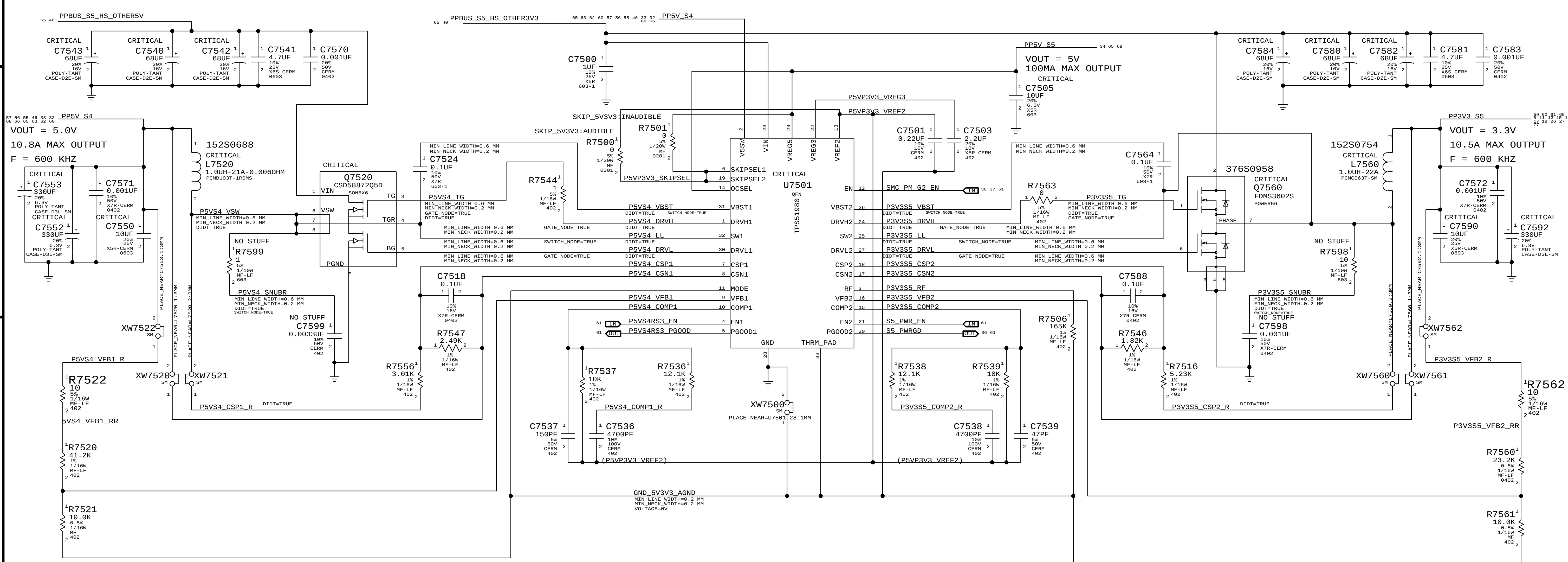

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

5V / 3.3V Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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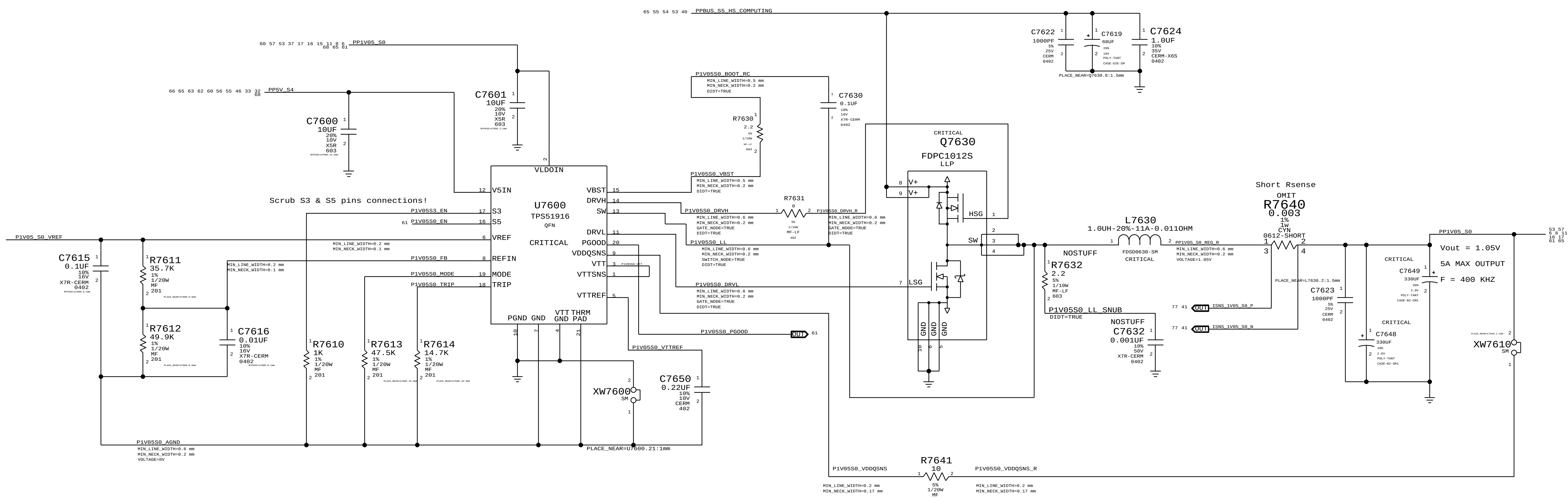
1.05V S0 Regulator



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	REVISION: <E4LABEL>
	BRANCH: <BRANCH>
	PAGE: 76 OF 120
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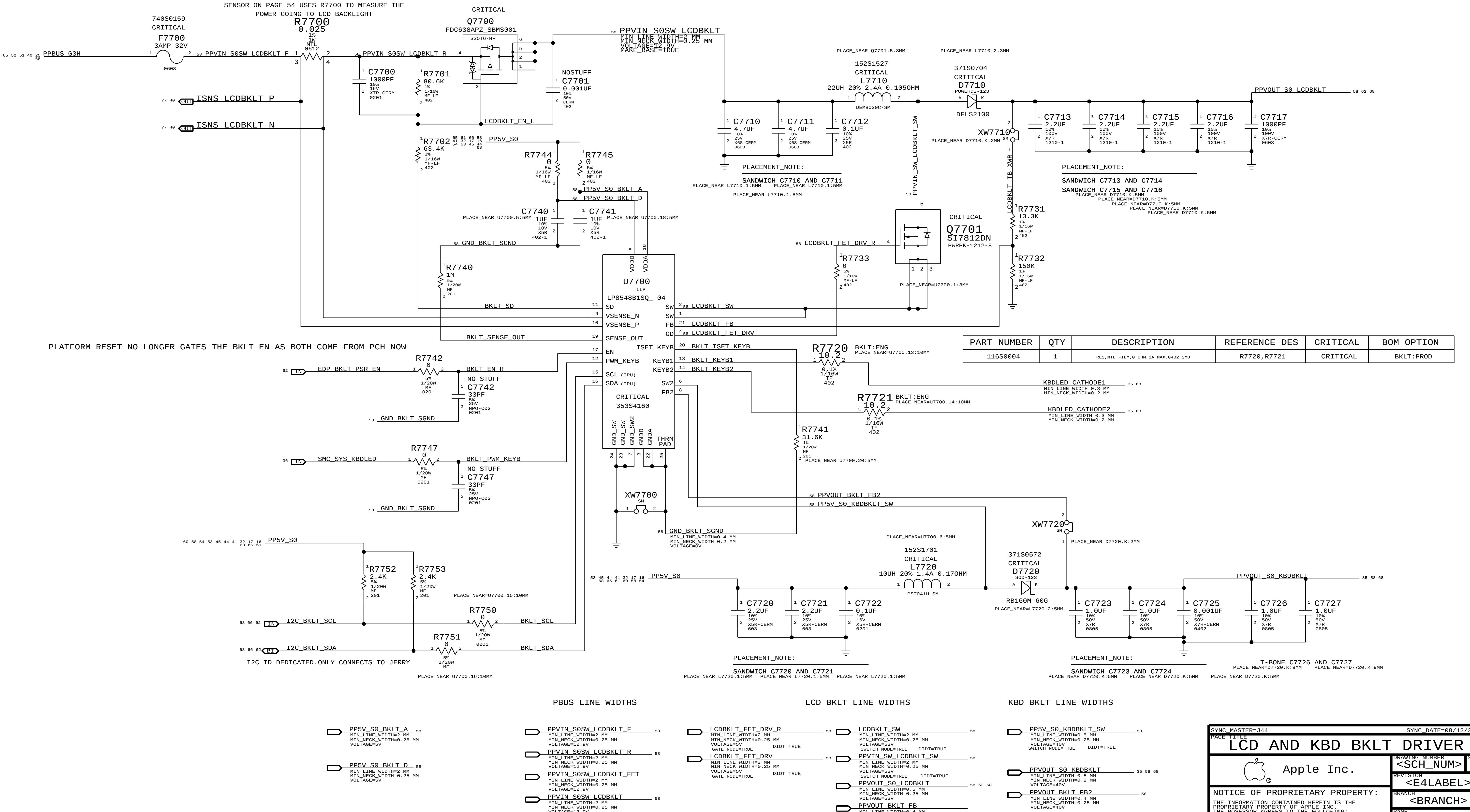
Page Notes

Power aliases required by this page:
- =PPVIN_S0SW_LCDBKLTFET (9-12.6V LCD BACKLIGHT INPUT)
- =PP5V_S0_BKLT (5V BACKLIGHT DRIVER INPUT)
- =PP5V_S0SW_KBDLED (5V KEYBOARD BACKLIGHT INPUT)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FILM,0 OHM,1A MAX,0402,SMD	R7720,R7721	CRITICAL	BKLT:PROD

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
LCD AND KBD BKLT DRIVER

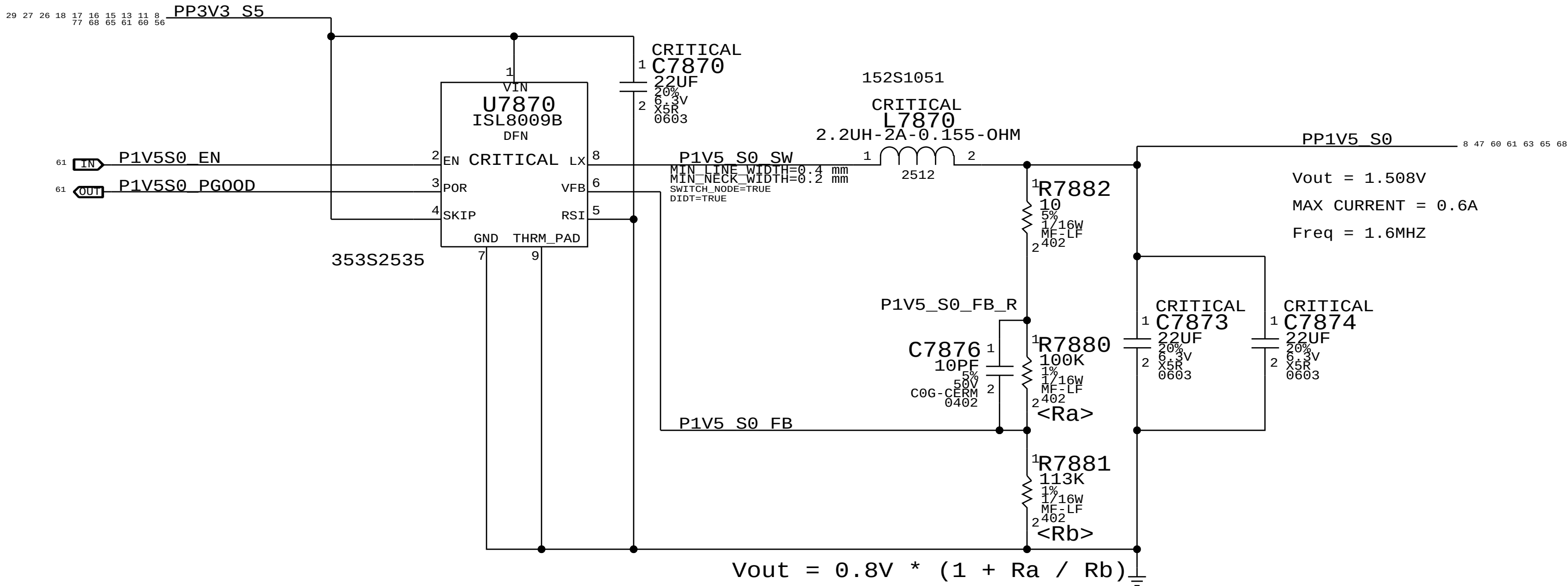
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

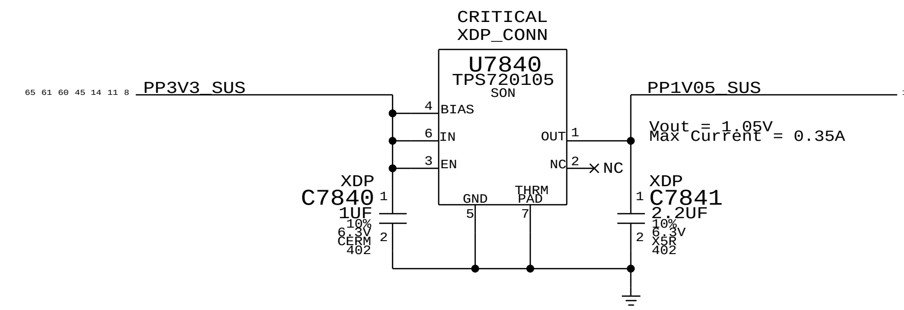
BRANCH <BRANCH>

1.5V S0 Switcher



1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Misc Power Supplies

Apple Inc.

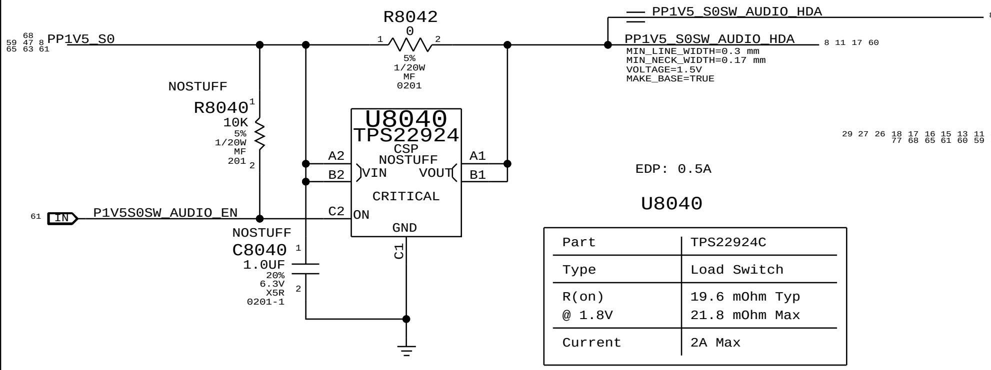
DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

1.5V S0 Audio Switch (BYPASSED)

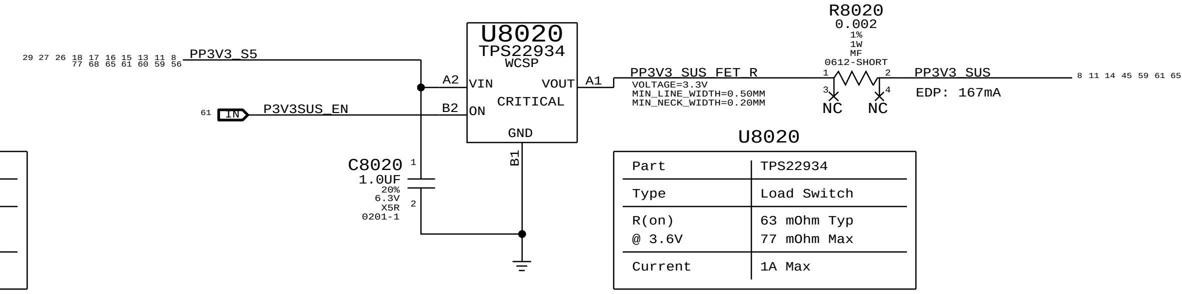
Loading specs per J41/43_PowerBudget_Riviera_rev0.99e



U8040

Part	TPS22924C
Type	Load Switch
R(on) @ 1.8V	19.6 mOhm Typ 21.8 mOhm Max
Current	2A Max

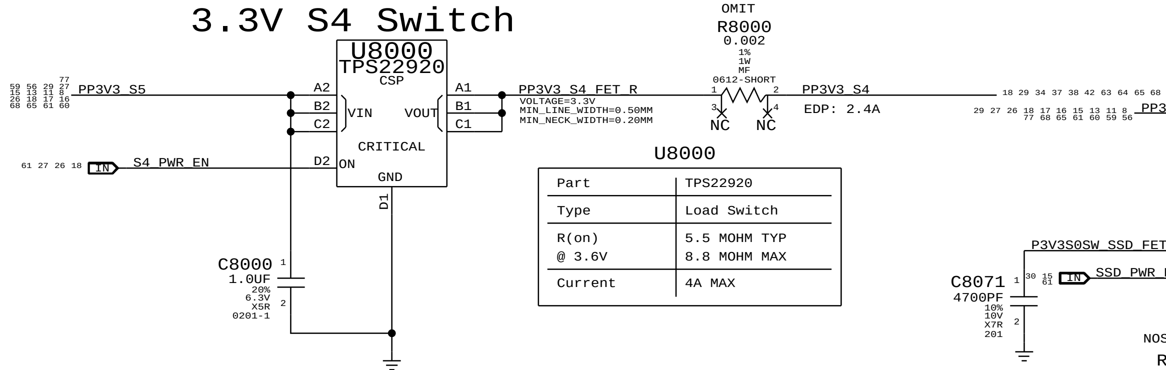
3.3V SUS Switch



U8020

Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

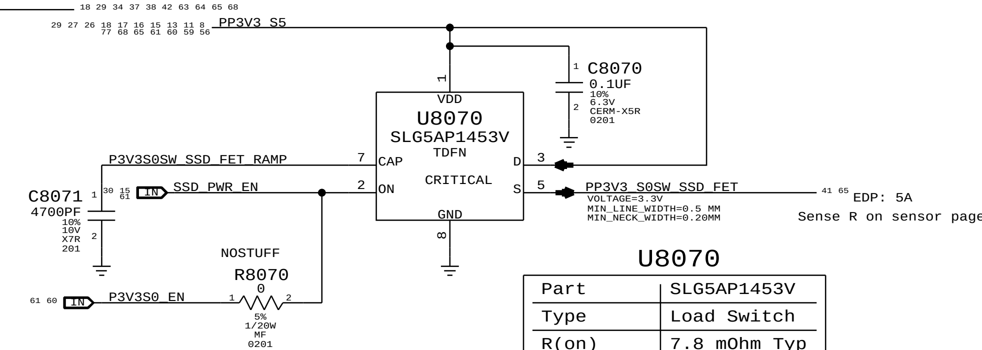
3.3V S4 Switch



U8000

Part	TPS22920
Type	Load Switch
R(on) @ 3.6V	5.5 MOHM TYP 8.8 MOHM MAX
Current	4A MAX

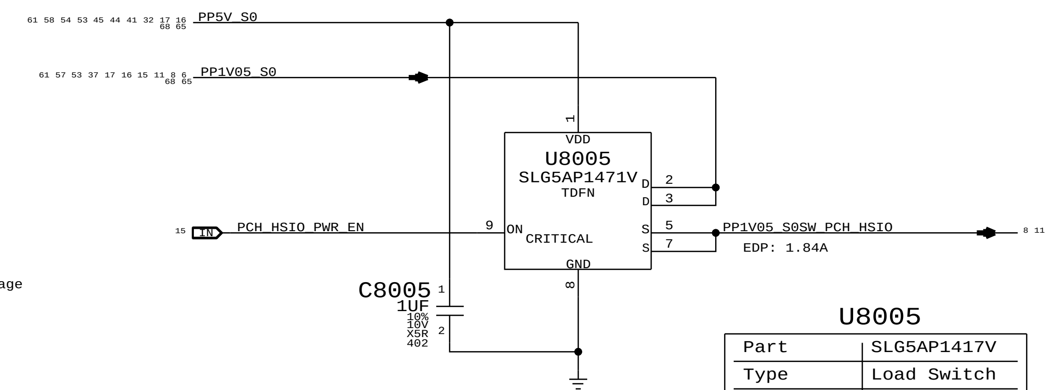
3.3V SSD Switch



U8070

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 25C	7.8 mOhm Typ 8.5 mOhm Max
Current	5.3A Max

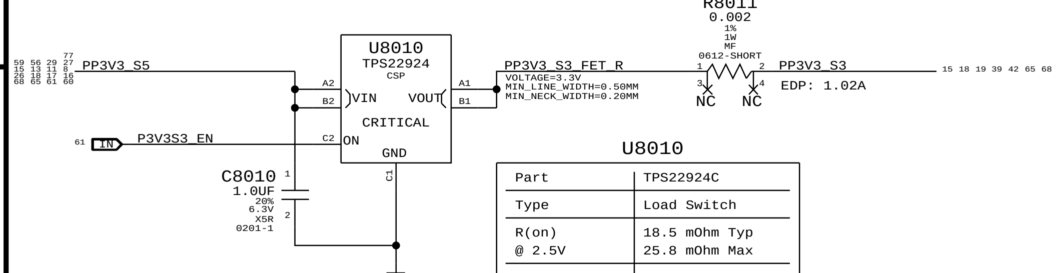
1.05V PCH HSIO Switch



U8005

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

3.3V S3 Switch

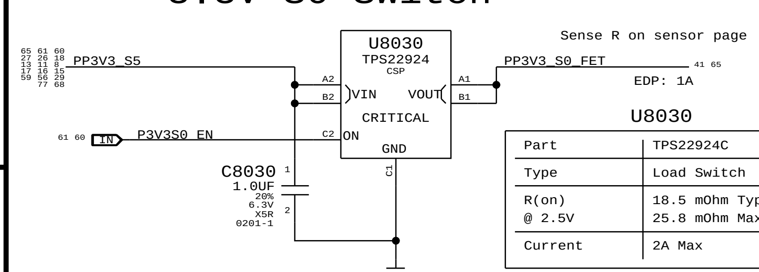


U8010

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

REMOVED THE ANALOG POWER GATE AS SLG5AP1471 SHOULD BE AVAILABLE BY THEN

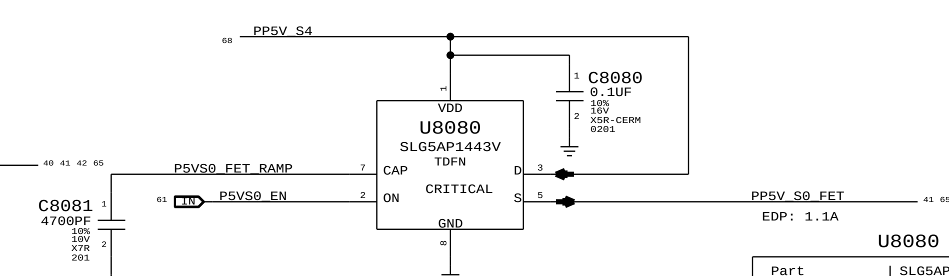
3.3V S0 Switch



U8030

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

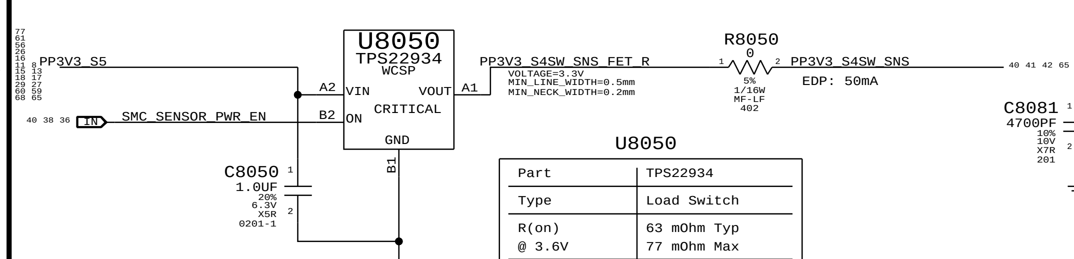
5V S0 Switch



U8080

Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

3.3V Sensor Switch



U8050

Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

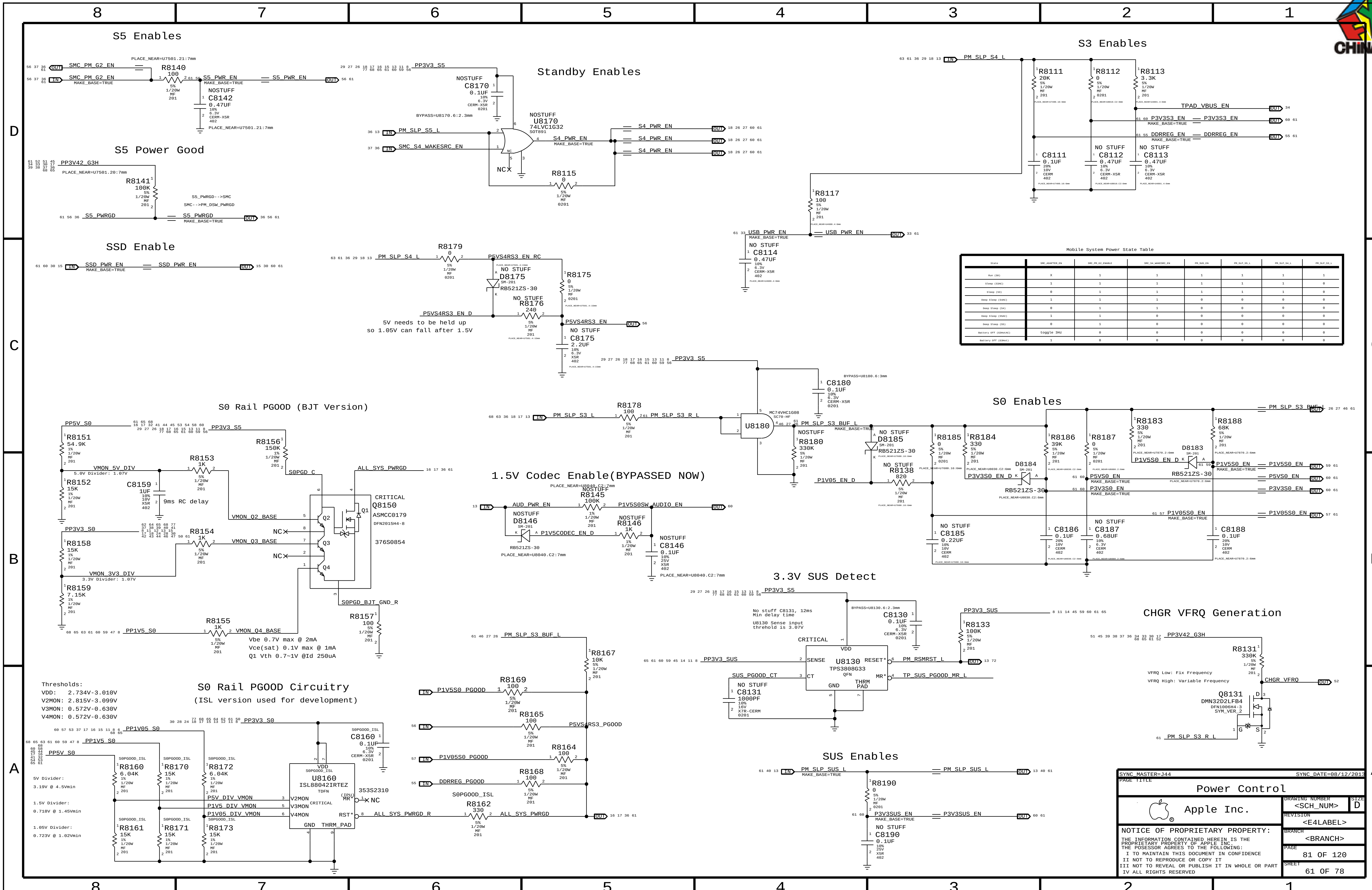
PAGE TITLE: Power FETs

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

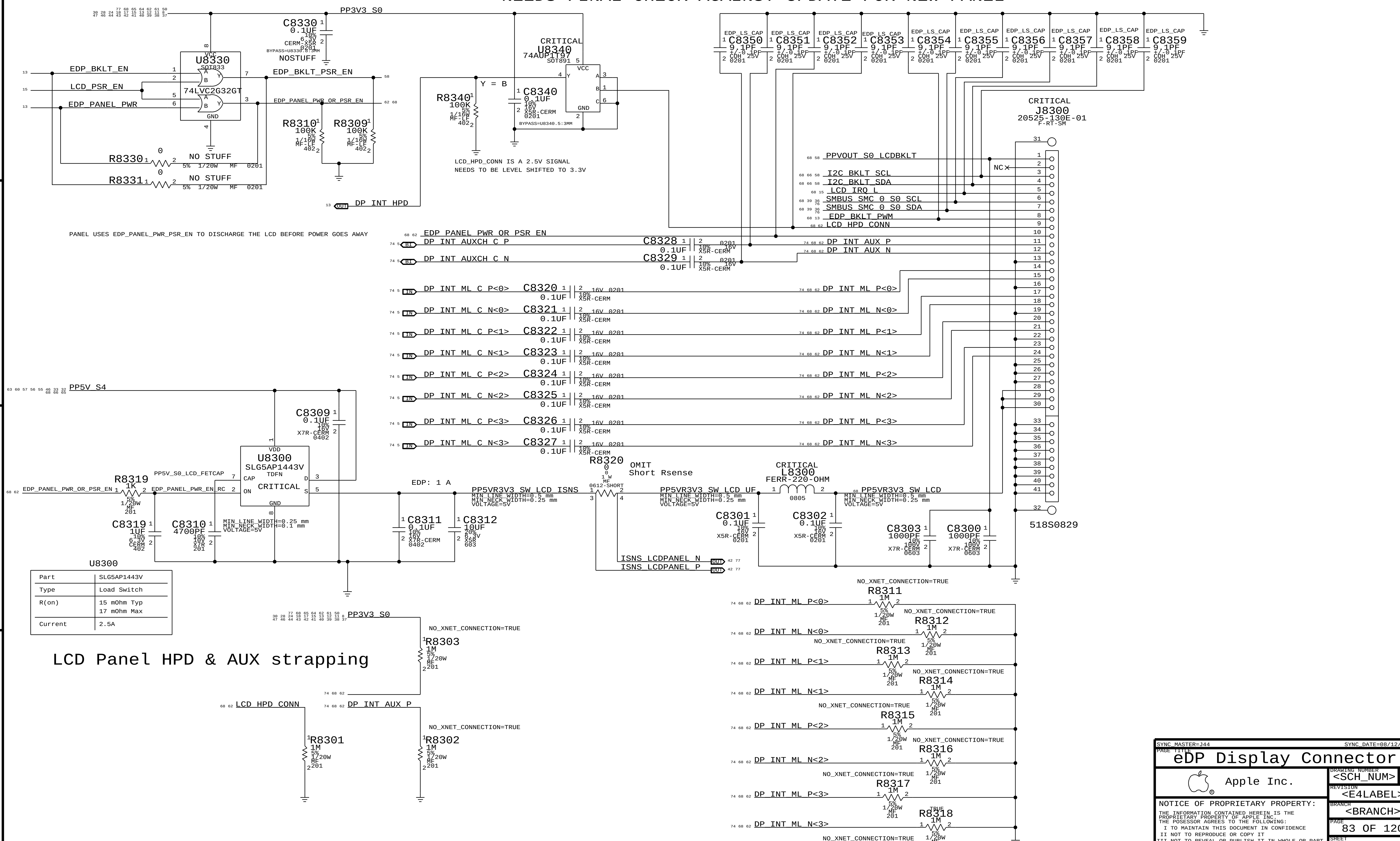


Mobile System Power State Table

State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (S4AC)	1	1	1	0	0	0	0
Deep Sleep (S4)	0	1	1	0	0	0	0
Deep Sleep (S5AC)	1	1	0	0	0	0	0
Deep Sleep (S5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC_MASTER=J44 SYNC_DATE=08/12/2013
PAGE TITLE
Power Control
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

LCD PANEL INTERFACE (eDP)
NEEDS FINAL CHECK AGAINST UPDATE FOR NEW PANEL



U8300

Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
eDP Display Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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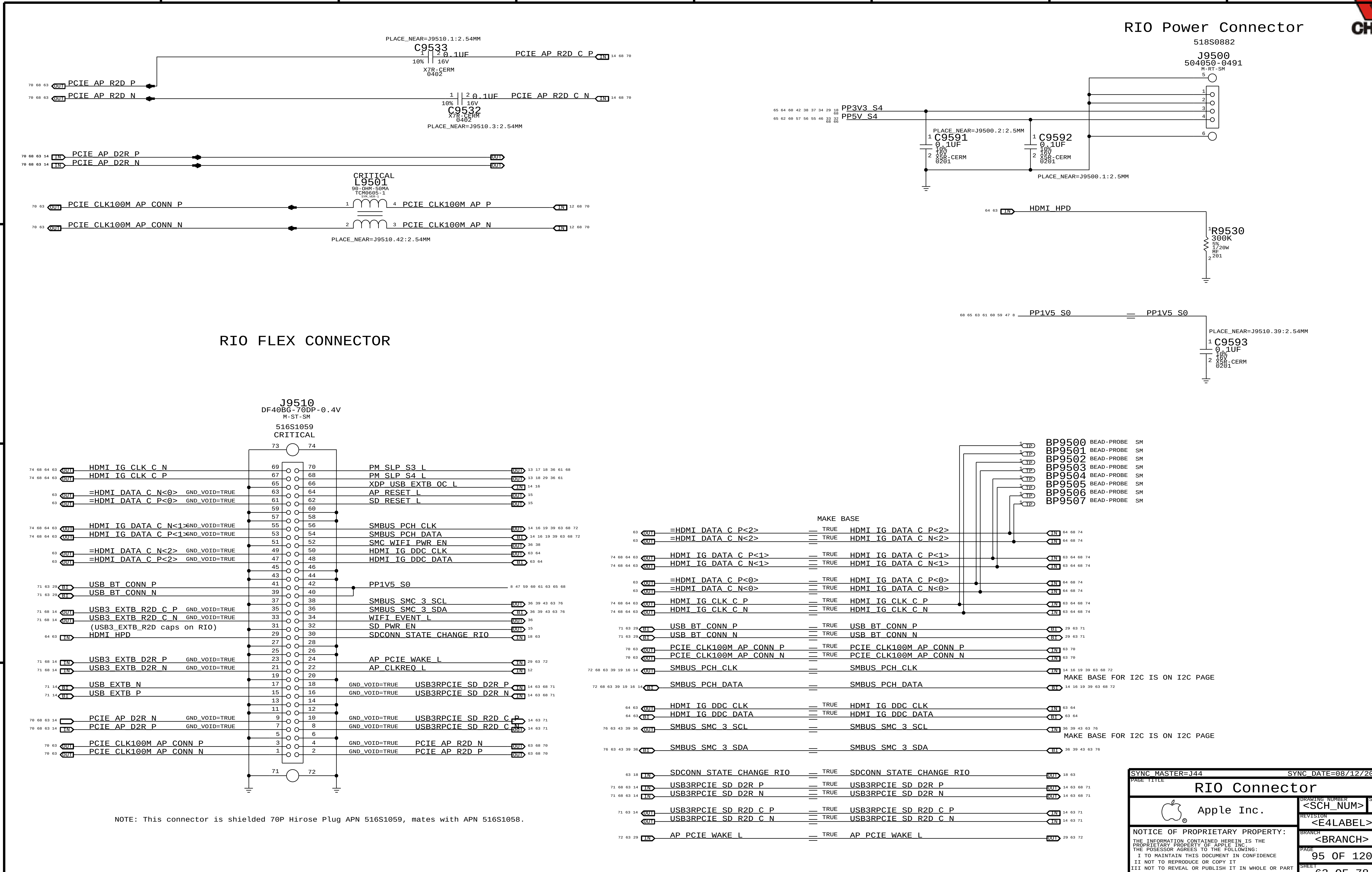
RIO Power Connector
518S0882
J9500
504050-0491
M-RT-SM
PP3V3_S4
PP5V_S4
C9591
0.1UF
C9592
0.1UF
PLACE_NEAR=J9500.2:2.5MM
PLACE_NEAR=J9500.1:2.5MM
HDMI_HPD
R9530
300K
PP1V5_S0
C9593
0.1UF
PLACE_NEAR=J9510.39:2.54MM
PLACE_NEAR=J9510.1:2.54MM
C9533
0.1UF
PCIE_AP_R2D_C_P
PCIE_AP_R2D_P
PCIE_AP_R2D_N
C9532
0.1UF
PCIE_AP_R2D_C_N
PLACE_NEAR=J9510.3:2.54MM
PCIE_AP_D2R_P
PCIE_AP_D2R_N
CRITICAL
L9501
90-OHM-50MA
TCM0605-1
PCIE_CLK100M_AP_CONN_P
PCIE_CLK100M_AP_P
PCIE_CLK100M_AP_CONN_N
PCIE_CLK100M_AP_N
PLACE_NEAR=J9510.42:2.54MM
RIO FLEX CONNECTOR
J9510
DF40BG-70DP-0.4V
M-ST-SM
516S1059
CRITICAL
HDMI_IG_CLK_C_N
HDMI_IG_CLK_C_P
=HDMI_DATA_C_N<0>
=HDMI_DATA_C_P<0>
HDMI_IG_DATA_C_N<1>
HDMI_IG_DATA_C_P<1>
=HDMI_DATA_C_N<2>
=HDMI_DATA_C_P<2>
USB_BT_CONN_P
USB_BT_CONN_N
USB3_EXTB_R2D_C_P
USB3_EXTB_R2D_C_N
(USB3_EXTB_R2D caps on RIO)
HDMI_HPD
USB3_EXTB_D2R_P
USB3_EXTB_D2R_N
USB_EXTB_N
USB_EXTB_P
PCIE_AP_D2R_N
PCIE_AP_D2R_P
PCIE_CLK100M_AP_CONN_P
PCIE_CLK100M_AP_CONN_N
PM_SLP_S3_L
PM_SLP_S4_L
XDP_USB_EXTB_OC_L
AP_RESET_L
SD_RESET_L
SMBUS_PCH_CLK
SMBUS_PCH_DATA
SMC_WIFI_PWR_EN
HDMI_IG_DDC_CLK
HDMI_IG_DDC_DATA
PP1V5_S0
SMBUS_SMC_3_SCL
SMBUS_SMC_3_SDA
WIFI_EVENT_L
SD_PWR_EN
SDCONN_STATE_CHANGE_RIO
AP_PCIE_WAKE_L
AP_CLKREQ_L
USB3RPCIE_SD_D2R_P
USB3RPCIE_SD_D2R_N
USB3RPCIE_SD_R2D_C_P
USB3RPCIE_SD_R2D_C_N
PCIE_AP_R2D_N
PCIE_AP_R2D_P
GND_VOID=TRUE
NOTE: This connector is shielded 70P Hirose Plug APN 516S1059, mates with APN 516S1058.
BP9500 BEAD-PROBE SM
BP9501 BEAD-PROBE SM
BP9502 BEAD-PROBE SM
BP9503 BEAD-PROBE SM
BP9504 BEAD-PROBE SM
BP9505 BEAD-PROBE SM
BP9506 BEAD-PROBE SM
BP9507 BEAD-PROBE SM
MAKE BASE
=HDMI_DATA_C_P<2> TRUE HDMI_IG_DATA_C_P<2>
=HDMI_DATA_C_N<2> TRUE HDMI_IG_DATA_C_N<2>
HDMI_IG_DATA_C_P<1> TRUE HDMI_IG_DATA_C_P<1>
HDMI_IG_DATA_C_N<1> TRUE HDMI_IG_DATA_C_N<1>
=HDMI_DATA_C_P<0> TRUE HDMI_IG_DATA_C_P<0>
=HDMI_DATA_C_N<0> TRUE HDMI_IG_DATA_C_N<0>
HDMI_IG_CLK_C_P TRUE HDMI_IG_CLK_C_P
HDMI_IG_CLK_C_N TRUE HDMI_IG_CLK_C_N
USB_BT_CONN_P TRUE USB_BT_CONN_P
USB_BT_CONN_N TRUE USB_BT_CONN_N
PCIE_CLK100M_AP_CONN_P TRUE PCIE_CLK100M_AP_CONN_P
PCIE_CLK100M_AP_CONN_N TRUE PCIE_CLK100M_AP_CONN_N
SMBUS_PCH_CLK SMBUS_PCH_CLK
MAKE BASE FOR I2C IS ON I2C PAGE
SMBUS_PCH_DATA SMBUS_PCH_DATA
HDMI_IG_DDC_CLK TRUE HDMI_IG_DDC_CLK
HDMI_IG_DDC_DATA TRUE HDMI_IG_DDC_DATA
SMBUS_SMC_3_SCL SMBUS_SMC_3_SCL
MAKE BASE FOR I2C IS ON I2C PAGE
SMBUS_SMC_3_SDA SMBUS_SMC_3_SDA
SDCONN_STATE_CHANGE_RIO TRUE SDCONN_STATE_CHANGE_RIO
USB3RPCIE_SD_D2R_P TRUE USB3RPCIE_SD_D2R_P
USB3RPCIE_SD_D2R_N TRUE USB3RPCIE_SD_D2R_N
USB3RPCIE_SD_R2D_C_P TRUE USB3RPCIE_SD_R2D_C_P
USB3RPCIE_SD_R2D_C_N TRUE USB3RPCIE_SD_R2D_C_N
AP_PCIE_WAKE_L TRUE AP_PCIE_WAKE_L
SYNC_MASTER=J44
SYNC_DATE=08/12/2013
PAGE TITLE
RIO Connector
DRAWING NUMBER <SCH_NUM>
SIZE D
Apple Inc.
REVISION <E4LABEL>
BRANCH <BRANCH>
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SHEET 63 OF 78
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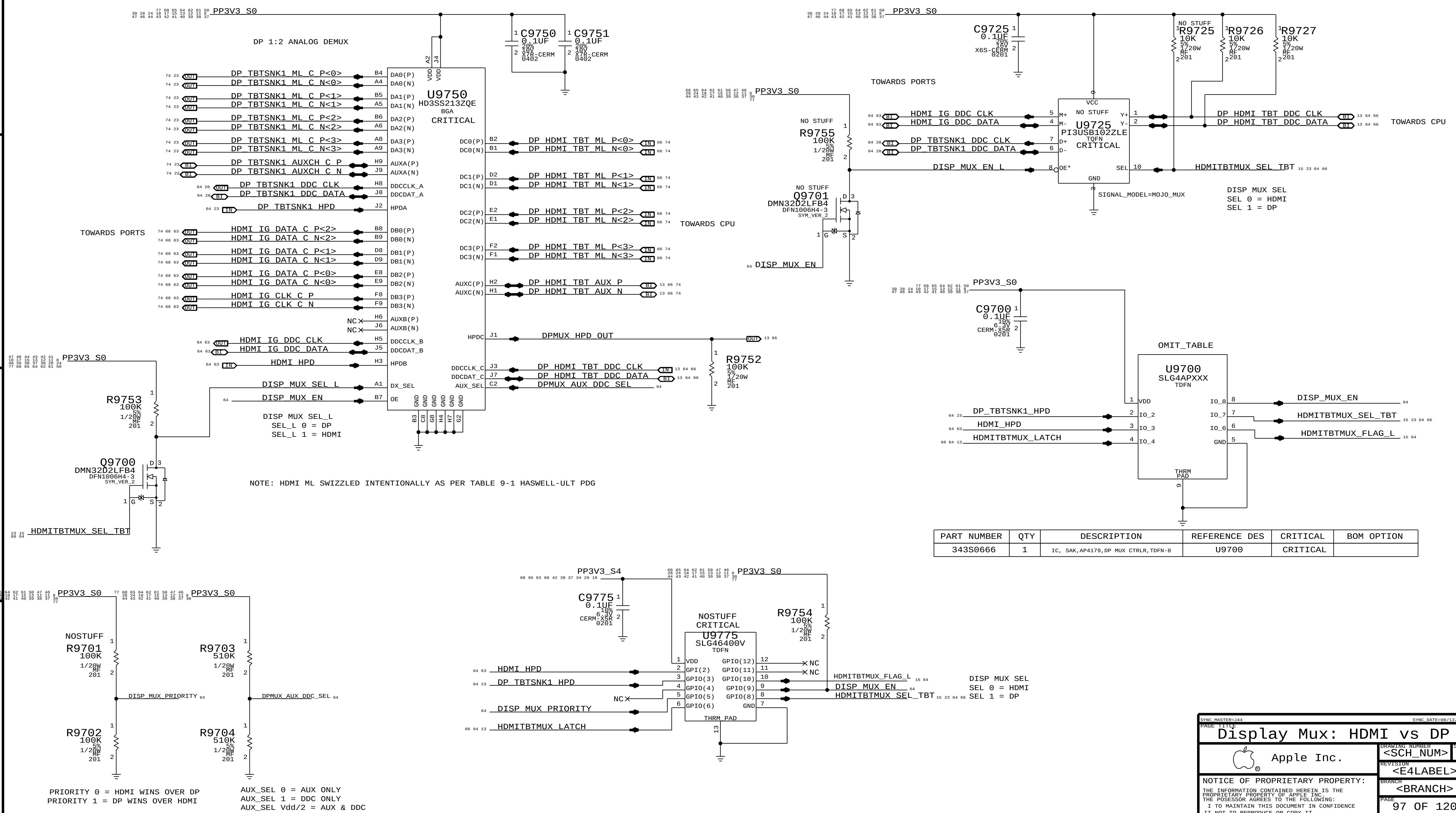
DISPLAY MUX: DP OR HDMI



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
343S0666	1	IC, SAK,AP4179,DP MUX CTRLR,TDFN-8	U9700	CRITICAL	

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Display Mux: HDMI vs DP

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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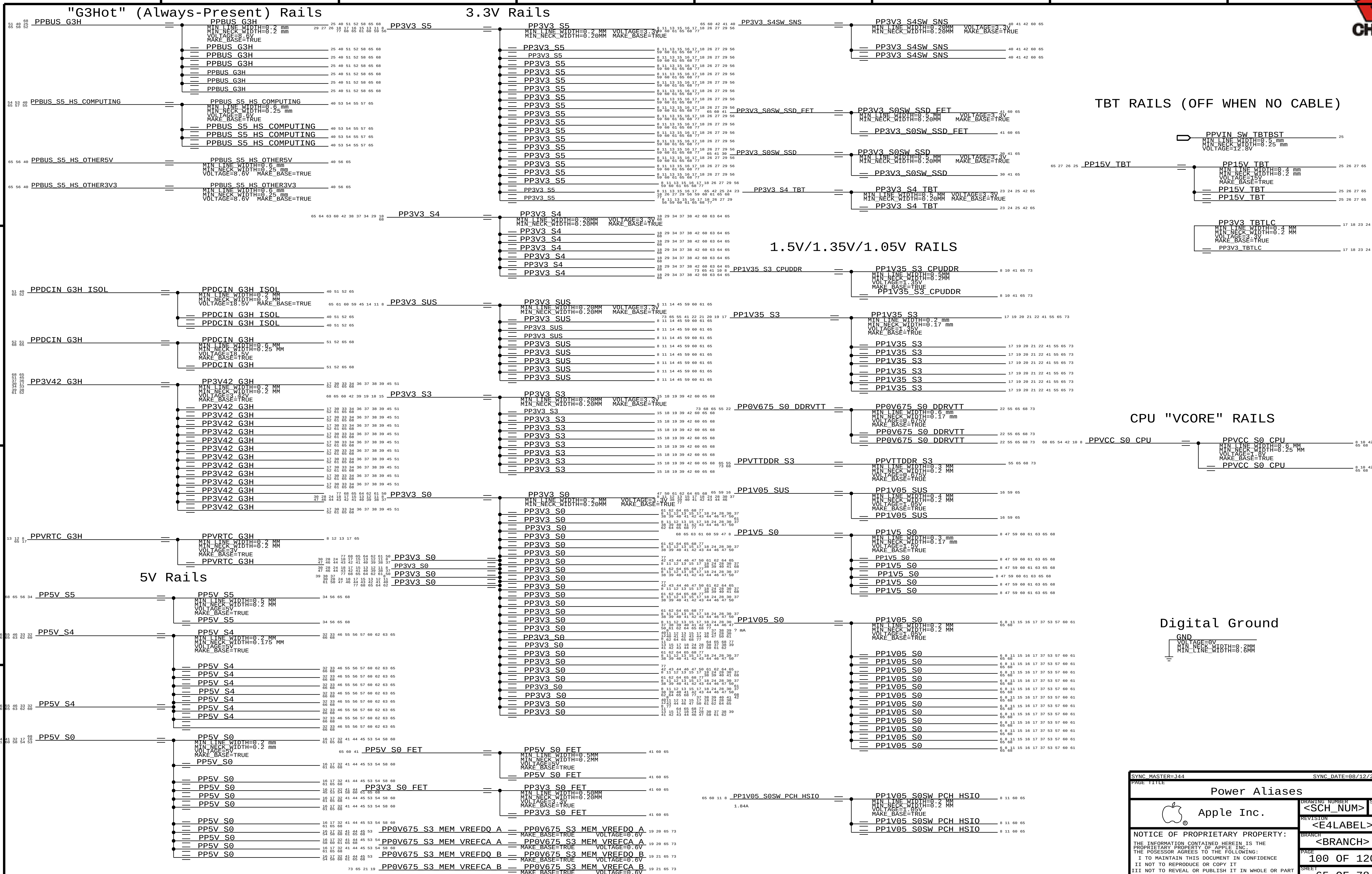

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE: Power Aliases

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

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MEMORY ADDRESS/CTRL

MAKE_BASE

7 =MEM_A_A<0>	TRUE	MEM_A_A<0> 20 22 73
7 =MEM_A_A<1>	TRUE	MEM_A_A<1> 20 22 73
7 =MEM_A_A<2>	TRUE	MEM_A_A<2> 20 22 73
7 =MEM_A_A<3>	TRUE	MEM_A_A<3> 20 22 73
7 =MEM_A_A<4>	TRUE	MEM_A_A<4> 20 22 73
7 =MEM_A_A<5>	TRUE	MEM_A_A<5> 20 22 73
66 22 20 7 73 MEM_A_A<6>	TRUE	MEM_A_A<6> 7 20 22 66 73
7 =MEM_A_A<7>	TRUE	MEM_A_A<7> 20 22 73
7 =MEM_A_A<8>	TRUE	MEM_A_A<8> 20 22 73
7 =MEM_A_A<9>	TRUE	MEM_A_A<9> 20 22 73
7 =MEM_A_A<10>	TRUE	MEM_A_A<10> 20 22 73
7 =MEM_A_A<11>	TRUE	MEM_A_A<11> 20 22 73
7 =MEM_A_A<12>	TRUE	MEM_A_A<12> 20 22 73
7 =MEM_A_A<13>	TRUE	MEM_A_A<13> 20 22 73
7 =MEM_A_A<14>	TRUE	MEM_A_A<14> 20 22 73

MAKE_BASE

7 =MEM_B_A<0>	TRUE	MEM_B_A<0> 21 22 73
7 =MEM_B_A<1>	TRUE	MEM_B_A<1> 21 22 73
7 =MEM_B_A<2>	TRUE	MEM_B_A<2> 21 22 73
7 =MEM_B_A<3>	TRUE	MEM_B_A<3> 21 22 73
7 =MEM_B_A<4>	TRUE	MEM_B_A<4> 21 22 73
7 =MEM_B_A<5>	TRUE	MEM_B_A<5> 21 22 73
66 22 21 7 73 MEM_B_A<6>	TRUE	MEM_B_A<6> 7 21 22 66 73
7 =MEM_B_A<7>	TRUE	MEM_B_A<7> 21 22 73
7 =MEM_B_A<8>	TRUE	MEM_B_A<8> 21 22 73
7 =MEM_B_A<9>	TRUE	MEM_B_A<9> 21 22 73
7 =MEM_B_A<10>	TRUE	MEM_B_A<10> 21 22 73
7 =MEM_B_A<11>	TRUE	MEM_B_A<11> 21 22 73
7 =MEM_B_A<12>	TRUE	MEM_B_A<12> 21 22 73
7 =MEM_B_A<13>	TRUE	MEM_B_A<13> 21 22 73
7 =MEM_B_A<14>	TRUE	MEM_B_A<14> 21 22 73

MAKE_BASE

66 22 7 MEM_A_ODT_CPU0	TRUE	MEM_A_ODT_CPU0 7 22 66
66 22 20 7 73 MEM_A_RAS_L	TRUE	MEM_A_RAS_L 7 20 22 66 73
66 22 20 7 73 MEM_A_WE_L	TRUE	MEM_A_WE_L 7 20 22 66 73
66 22 20 7 73 MEM_A_CAS_L	TRUE	MEM_A_CAS_L 7 20 22 66 73
7 =MEM_A_BA<0>	TRUE	MEM_A_BA<0> 20 22 73
66 22 20 7 73 MEM_A_BA<1>	TRUE	MEM_A_BA<1> 7 20 22 66 73
7 =MEM_A_BA<2>	TRUE	MEM_A_BA<2> 20 22 73

MAKE_BASE

66 22 7 MEM_B_ODT_CPU0	TRUE	MEM_B_ODT_CPU0 7 22 66
66 22 21 7 73 MEM_B_RAS_L	TRUE	MEM_B_RAS_L 7 21 22 66 73
66 22 21 7 73 MEM_B_WE_L	TRUE	MEM_B_WE_L 7 21 22 66 73
66 22 21 7 73 MEM_B_CAS_L	TRUE	MEM_B_CAS_L 7 21 22 66 73
7 =MEM_B_BA<0>	TRUE	MEM_B_BA<0> 21 22 73
66 22 21 7 73 MEM_B_BA<1>	TRUE	MEM_B_BA<1> 7 21 22 66 73
7 =MEM_B_BA<2>	TRUE	MEM_B_BA<2> 21 22 73

66 22 6 MEM_RESET_HSW_L — MAKE_BASE TRUE — MEM_RESET_HSW_L 6 22 66

HDMI VS TBT

MAKE_BASE

5 =DP_TBTSNK1_ML_C_P<0>	TRUE	DP_HDMI_TBT_ML_P<0> 64 74
5 =DP_TBTSNK1_ML_C_N<0>	TRUE	DP_HDMI_TBT_ML_N<0> 64 74
5 =DP_TBTSNK1_ML_C_P<1>	TRUE	DP_HDMI_TBT_ML_P<1> 64 74
5 =DP_TBTSNK1_ML_C_N<1>	TRUE	DP_HDMI_TBT_ML_N<1> 64 74
5 =DP_TBTSNK1_ML_C_P<2>	TRUE	DP_HDMI_TBT_ML_P<2> 64 74
5 =DP_TBTSNK1_ML_C_N<2>	TRUE	DP_HDMI_TBT_ML_N<2> 64 74
5 =DP_TBTSNK1_ML_C_P<3>	TRUE	DP_HDMI_TBT_ML_P<3> 64 74
5 =DP_TBTSNK1_ML_C_N<3>	TRUE	DP_HDMI_TBT_ML_N<3> 64 74
74 66 64 13 DP_HDMI_TBT_AUX_P	TRUE	DP_HDMI_TBT_AUX_P 13 64 66 74
74 66 64 13 DP_HDMI_TBT_AUX_N	TRUE	DP_HDMI_TBT_AUX_N 13 64 66 74
66 64 13 DP_HDMI_TBT_DDC_CLK	TRUE	DP_HDMI_TBT_DDC_CLK 13 64 66
66 64 13 DP_HDMI_TBT_DDC_DATA	TRUE	DP_HDMI_TBT_DDC_DATA 13 64 66
66 64 13 DPMUX_HPD_OUT	TRUE	DPMUX_HPD_OUT 13 64 66

66 64 23 15 HDMITBTMUX_SEL_TBT MAKE_BASE=TRUE — HDMITBTMUX_SEL_TBT 15 23 64 66

HDMITBTMUX_SEL_TBT 15 23 64 66

66 64 13 HDMITBTMUX_LATCH — HDMITBTMUX_LATCH MAKE_BASE=TRUE 13 64 66

EPD PANEL

MAKE_BASE

68 66 62 58 I2C_BKLT_SCL	TRUE	I2C_BKLT_SCL 58 62 66 68
68 66 62 58 I2C_BKLT_SDA	TRUE	I2C_BKLT_SDA 58 62 66 68

UNUSED SIGNALS

MAKE_BASE

66 12 NC_PCIE_CLK100M_FWP	TRUE	NO_TEST=TRUE	NC_PCIE_CLK100M_FWP 12 66
66 12 NC_PCIE_CLK100M_FWN	TRUE	NO_TEST=TRUE	NC_PCIE_CLK100M_FWN 12 66
66 14 NC_PCIE_FW_D2RP	TRUE	NO_TEST=TRUE	NC_PCIE_FW_D2RP 14 66
66 14 NC_PCIE_FW_D2RN	TRUE	NO_TEST=TRUE	NC_PCIE_FW_D2RN 14 66
66 14 NC_PCIE_FW_R2D_CP	TRUE	NO_TEST=TRUE	NC_PCIE_FW_R2D_CP 14 66
66 14 NC_PCIE_FW_R2D_CN	TRUE	NO_TEST=TRUE	NC_PCIE_FW_R2D_CN 14 66
66 12 NC_PCIE_CLK100M_ENETSDP	TRUE	NO_TEST=TRUE	NC_PCIE_CLK100M_ENETSDP 12 66
66 12 NC_PCIE_CLK100M_ENETSDN	TRUE	NO_TEST=TRUE	NC_PCIE_CLK100M_ENETSDN 12 66
71 66 14 NC_USB_IRP	TRUE	NO_TEST=TRUE	NC_USB_IRP 14 66 71
71 66 14 NC_USB_IRN	TRUE	NO_TEST=TRUE	NC_USB_IRN 14 66 71
71 66 14 NC_USB_CAMERAP	TRUE	NO_TEST=TRUE	NC_USB_CAMERAP 14 66 71
71 66 14 NC_USB_CAMERAN	TRUE	NO_TEST=TRUE	NC_USB_CAMERAN 14 66 71
71 66 14 NC_USB_SDP	TRUE	NO_TEST=TRUE	NC_USB_SDP 14 66 71
71 66 14 NC_USB_SDN	TRUE	NO_TEST=TRUE	NC_USB_SDN 14 66 71
66 12 NC_HDA_SDIN1	TRUE	NO_TEST=TRUE	NC_HDA_SDIN1 12 66
66 13 NC_PCI_PME_L	TRUE	NO_TEST=TRUE	NC_PCI_PME_L 13 66
66 14 NC_CLINK_CLK	TRUE	NO_TEST=TRUE	NC_CLINK_CLK 14 66
66 14 NC_CLINK_DATA	TRUE	NO_TEST=TRUE	NC_CLINK_DATA 14 66
66 14 NC_CLINK_RESET_L	TRUE	NO_TEST=TRUE	NC_CLINK_RESET_L 14 66

66 45 NC_SMC_TRST_L	TRUE	NO_TEST=TRUE	NC_SMC_TRST_L 45 66
NC_SMC_MD1	TRUE	NO_TEST=TRUE	NC_SMC_MD1 45 66

UNUSED MEMORY SIGNALS

NO_TEST

66 7 NC_MEM_A_CLKN<1>	TRUE MAKE_BASE=TRUE	NC_MEM_A_CLKN<1> 7 66
66 7 NC_MEM_A_CLKP<1>	TRUE MAKE_BASE=TRUE	NC_MEM_A_CLKP<1> 7 66
7 MEM_A_CKE<2>	TRUE MAKE_BASE=TRUE	NC_MEM_A_CKE<2>
66 7 NC_MEM_A_CKE<3>	TRUE MAKE_BASE=TRUE	NC_MEM_A_CKE<3> 7 66
66 7 NC_MEM_B_CLKN<1>	TRUE MAKE_BASE=TRUE	NC_MEM_B_CLKN<1> 7 66
66 7 NC_MEM_B_CLKP<1>	TRUE MAKE_BASE=TRUE	NC_MEM_B_CLKP<1> 7 66
7 MEM_B_CKE<2>	TRUE MAKE_BASE=TRUE	NC_MEM_B_CKE<2>
66 7 NC_MEM_B_CKE<3>	TRUE MAKE_BASE=TRUE	NC_MEM_B_CKE<3> 7 66

GND
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=0V

Digital Ground

65 63 62 60 57 56 55 46 33 32 68 PP5V_S4

XWA202 SM 1 2 PP5V_S0_AUDIO_AMP_L 48
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=5V

XWA203 SM 1 2 PP5V_S0_AUDIO_AMP_R 48
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=5V

SYNC_MASTER=MASTER SYNC_DATE=MASTER

PAGE TITLE

Signal Aliases

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Memory Bit/Byte Swizzle

MAKE_BASE

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MAKE_BASE

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73 67 21 7	TRUE	MEM_B_DQS_N<6>	MEM_B_DQS_N<6>	7 21 67 73
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Memory Bit/Byte Swizzle

Apple Inc.

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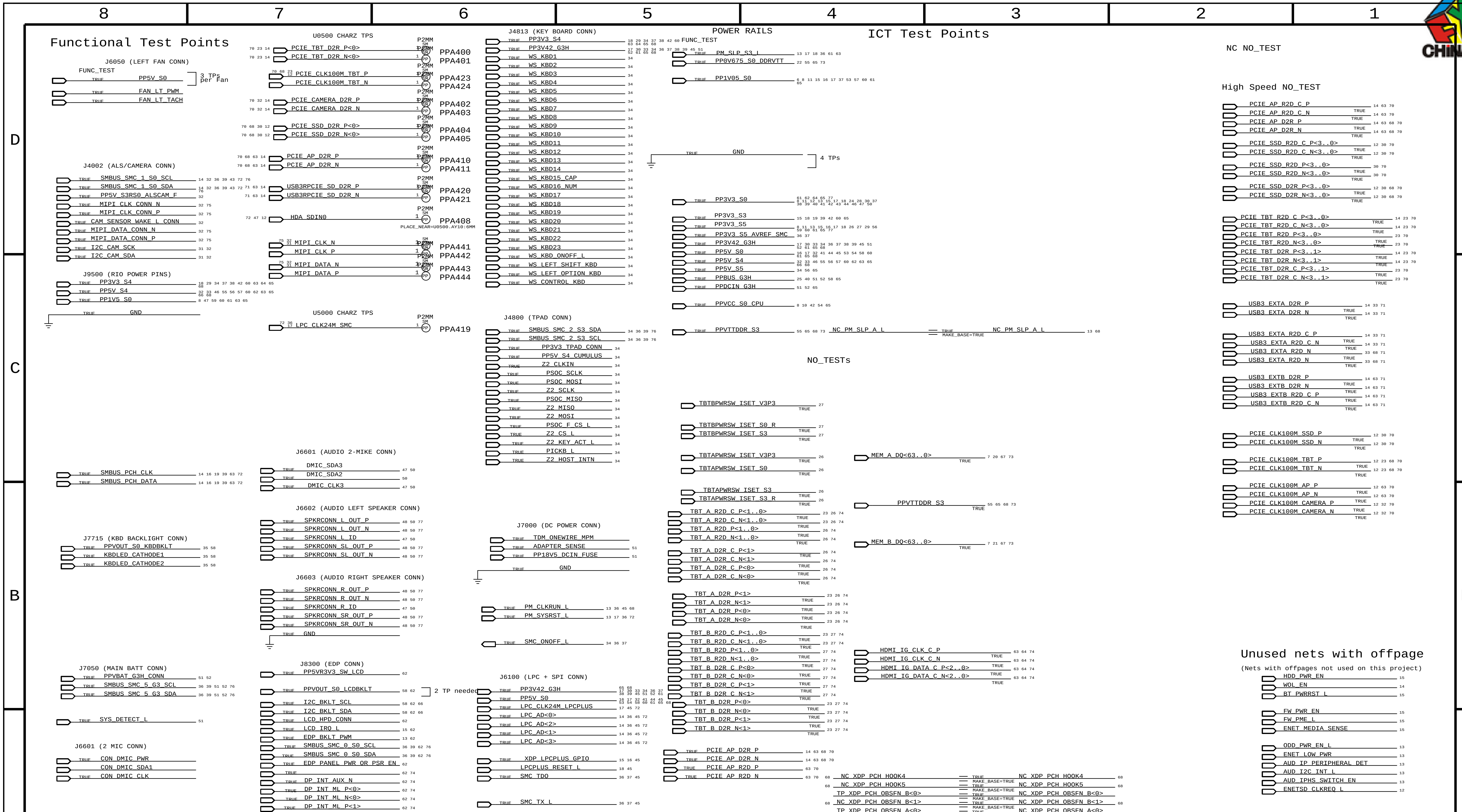

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PAGE TITLE

Functional / ICT Test

Apple Inc.

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J44 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,P65BGA,BGA_MEM	MM	16.5

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.125 MM		0.155 MM	0.155 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1x_DIELECTRIC	TOP,BOTTOM	0.058 MM	?
1x_DIELECTRIC	ISL3,ISL4,ISL9,ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2,ISL5,ISL6,ISL7,ISL8,ISL11	0.101 MM	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

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PCB Rule Definitions

Apple Inc.

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_08MIL	*	0.203 MM	?
CPU_12MIL	*	0.305 MM	?
CPU_18MIL	*	0.457 MM	?
CPU_25MIL	*	0.635 MM	?

PCI Express Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	*	=3X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?
PCIE_2OTHER	*	=4X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?
PCIECLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIE_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIECLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	*	*	PCIE_2OTHER
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_2OTHER
PCIE_TX	*_RX	*	PCIE_TXRX
PCIE_RX	*_TX	*	PCIE_TXRX

CPU Signal Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
XDP_TCK0	CPU_45S	CPU_18MIL	XDP_CPU_TCK	6 16
XDP_TCK0	CPU_45S	CPU_18MIL	PCH_JTAGX	12 16
XDP_TCK1	CPU_45S	CPU_18MIL	XDP_PCH_TCK	12 16
XDP_TDO	CPU_45S		XDP_CPU_TDO	6 16
XDP_TDO	CPU_45S		XDP_PCH_TDO	12 16
XDP_TDI	CPU_45S		XDP_CPU_TDI	6 16
XDP_TDI	CPU_45S		XDP_PCH_TDI	12 16
XDP_TMS	CPU_45S		XDP_CPU_TMS	6 16
XDP_TMS	CPU_45S		XDP_PCH_TMS	12 16
XDP_TRST_L	CPU_45S		XDP_TRST_L	16
XDP_TRST_L	CPU_45S		XDP_CPUPCH_TRST_L	6 12 16
XDP_PRDY_L	CPU_45S		XDP_CPU_PRDY_L	6 16
XDP_PREQ_L	CPU_45S		XDP_CPU_PREQ_L	6 16
CPU_VCCST_PWRGD	CPU_45S	CPU_08MIL	CPU_VCCST_PWRGD	8 16 17
CPU_VCCST_PWRGD	CPU_45S	CPU_08MIL	XDP_CPU_VCCST_PWRGD	16
CPU_BPM	CPU_45S	CPU_08MIL	XDP_BPM_L<1..0>	6 16
CPU_BPM_TP	CPU_45S		XDP_BPM_L<7..2>	6 16
CPU_RCOMP_SM	CPU_27P4S	CPU_25MIL	CPU_SM_RCOMP<2..0>	6
CPU_RCOMP_EDP	CPU_27P4S	CPU_25MIL	MCP_EDP_RCOMP	5
CPU_RCOMP_OPI	CPU_27P4S	CPU_12MIL	CPU_OPI_RCOMP	6
CPU_PROCHOT	CPU_45S	CPU_08MIL	CPU_PROCHOT_L	6 36 37 53
CPU_PROCHOT	CPU_45S	CPU_08MIL	CPU_PROCHOT_R_L	6
CPU_CATERR	CPU_45S	CPU_08MIL	CPU_CATERR_L	6 36
CPU_VIDALERT	CPU_45S	CPU_18MIL	CPU_VIDALERT_L	8 53
CPU_VIDALERT	CPU_45S	CPU_18MIL	CPU_VIDALERT_R_L	8
CPU_VIDSCLK	CPU_45S	CPU_18MIL	CPU_VIDSCLK	8 53
CPU_VIDSCLK	CPU_45S	CPU_18MIL	CPU_VIDSCLK_R	8
CPU_VIDSOUT	CPU_45S	CPU_18MIL	CPU_VIDSOUT	8 53
CPU_VIDSOUT	CPU_45S	CPU_18MIL	CPU_VIDSOUT_R	8
CPU_PECI	CPU_45S	CPU_18MIL	CPU_PECI	6 37
CPU_PECI	CPU_45S	CPU_18MIL	CPU_PECI_R	36 37
CPU_PECI_SMC	CPU_45S	CPU_18MIL	SMC_PECI_L	36 37
CPU_PECI_SMC	CPU_45S	CPU_18MIL	SMC_PECI_L_R	37
CPU_CFG	CPU_45S		CPU_CFG<19..11>	6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<10..8>	6 16
CPU_CFG	CPU_45S		CPU_CFG<7..5>	6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<4>	6 16
CPU_CFG_3	CPU_45S		CPU_CFG<3>	6 16
CPU_CFG	CPU_45S		CPU_CFG<2>	6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<1..0>	6 16
CPU_MEM_RESET	CPU_45S	CPU_08MIL	MEM_RESET_L	20 21 22
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	8 53
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	9 53

PCI Express Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<3..1>	12 30 68
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<3..1>	12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<0>	12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<0>	12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_P<3..0>	12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_N<3..0>	12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_P<3..0>	30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_N<3..0>	30 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<0>	14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<0>	14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<0>	23
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<0>	23
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<3..1>	14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<3..1>	14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<3..1>	23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<3..1>	23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_P<3..0>	23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_N<3..0>	23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_P<3..0>	14 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_N<3..0>	14 23 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_P	63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_N	63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_P	14 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_N	14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_P	14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_N	14 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P	12 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N	12 63 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	31 32
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	31 32
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P	12 30 68
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N	12 30 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P	12 23 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N	12 23 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_P	14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_N	14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_P	31 32
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_N	31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_P	31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_N	31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_P	14 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_N	14 32

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
CPU & PCIe Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

USB 2 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?
USB_RBIAS	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAS	TOP,BOTTOM	=10X_DIELECTRIC	?

D

USB 3 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?
USB3_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	*	*	USB3_2OTHER
USB3_*	=SAME	*	USB3_2SAME
USB3_TX	*_RX	*	USB3_TXRX
USB3_RX	*_TX	*	USB3_TXRX

C

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

B

SATA Interface Constraints (Not Used)

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	*	*	SATA_2OTHER
SATA_*	=SAME	*	SATA_2SAME
SATA_TX	*_RX	*	SATA_TXRX
SATA_RX	*_TX	*	SATA_TXRX

A

USB Constraints

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
USB_BT	USB_85D	USB	USB_BT_P	14 29
USB_BT	USB_85D	USB	USB_BT_N	14 29
USB_BT	USB_85D	USB	USB_BT_CONN_P	29 63
USB_BT	USB_85D	USB	USB_BT_CONN_N	29 63
USB_EXTA	USB_85D	USB	USB_EXTA_P	14 33
USB_EXTA	USB_85D	USB	USB_EXTA_N	14 33
	DEFAULT	DEFAULT	SMC_DEBUGPRT_RX_L	33 36 37
	DEFAULT	DEFAULT	SMC_DEBUGPRT_TX_L	33 36 37
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_P	33
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_N	33
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_F_P	33
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_F_N	33
USB_EXTA	USB_85D	USB	USB_LT1_P	68
USB_EXTA	USB_85D	USB	USB_LT1_N	68
USB_EXTB	USB_85D	USB	USB_EXTB_P	14 63
USB_EXTB	USB_85D	USB	USB_EXTB_N	14 63
USB_TPAD	USB_85D	USB	USB_TPAD_P	14 34
USB_TPAD	USB_85D	USB	USB_TPAD_N	14 34
I330 USB_TPAD	USB_85D	USB	USB_TPAD_R_P	34
I331 USB_TPAD	USB_85D	USB	USB_TPAD_R_N	34
USB3_EXTA_D2R	USB_85D	USB3_RX	USB3_EXTA_D2R_P	14 33 68
USB3_EXTA_D2R	USB_85D	USB3_RX	USB3_EXTA_D2R_N	14 33 68
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_P	33
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_N	33 68
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_C_P	14 33 68
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_C_N	14 33 68
USB3_EXTB_D2R	USB_85D	USB3_RX	USB3_EXTB_D2R_P	14 63 68
USB3_EXTB_D2R	USB_85D	USB3_RX	USB3_EXTB_D2R_N	14 63 68
USB3_EXTB_R2D	USB_85D	USB3_TX	USB3_EXTB_R2D_C_P	14 63 68
USB3_EXTB_R2D	USB_85D	USB3_TX	USB3_EXTB_R2D_C_N	14 63 68
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD_D2R_P	14 63 68
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD_D2R_N	14 63 68
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD_R2D_C_P	14 63
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD_R2D_C_N	14 63
USB_NC	USB_85D	USB	NC_USB_IRP	14 66
USB_NC	USB_85D	USB	NC_USB_IRN	14 66
USB_NC	USB_85D	USB	TP_USB_5P	14
USB_NC	USB_85D	USB	TP_USB_5N	14
USB_NC	USB_85D	USB	NC_USB_SDP	14 66
USB_NC	USB_85D	USB	NC_USB_SDN	14 66
USB_NC	USB_85D	USB	NC_USB_CAMERAP	14 66
USB_NC	USB_85D	USB	NC_USB_CAMERAN	14 66
PCH_USB_RBIAS	PCH_USB_RBIAS	USB_RBIAS	PCH_USB_RBIAS	14
	SATA_85D	SATA_RX	DUMMY_SATA_D2R_P	
	SATA_85D	SATA_RX	DUMMY_SATA_D2R_N	
	SATA_85D	SATA_TX	DUMMY_SATA_R2D_P	
	SATA_85D	SATA_TX	DUMMY_SATA_R2D_N	
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1	17
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2	17
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R	17
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	17 32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP	31 32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R	32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP	32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN	32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN	31 32
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	17 23
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	23

Notes:
This is here to keep the SATA rules.

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE
USB Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
PCH_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_12MIL	*	0.305 MM	?
PCH_15MIL	*	0.381 MM	?
PCH_18MIL	*	0.457 MM	?
PCH_20MIL	*	0.508 MM	?

PCH Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 36 45 68
LPC_AD	LPC_45S	LPC	LPC_FRAME_L	14 36 45 68
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 36 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS	17 45 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS_R	12 17
SMBUS_PCH	SMB_45S	SMB	SMBUS_PCH_CLK	14 16 19 39 63 68
SMBUS_PCH	SMB_45S	SMB	SMBUS_PCH_DATA	14 16 19 39 63 68
SML_PCH_0	SMB_45S	SMB	SML_PCH_0_CLK	14 39
SML_PCH_0	SMB_45S	SMB	SML_PCH_0_DATA	14 39
	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	14 32 36 39 43 68 76
	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	14 32 36 39 43 68 76
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 47
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 47
HDA_SYNC	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST	HDA_45S	HDA	HDA_RST_R_L	12
HDA_RST	HDA_45S	HDA	HDA_RST_L	12 47
HDA_SDIN	HDA_45S	HDA	HDA_SDIN0	12 47 68
HDA_SDIN	HDA_45S	HDA	CS4208_HDA_SDOUT0_R	47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12 47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT_R	12 17
SPI_MLB	SPI_45S	SPI	SPI_ALT_CLK	45
SPI_MLB	SPI_45S	SPI	SPI_CLK	45
SPI_MLB	SPI_45S	SPI	SPI_CLK_R	14 45
SPI_MLB	SPI_45S	SPI	SPI_MLB_CLK	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_CLK	36 45
SPI_MLB	SPI_45S	SPI	SPI_ALT_CS_L	45
SPI_MLB	SPI_45S	SPI	SPI_CS0_L	45
SPI_MLB	SPI_45S	SPI	SPI_CS0_R_L	14 45
SPI_MLB	SPI_45S	SPI	SPI_MLB_CS_L	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_CS_L	36 45
SPI_MLB	SPI_45S	SPI	SPI_ALT_MISO	45
SPI_MLB	SPI_45S	SPI	SPI_MISO	14 45
SPI_MLB	SPI_45S	SPI	SPI_MISO_R	45
SPI_MLB	SPI_45S	SPI	SPI_MLB_MISO	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_MISO	36 45
SPI_MLB	SPI_45S	SPI	SPI_ALT_MOSI	45
SPI_MLB	SPI_45S	SPI	SPI_MOSI	45
SPI_MLB	SPI_45S	SPI	SPI_MOSI_R	14 45
SPI_MLB	SPI_45S	SPI	SPI_MLB_MOSI	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_MOSI	36 45
SPI_MLB_IO2	SPI_45S	SPI	SPI_IO<2>	14 45
SPI_MLB_IO2	SPI_45S	SPI	SPIROM_WP_L	45
SPI_MLB_IO3	SPI_45S	SPI	SPI_IO<3>	14 45
SPI_MLB_IO3	SPI_45S	SPI	SPIROM_HOLD_L	45
SPI_MLB_IO3	SPI_45S	SPI	SPIROM_USE_MLB	15 45 68
PCH_RTCX	PCH_45S	PCH_15MIL	PCH_CLK32K_RTCX1	12 17
PCH_SRTCRST	PCH_45S	PCH_15MIL	PCH_SRTCRST_L	12
PCH_RTCRST	PCH_45S	PCH_15MIL	RTC_RESET_L	12
PCH_THRMTRIP	PCH_45S	PCH_18MIL	PM_THRMTRIP_L	15 37
PCH_THRMTRIP	PCH_45S	PCH_18MIL	PM_THRMTRIP_R_L	37
	PCH_45S	PCH_15MIL	PCH_INTRUDER_L	12
	PCH_45S	PCH_15MIL	PCH_INTVRMEN	12
	PCH_45S	PCH_15MIL	PCH_DSWVRMEN	13
	PCH_45S	PCH_15MIL	PM_RSMRST_L	13 61
	PCH_45S	PCH_15MIL	PM_SYSRST_L	13 17 36 68
	PCH_45S	PCH_15MIL	XDP_DBRESET_L	16 17
	PCH_45S	PCH_15MIL	PM_PCH_SYS_PWROK	13 16 17 36
	PCH_45S	PCH_15MIL	XDP_SYS_PWROK	16
	PCH_45S	PCH_15MIL	SYS_PWROK_R	17
	PCH_45S	PCH_15MIL	PM_PCH_PWROK	13 17
	PCH_45S	PCH_15MIL	PM_S0_PGOOD	17
	PCH_45S	PCH_15MIL	SMC_DELAYED_PWRGD	17 24 25 36 37
	PCH_45S	PCH_15MIL	PM_DSW_PWRGD	13 36
	PCH_45S	PCH_15MIL	PM_PWRBTN_L	13 16 36
	PCH_45S	PCH_15MIL	XDP_CPU_PWRBTN_L	16
	PCH_45S	PCH_15MIL	PCIE_WAKE_L	13 29 31
	PCH_45S	PCH_15MIL	AP_PCIE_WAKE_L	29 63
	PCH_45S	PCH_15MIL	CAM_PCIE_WAKE_L	31
	PCH_45S	PCH_15MIL	TBT_CIO_PLUG_EVENT_L	15 18 23
PCH_CLK24M_XTAL	PCH_45S	PCH_20MIL	PCH_CLK24M_XTALIN	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MIL	PCH_CLK24M_XTALOUT	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MIL	PCH_CLK24M_XTALOUT_R	17
PCH_RCOMP_PCIE	PCH_27P4S	PCH_12MIL	PCH_PCIE_RCOMP	14
PCH_RCOMP_OPI	PCH_27P4S	PCH_12MIL	PCH_OPI_COMP	15
PCH_RCOMP_SATA	PCH_27P4S	PCH_12MIL	PCH_SATA_RCOMP	12

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTL	*	=2x_DIELECTRIC	?
MEM_CTL2CTL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?
MEM_CMD2CMD_BM	*	=2x_DIELECTRIC	?
MEM_CMD2CTL_BM	*	=2x_DIELECTRIC	?
MEM_CTL2CTL_BM	*	=2x_DIELECTRIC	?
MEM_12MIL	*	0.305 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTL2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2OTHERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CMD_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CMD2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CTL2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	*	*	MEM_2OTHER
MEM_*_DQS_*	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTL	*	MEM_CMD2CTL
MEM_CTL	MEM_CTL	*	MEM_CTL2CTL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_2OTHERMEM
MEM_CMD	MEM_CMD	BGA_MEM	MEM_CMD2CMD_BM
MEM_CMD	MEM_CTL	BGA_MEM	MEM_CMD2CTL_BM
MEM_CTL	MEM_CTL	BGA_MEM	MEM_CTL2CTL_BM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DQBYTE_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DQBYTE_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DQBYTE_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DQBYTE_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DQBYTE_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DQBYTE_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DQBYTE_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DQBYTE_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DQBYTE_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DQBYTE_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DQBYTE_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DQBYTE_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DQBYTE_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DQBYTE_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DQBYTE_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DQBYTE_7	*	MEM_DQS2OWNDATA

Haswell ULT Memory Down DDR3L 1x8 Length Matching

DDR3 Signal Group	Unit	Min Length	Max Length
CTLmax - CTLmin	mils	0	100
CTL to CLK	mils	CLK - 500	CLK + 500
CMDi to CMDj	mils	CMDj - 100	CMDj + 100
CMD to CLK	mils	CLK - 500	CLK + 500
(DQmax - DQmin) per byte	mils	0	250
(DQS - DQmax) per byte	mils	-100	150
DQS to DQS#	mils	-5	5
Max(CLK-DQS) - Min(CLK-DQS)	mils	0	5500
CLK to CLK#	mils	-5	5

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
MEM_A_CLK	MEM_72D	MEM_CLK	MEM_A_CLK_P<0>	7 20 22
MEM_A_CLK	MEM_72D	MEM_CLK	MEM_A_CLK_N<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A_CKE<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A_CS_L<0>	7 20 22
MEM_A_ODT0	MEM_40S	MEM_CTL	MEM_A_ODT<0>	20 22
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_A<15..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_BA<2..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_RAS_L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_CAS_L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_WE_L	7 20 22 66
MEM_A_DQBYTE0	MEM_45S	MEM_A_DQBYTE_0	MEM_A_DQ<7..0>	7 67 68
MEM_A_DQBYTE1	MEM_45S	MEM_A_DQBYTE_1	MEM_A_DQ<15..8>	7 67 68
MEM_A_DQBYTE2	MEM_45S	MEM_A_DQBYTE_2	MEM_A_DQ<23..16>	7 67 68
MEM_A_DQBYTE3	MEM_45S	MEM_A_DQBYTE_3	MEM_A_DQ<31..24>	7 67 68
MEM_A_DQBYTE4	MEM_45S	MEM_A_DQBYTE_4	MEM_A_DQ<39..32>	7 20 67 68
MEM_A_DQBYTE5	MEM_45S	MEM_A_DQBYTE_5	MEM_A_DQ<47..40>	7 67 68
MEM_A_DQBYTE6	MEM_45S	MEM_A_DQBYTE_6	MEM_A_DQ<55..48>	7 67 68
MEM_A_DQBYTE7	MEM_45S	MEM_A_DQBYTE_7	MEM_A_DQ<63..56>	7 67 68
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM_A_DQS_P<0>	7 67
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM_A_DQS_N<0>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM_A_DQS_P<1>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM_A_DQS_N<1>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM_A_DQS_P<2>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM_A_DQS_N<2>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM_A_DQS_P<3>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM_A_DQS_N<3>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM_A_DQS_P<4>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM_A_DQS_N<4>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM_A_DQS_P<5>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM_A_DQS_N<5>	7 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM_A_DQS_P<6>	7 20 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM_A_DQS_N<6>	7 20 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM_A_DQS_P<7>	7 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM_A_DQS_N<7>	7 67
MEM_B_CLK	MEM_72D	MEM_CLK	MEM_B_CLK_P<0>	7 21 22
MEM_B_CLK	MEM_72D	MEM_CLK	MEM_B_CLK_N<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B_CKE<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B_CS_L<0>	7 21 22
MEM_B_ODT0	MEM_40S	MEM_CTL	MEM_B_ODT<0>	21 22
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_A<15..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_BA<2..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_RAS_L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_CAS_L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_WE_L	7 21 22 66
MEM_B_DQBYTE0	MEM_45S	MEM_B_DQBYTE_0	MEM_B_DQ<7..0>	7 67 68
MEM_B_DQBYTE1	MEM_45S	MEM_B_DQBYTE_1	MEM_B_DQ<15..8>	7 67 68
MEM_B_DQBYTE2	MEM_45S	MEM_B_DQBYTE_2	MEM_B_DQ<23..16>	7 67 68
MEM_B_DQBYTE3	MEM_45S	MEM_B_DQBYTE_3	MEM_B_DQ<31..24>	7 67 68
MEM_B_DQBYTE4	MEM_45S	MEM_B_DQBYTE_4	MEM_B_DQ<39..32>	7 21 67 68
MEM_B_DQBYTE5	MEM_45S	MEM_B_DQBYTE_5	MEM_B_DQ<47..40>	7 67 68
MEM_B_DQBYTE6	MEM_45S	MEM_B_DQBYTE_6	MEM_B_DQ<55..48>	7 67 68
MEM_B_DQBYTE7	MEM_45S	MEM_B_DQBYTE_7	MEM_B_DQ<63..56>	7 67 68
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM_B_DQS_P<0>	7 67
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM_B_DQS_N<0>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM_B_DQS_P<1>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM_B_DQS_N<1>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM_B_DQS_P<2>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM_B_DQS_N<2>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM_B_DQS_P<3>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM_B_DQS_N<3>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM_B_DQS_P<4>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM_B_DQS_N<4>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM_B_DQS_P<5>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM_B_DQS_N<5>	7 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM_B_DQS_P<6>	7 21 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM_B_DQS_N<6>	7 21 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM_B_DQS_P<7>	7 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM_B_DQS_N<7>	7 67
		MEM_PWR	PP1V35_S3	17 19 20 21 22 41 55 65
		MEM_PWR	PP1V35_S3_CPUDDR	8 10 41 65
		MEM_PWR	PP0V675_S0_DDRVTT	22 55 65 68
		MEM_PWR	PPVTTDDR_S3	55 65 68
		MEM_12MIL	CPU_DIMMA_VREFDQ	7 19
		MEM_12MIL	CPU_DIMMA_VREFDQ_A_ISOL	19
		MEM_12MIL	CPU_DIMMB_VREFDQ	7 19
		MEM_12MIL	CPU_DIMMB_VREFDQ_B_ISOL	19
		MEM_12MIL	CPU_DIMM_VREFCA	7 19
		MEM_12MIL	CPU_DIMM_VREFCA_A_ISOL	19
I146		MEM_12MIL	CPU_DIMM_VREFCA_B_ISOL	19
		MEM_12MIL	PP0V675_S3_MEM_VREFDQ_A	19 20 65
		MEM_12MIL	PP0V675_S3_MEM_VREFDQ_B	19 21 65
		MEM_12MIL	PP0V675_S3_MEM_VREFCA_A	19 20 65
		MEM_12MIL	PP0V675_S3_MEM_VREFCA_B	19 21 65

SYNC_MASTER=J44 SYNC_DATE=01/03/2013

PAGE TITLE
Memory Constraints

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Thunderbolt, DP, HDMI Constraints

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt & DisplayPort Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	*	=3X_DIELECTRIC	?
TBTDP_TXRX	*	=6X_DIELECTRIC	?
TBTDP_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
TBTDP_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
TBTDP_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_*	*	*	TBTDP_2OTHER
TBTDP_*	=SAME	*	TBTDP_2SAME
TBTDP_TX	*_RX	*	TBTDP_TXRX
TBTDP_RX	*_TX	*	TBTDP_TXRX

DisplayPort & HDMI Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?
DP_2OTHER	*	=4x_DIELECTRIC	?
HDMICLK_2OTHER	*	=7x_DIELECTRIC	?
HDMICLK_2DPHDMI	*	=4x_DIELECTRIC	?
HDMIDATA_2SAME	*	=3x_DIELECTRIC	?
HDMIDATA_2OTHER	*	=4x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICLK_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICLK_2DPHDMI	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMIDATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMIDATA_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDMI_DATA	*	*	HDMIDATA_2OTHER
HDMI_DATA	=SAME	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_TX	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_RX	*	TBTDP_TXRX
HDMI_CLK	*	*	HDMICLK_2OTHER
HDMI_CLK	HDMI_DATA	*	HDMICLK_2DPHDMI
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DPHDMI
HDMI_CLK	TBTDP_TX	*	HDMICLK_2DPHDMI

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_2OTHER
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	HDMI_DATA	*	DP_2SAME
DISPLAYPORT	TBTDP_TX	*	DP_2SAME
DISPLAYPORT	TBTDP_RX	*	TBTDP_TXRX

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.
MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_P	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_N	
			TBT_SPI_CLK	
SPI_TBT_MOSI	TBT_SPI_45S	TBT_SPI	TBT_SPI_MOSI	23
SPI_TBT_MISO	TBT_SPI_45S	TBT_SPI	TBT_SPI_MISO	23
SPI_TBT_CS_L	TBT_SPI_45S	TBT_SPI	TBT_SPI_CS_L	23
I315 DP_HDMI_TBT_ML	DP_85D	DISPLAYPORT	DP_HDMI_TBT_ML_P<3..0>	64 66
I314 DP_HDMI_TBT_ML	DP_85D	DISPLAYPORT	DP_HDMI_TBT_ML_N<3..0>	64 66
I313 DP_HDMI_TBT_AUX	DP_85D		DP_HDMI_TBT_AUX_P	13 64 66
I312 DP_HDMI_TBT_AUX	DP_85D		DP_HDMI_TBT_AUX_N	13 64 66
I309 HDMI_CLOCK	HDMI_85D	HDMI_CLK	HDMI_IG_CLK_C_P	63 64 68
I308 HDMI_CLOCK	HDMI_85D	HDMI_CLK	HDMI_IG_CLK_C_N	63 64 68
I311 HDMI_DATA	HDMI_85D	HDMI_DATA	HDMI_IG_DATA_C_P<2..0>	63 64 68
I310 HDMI_DATA	HDMI_85D	HDMI_DATA	HDMI_IG_DATA_C_N<2..0>	63 64 68

Only used on hosts supporting Thunderbolt video-in

Thunderbolt, DP, HDMI Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_C_P<1..0>	23 26 68
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_C_N<1..0>	23 26 68
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_P<1..0>	26 68
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_N<1..0>	26 68
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	23 26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	23 26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	26
I355 DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	23 26
I354 DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	23 26
I353 DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	26
I352 DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	26
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_P<0>	26 68
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_N<0>	26 68
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_P<0>	23 26 68
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_N<0>	23 26 68
I343 TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_P<1>	26 68
I342 TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_N<1>	26 68
I341 TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_P<1>	23 26 68
I340 TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_N<1>	23 26 68
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R1_AUXDDC_P	26
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R1_AUXDDC_N	26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_P	23 26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_N	23 26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_P	26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_N	26
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_C_P<1..0>	23 27 68
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_C_N<1..0>	23 27 68
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_P<1..0>	27 68
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_N<1..0>	27 68
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	23 27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	23 27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	27
I351 DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	23 27
I350 DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	23 27
I348 DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	27
I349 DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	27
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_P<0>	27 68
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_N<0>	27 68
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_P<0>	23 27 68
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_N<0>	23 27 68
I347 TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_P<1>	27 68
I346 TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_N<1>	27 68
I344 TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_P<1>	23 27 68
I345 TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_N<1>	23 27 68
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R1_AUXDDC_P	27
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R1_AUXDDC_N	27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_P	23 27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_N	23 27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_P	27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_N	27
I338 DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	5 23
I339 DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	5 23
I336 DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	23
I337 DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	23
I335 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_C_P	13 23
I334 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_C_N	13 23
I324 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_P	23
I333 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_N	23
I332 DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	23 64
I330 DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	23 64
I331 DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	23
I328 DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	23
I329 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_C_P	23 64
I327 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_C_N	23 64
I325 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_P	23
I326 DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_N	23
I316 DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_C_P<3..0>	5 62
I317 DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_C_N<3..0>	5 62
I318 DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_P<3..0>	62 68
I319 DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_N<3..0>	62 68
I320 DP_INT_AUXCH	DP_85D		DP_INT_AUXCH_C_P	5 62
I321 DP_INT_AUXCH	DP_85D		DP_INT_AUXCH_C_N	5 62
I322 DP_INT_AUXCH	DP_85D		DP_INT_AUX_P	62 68
I323 DP_INT_AUXCH	DP_85D		DP_INT_AUX_N	62 68

Notes:
AUX and DDC was removed from DISPLAYPORT or TBTDP_RX/TX because it's not high speed, and to save routing space.

Only used on dual-port hosts.

SYNC_MASTER=J44 SYNC_DATE=08/12/2013
PAGE TITLE
TBT,DP,HDMI Constraints
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_2OTHER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_2OTHER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2x_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2x_DIELECTRIC	?
S2_CMD2CMD	*	=2x_DIELECTRIC	?
S2_CMD2CTRL	*	=2x_DIELECTRIC	?
S2_CTRL2CTRL	*	=2x_DIELECTRIC	?
S2_2OTHERMEM	*	=4x_DIELECTRIC	?
S2MEM_2PWR	*	=2x_DIELECTRIC	?
S2MEM_2GND	*	=2x_DIELECTRIC	?
S2MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS2OWNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_2OTHERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_2OTHER
S2_MEM_DQS*	*	*	S2MEM_2OTHER
S2_MEM_CMD	*	*	S2MEM_2OTHER
S2_MEM_CTRL	*	*	S2MEM_2OTHER
S2_MEM_CLK	*	*	S2MEM_2OTHER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS2OWNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS2OWNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	31 32
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	31 32
S2_MEM_CKE	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	31 32
S2_MEM_CS	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	31 32
	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	31 32
S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	31 32
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	32 68
		S2_MEM_PWR	PP1V35_CAM	31 32
		S2_MEM_PWR	PP0V675_CAM_VREF	31 32
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	32
I149		S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	32

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

Camera Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

SMC SMBus & Charger Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
SMBUS_SMC_2	SMB_45S	SMB	SMBUS_SMC_2_S3_SCL	34 36 39 68
SMBUS_SMC_2	SMB_45S	SMB	SMBUS_SMC_2_S3_SDA	34 36 39 68
SMBUS_SMC_1	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	14 32 36 39 43 68 72
SMBUS_SMC_1	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	14 32 36 39 43 68 72
SMBUS_SMC_0	SMB_45S	SMB	SMBUS_SMC_0_S0_SCL	36 39 62 68
SMBUS_SMC_0	SMB_45S	SMB	SMBUS_SMC_0_S0_SDA	36 39 62 68
SMBUS_SMC_5	SMB_45S	SMB	SMBUS_SMC_5_G3_SCL	36 39 51 52 68
SMBUS_SMC_5	SMB_45S	SMB	SMBUS_SMC_5_G3_SDA	36 39 51 52 68
SMBUS_SMC_3	SMB_45S	SMB	SMBUS_SMC_3_SCL	36 39 43 63
SMBUS_SMC_3	SMB_45S	SMB	SMBUS_SMC_3_SDA	36 39 43 63

SYNC_MASTER=J44 SYNC_DATE=08/12/2013

PAGE TITLE

SMC Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
THERM_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
DIG_AUDIO	*	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	0.1 MM	0.1 MM
ANL_AUDIO	*	=1TO1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
ANL_AUDIO_WIDE	*	=1TO1_DIFFPAIR	0.3 MM	0.3 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
GND	PCIE_*	*	GND_P2MM
GND	SATA_*	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SB_POWER	SATA_*	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_45S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.070 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.100 MM	500 MIL		
CPU_27P4S	BOTTOM			0.230 MM	100 MIL		
USB3_85D	TOP			0.100 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

DDR3 Loaded Segment Constraint Relaxations

Alternate single ended and differential impedances between devices.

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_40S	BGA_MEM	MEM_45S
MEM_72D	BGA_MEM	MEM_85D

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.100 MM	6.35 MM		

DP, SATA, HDMI, PCIE CONSTRAINT RELAXATIONS

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_85D	BGA	P65_BGA
CLK_PCIE_85D	BGA	P65_BGA
HDMI_85D	BGA	P65_BGA

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SENSE_45S	*	SENSE_45S
THERM_45S	*	THERM_45S
DIG_AUDIO	*	DIG_AUDIO
ANL_AUDIO	*	ANL_AUDIO
1TO1_DIFFPAIR	*	1TO1_DIFFPAIR

J44 Specific Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
THERM_DP_TBT_D1	THERM_45S	THERM	TBTTHMSNS_D1_P	23 43
THERM_DP_TBT_D1	THERM_45S	THERM	TBTTHMSNS_D1_N	43
THERM_DP_CPU_D1	THERM_45S	THERM	CPUTHMSNS_D1_P	43
THERM_DP_CPU_D1	THERM_45S	THERM	CPUTHMSNS_D1_N	43
THERM_DP_CPU_D2	THERM_45S	THERM	CPUTHMSNS_D2_P	43
THERM_DP_CPU_D2	THERM_45S	THERM	CPUTHMSNS_D2_N	43
	SENSE_45S	SENSE	NC_ISNS_CAMERAP	42
	SENSE_45S	SENSE	NC_ISNS_CAMERAN	42
SENSE_DP	SENSE_45S	SENSE	ISNS_CPUDDR_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_CPUDDR_N	41
SENSE_DP_LCDBKLT	SENSE_45S	SENSE	ISNS_LCDBKLT_P	40 58
SENSE_DP_LCDBKLT	SENSE_45S	SENSE	ISNS_LCDBKLT_N	40 58
SENSE_DP_TBT	SENSE_45S	SENSE	ISNS_TBT_P	42
SENSE_DP_TBT	SENSE_45S	SENSE	ISNS_TBT_N	42
SENSE_DP	SENSE_45S	SENSE	ISNS_LCDPANEL_P	42 62
SENSE_DP	SENSE_45S	SENSE	ISNS_LCDPANEL_N	42 62
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_COMPUTING_P	40 42
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_COMPUTING_N	40 42
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER5V_P	40
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER5V_N	40
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER3V3_P	40
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER3V3_N	40
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_P	41
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_N	41
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_R_P	41
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_R_N	41
	SENSE_45S	SENSE	CPUVR_ISNS1_P	41 54
	SENSE_45S	SENSE	CPUVR_ISNS1_N	41 54
	SENSE_45S	SENSE	CPUVR_ISNS2_P	41 54
	SENSE_45S	SENSE	CPUVR_ISNS2_N	41 54
SENSE_DP	SENSE_45S	SENSE	ISNS_1V05_S0_P	41 57
SENSE_DP	SENSE_45S	SENSE	ISNS_1V05_S0_N	41 57
SENSE_DP	SENSE_45S	SENSE	ISNS_SSD_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_SSD_N	41
SENSE_DP	SENSE_45S	SENSE	ISNS_TPAD_P	
SENSE_DP	SENSE_45S	SENSE	ISNS_TPAD_N	
	SENSE_45S	SENSE	NC_ISNS_DDR_S3P	41 55
	SENSE_45S	SENSE	NC_ISNS_DDR_S3N	41 55
SENSE_DP	SENSE_45S	SENSE	ISNS_PP3V3S0_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_PP3V3S0_N	41
SENSE_DP	SENSE_45S	SENSE	ISNS_PP5VS0_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_PP5VS0_N	41
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_P	42 43
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_N	42 43
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_R_P	42
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_R_N	42
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_P	52
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_N	52
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_R_P	52
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_R_N	52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_P	52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_N	52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_R_P	42 52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_R_N	42 52

J44 Specific Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_L_P	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_L_N	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LIN_N	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_LIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_LIN_N	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_R_P	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_R_N	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RIN_N	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_RIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_RIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_L_P	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_L_N	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LSUBIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	LSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	LSUBIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_R_P	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_R_N	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RSUBIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	RSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	RSUBIN_N	48
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SL_OUT_P	48 50 68
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SL_OUT_N	48 50 68
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SR_OUT_P	48 50 68
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SR_OUT_N	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_L_OUT_P	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_L_OUT_N	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_R_OUT_P	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_R_OUT_N	48 50 68
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CH_HS_GND	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_HS_MIC_P	
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_SLEEVE	50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_SLEEVE_XW	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HP_PORT_REFCH	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HS_MIC_P	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	CODEC_HS_MIC_P	46
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	HS_MIC_P	46 49
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_HS_MIC_N	
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_RING2	50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_RING2_XW	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HP_PORT_REFUS	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HS_MIC_N	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_US_HS_GND	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	HS_MIC_N	46 49
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	CODEC_HS_MIC_N	46
		SB_POWER	PP3V3_S5	8 11 13 15 16 17 18 26 27 29 56 59 60 61 65 68
		SB_POWER	PP3V3_S0	61 62 64 65 68 8 11 12 13 15 17 18 24 28 30 37 38 39 40 41 42 43 44 46 47 50
		GND	GND	

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